Exact Generating-Function Analysis of a Two-Class Non-Preemptive Priority $M/M/1$ Queue with Jockeying and Abandonment

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Abstract

We study the stationary queue-length distribution of a two-class, non-preemptive priority $M/M/1$ queue augmented with two waiting-room mechanisms: *jockeying*, in which a waiting customer switches to the other queue, and *abandonment*, in which a waiting customer leaves the system. Jockeying conserves the total number of customers, whereas abandonment does not, and combining both yields a model (Model-X) that is analytically intractable. We therefore present a collection of simpler models: Model-A (baseline), Model-B (bidirectional jockeying), Model-B_2 (jockeying from the priority class to the non-priority class) and Model-C_2 (abandonments in the priority class). We also study two head-of-line models: Model-B_2^{H} (only the first customer in the priority queue can jockey) and Model-C_2^{H} (only the first customer in the priority queue can abandon).

The central object is the bivariate Probability Generating Function (PGF) of the two queue lengths, obtained by analytical and, where the structure allows, probabilistic arguments. The purpose of this work is to present a unified framework for deriving the functional equation of each model and to expose the mathematical challenges intrinsic to each. Model-A is classical and reduces to an algebraic expression by finding a kernel root inside the unit disc; Model-B_2 is solved by imposing analyticity of the PGF at an interior point and Model-C_2 by imposing boundedness of the PGF on the boundary of the unit disc. The head-of-line models, like the baseline Model-A, also yield an algebraic expression for the PGF. Two models are left open: Model-B reduces to an integral expression for the PGF involving a Volterra-type integral equation, and Model-X compounds this with nonlinear characteristic curves.

Every closed form is verified against a truncated continuous-time Markov Chain (CTMC) that recovers the baseline model as the mechanism rates vanish. The comparison reveals a key difference between jockeying and abandonment: the former merely redistributes delay between classes, whereas the latter substantially shortens the queue lengths.

Contents

1 Introduction

Priority queues arise naturally wherever a common server must satisfy demand of heterogeneous urgency. The primary motivation for this thesis is healthcare operations, specifically the management of waiting lists for elderly-care facilities, where medical severity dictates a user's —hereafter *customer*— priority class. Here prolonged waiting may itself alter the system's dynamics: a lack of appropriate care can deteriorate a waiting customer's health, leading them to change priority status (jockeying)[1] or leave the system (abandonment), whether to seek alternative care or through death. Both mechanisms are documented in the healthcare-operations literature: Zenios [16] models the organ-transplant waiting list as a queue whose abandonment mechanism is patient mortality, and Hu, Chan & Dong [9] model patients who transition between a moderate and an urgent class as their condition deteriorates while they wait. Despite this motivation, the work is fundamentally modelling-driven: we characterise the long-run behaviour of queueing systems in which jockeying and/or abandonment play a role.

This thesis analyses a 2-class $M/M/1$ queue with non-preemptive priority: each class arrives according to a Poisson process (rates λ_1 and λ_2), service times are exponential at a common rate μ, and the single server completes the current service even when a higher-priority customer arrives. To this we add the mechanisms of jockeying and abandonment. The parent model in which both mechanisms act simultaneously is called Model-X; it is described in detail in §4, and the models studied in this thesis are its specialisations, obtained by activating one mechanism at a time (Table 1 below). How these two mechanisms interact within a priority discipline, and how that interaction affects the difficulty of obtaining a closed-form Probability Generating Function (PGF) $P(x,y) = \mathbb{E}[x^{N_1} y^{N_2}]$ —where N_1 and N_2 count the class-1 and class-2 customers *waiting* in their queues— is the main focus of this thesis.

The analytical challenge

For the general Model-X, the bivariate PGF satisfies a functional equation with terms of three kinds: algebraic coefficients, partial derivatives in x —from jockeying and abandonment in class-1— and partial derivatives in y —from the same mechanisms in class-2. When both kinds of derivative terms are present, the equation belongs to a family of integro-differential equations whose solution would require more advanced tools. The problem is moreover asymmetric: derivative terms from the non-priority class complicate the equation more than those from the priority class alone. The reason, developed in §6, lies in the equation's unknown boundary function, which is a function of y: when the derivatives act only in x, the equation is an ODE in x and the unknown function rides along as a parameter, whereas a derivative in y forces an integration that runs *through* the unknown function's argument, turning the problem into an integral equation.

We therefore solve less complex models obtained by setting some of the jockeying and abandonment parameters to zero. Table 1 lists the models studied here and indicates whether a PGF expression is obtained for each:

Model	**Active mechanism**	**Equation type**	**Status**	**Section**
X	jockeying and abandonment	PDE, transcendental characteristics	open	§4
A	none	algebraic PGF equation	solved	§5
B	bidirectional jockeying	PDE, Volterra family	open	§6
B_2	one-way jockeying ($1 \to 2$)	ODE in x, Kummer	solved	§7
C_2	class-1 abandonment	ODE in x, Kummer	solved	§8
B_2^{H}	one-way jockeying, head-of-line	algebraic PGF equation	solved	§9
C_2^{H}	class-1 abandonment, head-of-line	algebraic PGF equation	solved	§10

Table 1: Models studied in this thesis, with the active mechanism, the resulting functional-equation type, and the solution status. The head-of-line variants B_2^{H} and C_2^{H} restrict the mechanism to the customer at the head of Queue 1, which collapses the derivative term to an algebraic one and renders the model solvable in closed form.

The two open entries, Model-X and Model-B, are exactly those whose equations involve derivative terms in y; potential variants such as Model-B_1, Model-C or Model-C_1 are not considered for

[1] In much of the queueing literature, *jockeying* denotes a waiting customer moving between *parallel* queues, typically to a shorter line. In this thesis the two queues are priority classes at a single server, so jockeying denotes a change of priority class while waiting —what the priority-change literature calls a *priority change* [3, 9]. We keep the shorter term throughout.

the same reason, as they all reduce to a Volterra integral equation lying beyond the scope of this thesis. Model-B serves as our representative example of how such problems reduce to that integral equation; the remaining models are solvable. The last two entries, Models B_2^{H} and C_2^{H} (§§9 and 10), are *head-of-line* simplifications in which only the customer at the head of Queue 1 may jockey or abandon. The mechanism then acts at the constant rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$ rather than the length-proportional $\gamma_1 n_1$ or $\theta_1 n_1$, so the fundamental equation stays algebraic and is solved by the Kernel Method exactly as in Model-A.

Methods

Three complementary proof strategies are used throughout. All of them act on a functional equation of the common shape

$$K(x,y)\,P(x,y) \;+\; A_1(x,y)\,\frac{\partial P}{\partial x} \;+\; A_2(x,y)\,\frac{\partial P}{\partial y} \;=\; B_1(x,y)\,P_y(y) \;+\; B_0(x,y)\,\pi(0,0),$$

in which the left-hand side involves the unknown PGF and the right-hand side two boundary unknowns: the boundary function $P_y(y) = P(0,y)$ —the transform of the states with an empty priority queue— and the scalar $\pi(0,0)$ —the probability that the server is busy while both queues are empty (Lemma 1). Solving a model means determining these boundary unknowns and recovering $P(x,y)$ from them.

Kernel method / ODE with integrating factor. For Model-A, which has no derivative terms, the algebraic coefficient of $P(x,y)$ vanishes on a curve (the *kernel*); evaluating the equation there determines the remaining unknowns. Model-B_2 and Model-C_2, which do carry derivative terms in x, reduce to a first-order linear ODE in x with an explicit integrating factor, and their unknowns follow from a regularity argument at a singularity of that factor. We present both strategies together, as they share the same underlying idea: the latter extends the former to the case where the unknowns are independent of the derivative terms and can be taken outside the integral. Both deliver the complete transform $P(x,y)$.

Probabilistic argument via Pollaczek–Khinchine and the effective service time. For Model-A and Model-C_2, class-2 arrivals are Poisson, so the PASTA property holds and we can apply the Pollaczek–Khinchine formula together with the *effective service time* seen by a class-2 customer. This is an $M/G/1$ queue whose service times equal the busy period of the class-1 subsystem —an $M/M/1$ queue for Model-A and an $M/M/1+M$ system for Model-C_2. Unlike the first strategy, this route determines only one of the boundary unknowns, the boundary function $P_y(y)$, from which $P(x,y)$ is then recovered through the fundamental equation; in both cases it independently confirms the analytic results. Since jockeying destroys the Poisson structure, it does not apply to the other models.

Partial PGF (PPGF). Cohen's recursion argument [2] transforms only one queue-length index while keeping the other discrete; on the regular state space it delivers the complete transform for Model-A through a third, independent proof. For the same baseline model we also introduce an alternative state space indexed by the number of class-2 customers —whose distribution is hard— and the total number of customers —which is easy, since the total is conserved and behaves like a regular $M/M/1$ queue. This yields a PPGF with a non-homogeneous term; we explore, with limited success, the challenges of carrying Cohen's argument over to this alternative space and assess whether it can deliver a feasible rough approximation.

Structure of the thesis

Section 2 reviews the relevant literature and Section 3 collects the probabilistic and analytic background used throughout this work. Section 4 defines the full model, Model-X, and derives the general balance equations and the fundamental PGF equations, both for the regular and the alternative state space. The analysis then proceeds from baseline to synthesis: Section 5 solves the baseline Model-A in closed form; Section 6 shows that bidirectional jockeying (Model-B) reduces the problem to a Volterra-type integral equation and is left open. Sections 7 and 8 then recover tractability in the one-way-jockeying (Model-B_2) and class-1-abandonment (Model-C_2) special cases, while Sections 9 and 10 show how the head-of-line variants, Models B_2^{H} and C_2^{H}, collapse the functional equation to an algebraic one solved in closed form. Finally, Section 11 verifies every closed form numerically against a truncated CTMC and compares the solved models, and Section 12 concludes by abstracting the obstruction taxonomy and setting out the future work.

2 Literature

The study of two-class priority queues with jockeying and abandonment sits at the intersection of three well-developed streams: the exact generating-function analysis of non-preemptive priority queues, the theory of queues with customer impatience (abandonment), and the theory of jockeying and dynamic priority changes. The applied motivation throughout is the management of access to care —elderly-care waiting lists, emergency departments, and transplant registries— in which the two mechanisms studied here carry direct clinical readings: a waiting patient whose condition deteriorates changes priority class (jockeying), while one who leaves the list by death or transfer abandons the system (abandonment). This thesis contributes an exact, single-server treatment that admits jockeying and abandonment within a unified bivariate-PGF framework, filling a gap that the multi-server and approximate literature has so far left open.

Non-preemptive priority queues: exact queue-length analysis

The expected waiting times of the non-preemptive priority $M/M/1$ queue are standard textbook material [1]; the exact *queue-length* analysis is where the generating-function tradition begins. The direct methodological ancestor of this work is Cohen [2], who studied a single server loaded by two independent Poisson streams under non-preemptive priority and introduced a partial-generating-function recursion that resolves the lower-priority queue length by transforming only one coordinate of the joint distribution while keeping the other discrete; we refer to this device as the "Cohen trick". It is exactly the device reused, in modern notation, in the partial-PGF analysis of Model-A on the total-count state space (§5.2). The stationary queue-length law of the two-class non-preemptive $M/M/1$ priority queue —the content of Theorem 2 (Model-A)— is in this sense classical, recoverable by several routes.

The most recent exact closed-form treatment of the non-preemptive priority queue is that of Zuk & Kirszenblat [17], who obtain the joint and marginal queue-length distributions of a non-preemptive $M/M/c$ queue with two priority classes and a common service rate by generating-function methods and a quadratic recurrence; their single-server specialisation ($c = 1$) coincides with Model-A.

Priority queues with abandonment

Abandonment enters queueing theory through the Erlang-A ($M/M/c + M$) model, in which each waiting customer holds an independent exponential patience clock; the modern treatment and its heavy-traffic scaling are due to Garnett et al. [7]. The analytic signature of impatience is well documented: per-customer patience clocks make the transition rates state-inhomogeneous, so the generating-function solutions, where they close at all, close in terms of hypergeometric series.

Kapodistria [11] exhibits this pattern most clearly in a single-server queue with *synchronised* abandonments, in which all waiting customers abandon simultaneously, each with probability p, at Poisson opportunity epochs. The stationary law is solved exactly through basic (q-)hypergeometric series, and in the no-synchronisation limit $p \to 0^+$ it recovers the $M/M/1 + M$ queue-length PGF as a ratio of Kummer functions: precisely the special-function structure that carries the boundary function of Model-C_2.

Exact stationary results for priority queues *with* abandonment are scarce, and those available concern multi-server systems. Jouini & Roubos [10] treat an $M/M/s + M$ queue with two non-preemptive classes in which the classes are statistically identical —a common service rate and a common patience distribution, so that the classes differ only in their priority level. They derive Laplace–Stieltjes transforms for conditional and unconditional waiting times together with the probabilities of service and of abandonment; this is the multi-server analogue of Model-C_2, but with the waiting time, not the queue-length PGF, as its object. Where exact transforms are unavailable the literature turns to fluid and diffusion approximations or to many-server asymptotics [15]; the exact Erlang-A stationary law itself is expressed through confluent hypergeometric and incomplete-gamma functions, the same special-function structure that appears, at finite scale, in the closed form for Model-C_2.

The exact multi-server priority results that do exist involve no abandonment at all: Wang et al. [14] analyse a preemptive $M/M/c$ queue with class-dependent service rates and no impatience, and Zuk & Kirszenblat [17] a non-preemptive $M/M/c$ queue, likewise without impatience. Model-C_2 therefore appears to be the first exact, single-server, queue-length treatment of *class-asymmetric* abandonment, with impatience acting on the priority class alone. The clinical reading that licenses this asymmetry is long-standing: Zenios [16] models the organ-transplant waiting list as a queue whose abandonment mechanism is patient mortality, precisely the regime in which losing high-priority demand is the outcome of interest.

Jockeying and priority changes

The closest predecessor on the jockeying side is de Waal [3], who studies an $M/G/1$ priority queue in which impatient customers *may change priority class* while waiting. de Waal's analysis is approximate by necessity rather than by choice: with general service times an exact analysis is out of reach, and the paper delivers Laplace–Stieltjes-based *approximations* of the waiting-time distributions rather than an exact stationary law. The healthcare incarnation of the same idea is the model of Hu, Chan & Dong [9], who let customers transition between a moderate and an urgent class as their condition deteriorates or improves and characterise a near-optimal $c\mu/\theta$-type scheduling index in heavy traffic; the system is multi-server and its results are asymptotic, so it too is approximate in the sense of Table 2.

A complementary strand studies dynamic rather than static priority: Stanford et al. [13] analyse the *accumulating-priority* $M/G/1$ queue, in which each class earns priority linearly in its waiting time so that older customers eventually overtake newer ones, and derive exact waiting-time distributions through a maximum-priority process and Laplace–Stieltjes transforms. Accumulating priority is a continuous-time analogue of the discrete, rate-γ_i jockeying studied here; like the present work it is exact and single-server, but its object is the sojourn time rather than the bivariate queue-length PGF.

Model-B_2 is the exact counterpart of this "priority-change" literature. It retains the deterioration mechanism —a waiting class-1 customer relocates to the class-2 queue at rate γ_1— and pays for exactness with the single-server, Markovian, one-directional restriction. In return it produces a closed-form boundary function in place of an approximation.

Methodological lineage

The functional-equation technique used in Sections 7 and 8 belongs to the kernel-method and boundary-value tradition for two-dimensional Markovian queues. The systematic theory —reducing a bivariate balance equation to the curve on which its kernel vanishes, and from there to a boundary-value problem for the unknown boundary functions— is developed by Fayolle, Iasnogorodski & Malyshev [6] for random walks in the quarter-plane. That approach still delivers explicit two-class results: Devos, Walraevens, Fiems & Bruneel [4] obtain the closed-form joint queue-length distribution of a discrete-time two-class queue with randomly alternating service by analysing the kernel of just such a functional equation and determining its two unknown boundary functions through analytic continuation.

The arguments of this thesis are elementary, one-dimensional instances of that method. In Model-B_2 the boundary function is fixed by demanding *analyticity* of the PGF at the interior point where the kernel root meets the diagonal (§7), and in Model-C_2 by demanding mere *boundedness* at the unit-disc boundary (§8); both select the analytic solution of the reduced equation, exactly the role the boundary conditions play in the general theory.

Two narrower tools recur. The confluent hypergeometric (Kummer) function that carries the Model-C_2 boundary function is the same special function through which the Erlang-A and reneging-queue stationary laws are expressed [15, 11, 5]; and the selection of the decaying solution of the three-term recurrence behind the partial-PGF analysis is governed by Pincherle's theorem, for which we cite the recurrence-relation treatment of Gautschi [8] alongside the DLMF [5].

Position of this thesis

Table 2: Literature overview: models and mechanisms. ✓ = present, – = absent, ≈ = approximate only.

Reference	Servers	Non-preem.	Jockeying	Aband.	Queue-len.	Exact
Cohen (1956)	1	✓	–	–	✓	✓
de Waal (1992)	1	✓	✓	–	–	≈
Jouini & Roubos (2014)	c	✓	–	✓	–	✓
Wang et al. (2015)	c	–	–	–	✓	✓
Stanford et al. (2014)	1	✓	✓	–	–	✓
Zuk & Kirszenblat (2023)	c	✓	–	–	✓	✓
Hu, Chan & Dong (2022)	c	✓	✓	✓	–	≈
This thesis (Model-A)	**1**	✓	–	–	✓	✓
This thesis (Model-B_2)	**1**	✓	✓	–	✓	✓
This thesis (Model-C_2)	**1**	✓	–	✓	✓	✓

Table 2 places this work against its closest neighbours, and reading it by row makes the gap precise. The Cohen (1956) row is the single-server, exact, non-preemptive queue-length baseline with neither extra mechanism —the ancestor of Model-A. de Waal (1992) adds jockeying (priority change) at a single server, but its analysis is approximate and concerns waiting times rather than queue lengths. Jouini & Roubos (2014) add abandonment exactly, but at c servers, for statistically identical classes, and again for waiting times. Wang et al. (2015) is exact and queue-length-focused, yet multi-server, preemptive, and mechanism-free. Stanford et al. (2014) is single-server and exact with a dynamic-priority (jockeying-like) mechanism —the closest analogue of Model-B_2 in the table— but its object is the sojourn time. Zuk & Kirszenblat (2023) is the exact multi-server queue-length result whose $c = 1$ case is Model-A, with neither jockeying nor abandonment. Hu, Chan & Dong (2022) carries both mechanisms —jockeying-as-deterioration and abandonment— and is in that sense the richest model in the table, but it is multi-server and offers no exact results. The pattern is symptomatic of the multi-server literature as a whole: once $c > 1$ is combined with jockeying or abandonment, the analysis moves to approximations and asymptotics, so that exact, mechanism-rich analysis is essentially a single-server phenomenon.

No row above combines the exact bivariate queue-length PGF with either jockeying or abandonment —which is precisely the column pattern of the three "This thesis" rows. The gap can therefore be stated in one sentence: prior to this work there is no exact, single-server, queue-length analysis that places non-preemptive priority together with jockeying or abandonment inside a single PGF framework.

The limits of these claims must be stated with equal precision. The framework is unified but not complete: the full model (Model-X, Remark 2), in which jockeying and abandonment act together, and the bidirectional-jockeying model (Model-B, §6) are left open. For each of these, this thesis characterises the analytical obstruction and reduces the problem to a Volterra-type integral equation, providing partial results rather than a closed form. The exact contributions are confined to the three "This thesis" rows of Table 2 —the baseline Model-A, the one-way jockeying Model-B_2, and the class-1 abandonment Model-C_2, together with their head-of-line simplifications. The claim is exactness and unification at a single server with one mechanism active at a time; it is explicitly not a solution of the combined or bidirectional problems, which remain, as in the multi-server literature, beyond exact reach.

3 Preliminaries

This section collects the background material used throughout the derivations, so that the later arguments can stay focused on their primary objectives.

The central object throughout is the (P)PGF of the numbers of waiting customers, together with the system's idle probability.

3.1 $M/M/1$ queue

The canonical $M/M/1$ queue is the foundation of queueing theory and is treated in many textbooks; we follow [1]. Customers arrive according to a Poisson process —i.e. the inter-arrival times are exponentially distributed with parameter λ—, service times are exponentially distributed with parameter μ, and a single server operates in first-come-first-served (FCFS) order. The system is stable if and only if the load $\rho = \lambda/\mu < 1$.

Let π_n be the limiting probability of finding n customers in the system. Under stability, the detailed balance equations $\lambda\,\pi_n = \mu\,\pi_{n+1}$ ($n \geq 0$) together with the normalisation $\sum_{n=0}^{\infty} \pi_n = 1$ give us the geometric steady-state distribution of the number of customers in the system:

$$\pi_n = (1-\rho)\,\rho^n, \quad n \geq 0.$$

By the PASTA property (§3.2), this also corresponds to the distribution seen by arriving customers. Let L be the number of customers in the system, including any in service. Its expectation is

$$\mathbb{E}[L] = \frac{\rho}{1-\rho}.$$

From this, the expected sojourn time W—the time between a customer's arrival and their departure after service—follows by Little's Law ($\mathbb{E}[L] = \lambda\mathbb{E}[W]$):

$$\mathbb{E}[W] = \frac{\mathbb{E}[L]}{\lambda} = \frac{1/\mu}{1-\rho}.$$

A less common but highly relevant concept is the Busy Period (BP): the time between a customer arriving to an idle system and the subsequent departure that leaves the system empty. Its Laplace–Stieltjes Transform (LST) and expectation are known:

$$\widetilde{BP}(s) = \frac{(\mu+\lambda+s) - \sqrt{(\mu+\lambda+s)^2 - 4\mu\lambda}}{2\lambda}, \tag{3.1}$$

$$\mathbb{E}[BP] = \frac{1}{\mu(1-\rho)}. \tag{3.2}$$

3.2 The PASTA property

We invoke repeatedly the *PASTA* property (*Poisson Arrivals See Time Averages*) [1]: if a system is fed by a Poisson arrival process that satisfies the *lack-of-anticipation assumption* (at each instant the future increments of the arrival process are independent of the current system state), then the long-run fraction of arrivals that find the system in a given state equals the stationary probability of that state, so the distribution seen by an arriving customer coincides with the time-stationary one. Two standard companions of PASTA are also used: by Poisson *superposition* the merger of independent Poisson streams is itself Poisson, and by i.i.d. *thinning* each component stream of such a merger is Poisson as well. In this thesis these properties are applied to the two class arrival streams of the priority queue; the verification of the hypotheses is given once, when the model is defined (§4).

3.3 Confluent hypergeometric function

The *Kummer confluent hypergeometric function of the first kind* appears in several derivations in this thesis; it reads as follows:

$${}_1F_1(a;\, b;\, z) \;=\; \sum_{n=0}^{\infty} \frac{a^{(n)}}{b^{(n)}\, n!}\, z^n, \qquad a^{(n)} = \prod_{k=0}^{n-1}(a+k), \quad a^{(0)} = 1, \tag{3.3}$$

with $b^{(n)}$ defined analogously; here b is not 0 or a negative integer —i.e. $b \notin \{0, -1, -2, \dots\}$— and the series converges for all $z \in \mathbb{C}$. The function ${}_1F_1(a;\, b;\, z)$ is the unique solution of *Kummer's equation* [5]:

$$z\,u'' + (b-z)\,u' - a\,u = 0,$$

normalised by ${}_1F_1(a;\, b;\, 0) = 1$.

For this thesis, two identities related to it are used. First, the *Euler integral representation*, for $a, b > 0$ we have:

$${}_1F_1(a;\, a+b;\, z) \;=\; \frac{1}{B(a,b)} \int_0^1 t^{a-1}(1-t)^{b-1} e^{zt}\, dt, \tag{3.4}$$

where B is the *Beta function* $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, and Γ is the *Gamma function*. Secondly, *contiguous ratios*: ratios of two ${}_1F_1$ values whose parameters differ by integer shifts. Two patterns arise in this thesis. The *simultaneous shift*

$$\frac{{}_1F_1(a+1;\, a+1+b;\, z)}{{}_1F_1(a;\, a+b;\, z)}, \tag{3.5}$$

in which both parameters move by one so that their difference b is preserved, is by the Euler representation (3.4) a ratio of two weighted Beta integrals differing only in the exponent of t; it appears as the closed-form value of the Erlang-A busy-period continued fraction (§3.4, Equation (3.10)) and hence in the boundary function of Model-C_2. The *first-parameter shift* ${}_1F_1(a{+}1;\, b;\, z)\big/{}_1F_1(a;\, b;\, z)$, with the second parameter fixed, appears instead in the boundary function of Model-B_2 (§7).

3.4 $M/M/1{+}M$ queue (Erlang-A)

Another model relevant to this thesis is the $M/M/1{+}M$ queue, also known as the *Erlang-A* model. It extends the $M/M/1$ queue with customer impatience: each *waiting* customer (the one in service is not impatient) independently abandons after an exponentially distributed patience time with rate θ. Crucially, unlike the $M/M/1$ it does not require $\rho = \lambda/\mu < 1$ for stability: it is positive recurrent for all $\lambda, \mu, \theta > 0$.

The steady-state distribution follows from its balance equations, where n denotes the number of customers in the *system*:

$$\lambda\, \pi_n = (\mu + n\theta)\, \pi_{n+1}, \qquad n \geq 0.$$

Iterating yields

$$\pi_n = \pi_0\, \frac{(\lambda/\theta)^n}{(\mu/\theta)^{(n)}} \quad n \geq 0; \quad \text{where} \quad a^{(n)} = a(a+1)\cdots(a+n-1)$$

Normalisation gives the non-trivial idle probability $\pi_0 = [{}_1F_1(1;\, \mu/\theta;\, \lambda/\theta)]^{-1}$ (see §3.3 for the Kummer function ${}_1F_1$), and summing the geometric-like terms shows that the queue-length PGF $\Pi(z) = \sum_{n\geq 0} \pi_n z^n$ is correspondingly a *ratio* of Kummer functions,

$$\Pi(z) = \frac{{}_1F_1(1;\, \mu/\theta;\, (\lambda/\theta)z)}{{}_1F_1(1;\, \mu/\theta;\, \lambda/\theta)}.$$

The same closed form is derived by Kapodistria [11, Thm. 3] as the no-synchronisation limit of a queue with synchronised abandonments, there in the sojourn-time-impatience convention (the customer in service also abandons), which amounts to replacing μ by $\mu + \theta$.

Unlike the textbook facts collected above, the busy-period identity that follows is the one preliminary we derive from scratch, as Model-C_2 rests on it directly. The *Busy Period* of this $M/M/1+M$ is far more complex than its $M/M/1$ counterpart. We write B_C for it —the notation anticipates its use in Model-C_2— and record its LST and expectation, both derived below:

$$\widetilde{B}_C(s) = \cfrac{\mu}{\lambda + \mu + s - \cfrac{\lambda(\mu+\theta)}{\lambda + \mu + \theta + s - \cfrac{\lambda(\mu + 2\theta)}{\lambda + \mu + 2\theta + s - \cdots}}}, \tag{3.6}$$

$$\mathbb{E}[B_C] = \frac{1}{\lambda}\Big({}_1F_1\big(1;\, \mu/\theta;\, \lambda/\theta\big) - 1\Big) \tag{3.7}$$

The continued fraction (3.6) is generated by a first-passage recursion. Let $\phi_n(s)$ denote the LST of the first-passage (sub-busy-period) time from n customers in the system down to $n-1$. Out of state n, the first transition is either an arrival (rate λ) to state $n+1$, or a departure to state $n-1$ by service completion or abandonment (combined rate $\mu + (n-1)\theta$: one customer in service plus $n-1$ waiting, each impatient at rate θ). Conditioning on this first transition, the strong Markov property gives

$$\phi_n(s) \;=\; \frac{\mu + (n-1)\theta}{\lambda + \mu + (n-1)\theta + s - \lambda\, \phi_{n+1}(s)}, \qquad n \geq 1, \qquad \widetilde{B}_C(s) = \phi_1(s). \tag{3.8}$$

Unrolling Equation (3.8) from $n = 1$ reproduces the continued fraction (3.6) term by term. The continued fraction itself admits a closed-form evaluation —Equation (3.10) below expresses it as a ratio of Kummer functions— and the argument is short enough to give in full. Fix $s > 0$ and scale the rates by θ, setting

$$a = \frac{\mu}{\theta}, \qquad b = \frac{s}{\theta}, \qquad c = \frac{\lambda}{\theta},$$

so that Equation (3.8) reads $\phi_n(s) = (a+n-1)\big/\big((a+n-1) + b + c - c\,\phi_{n+1}(s)\big)$. Consider the integrals

$$I(p) \;=\; \int_0^1 t^{\,p}\, (1-t)^{\,b-1}\, e^{-ct}\, dt, \qquad p > -1,$$

where the dependence of I on the fixed parameters b and c is suppressed from the notation. By the Euler representation (3.4) these integrals are Beta-function multiples of Kummer values, $I(p) = B(p+1,\, b)\,{}_1F_1(p+1;\, p+1+b;\, -c)$. The derivative $\frac{d}{dt}\big[t^{\,p}(1-t)^{\,b}e^{-ct}\big]$ integrates to zero over $[0,1]$ by the fundamental theorem of calculus, since the boundary terms vanish for $p, b > 0$. Expanding this derivative by the product rule and splitting $(1-t)^b = (1-t)^{b-1} - t\,(1-t)^{b-1}$ in the two terms where it appears yields the three-term recurrence

$$(p+b+c)\,I(p) \;=\; p\,I(p-1) \;+\; c\,I(p+1), \tag{3.9}$$

a relation of contiguous type for ${}_1F_1$ (obtainable equivalently by combining the contiguous relations of [5, §13.3]). Dividing Equation (3.9) at $p = a+n-1$ by $I(a+n-2)$ shows that the ratios $R_n := I(a+n-1)/I(a+n-2)$ satisfy exactly the recursion of ϕ_n. Two further facts pin the identification. First, by Laplace's method at the endpoint $t = 1$, $I(p) \sim e^{-c}\,\Gamma(b)\,p^{-b}$ as $p \to \infty$: the solution $\{I(p)\}$ decays only polynomially, whereas the dominant solution of Equation (3.9) grows superexponentially (its consecutive ratio grows like p/c), so $\{I(p)\}$ is the *minimal (recessive)* solution. Second, by *Pincherle's theorem* the convergent continued fraction equals precisely the ratio built from the minimal solution. Hence

$$\widetilde{B}_C(s) \;=\; \phi_1(s) \;=\; \frac{I(a)}{I(a-1)} \;=\; \frac{B(a+1,\, b)\,{}_1F_1\big(a+1;\, a+1+b;\, -c\big)}{B(a,\, b)\,{}_1F_1\big(a;\, a+b;\, -c\big)}, \tag{3.10}$$

a contiguous ratio of the form (3.5). This identity is used in the boundedness route for Model-C_2 (§8, Equation (8.11)), where the $M/M/1+M$ queue in question is the class-1 subsystem, with parameters $(\lambda, \theta) = (\lambda_1, \theta_1)$ and argument $s = \lambda_2(1-y)$.

3.5 Pollaczek–Khinchine formula

The $M/G/1$ queue [1] is a single-server queue with Poisson arrivals at rate λ and generally distributed i.i.d. service times B with LST $\widetilde{B}(s) = \mathbb{E}[e^{-sB}]$. For stability purposes, the system load $\rho_B = \lambda\,\mathbb{E}[B] < 1$ is required (written ρ_B to keep the symbol ρ for the two-class load of this thesis). Under these two conditions —Poisson arrivals and stability— the Pollaczek–Khinchine (PK) formula gives the PGF of the stationary number of customers in the system:

$$L(z) \;=\; \frac{(1-\rho_B)(1-z)\,\widetilde{B}(\lambda(1-z))}{\widetilde{B}(\lambda(1-z)) - z}. \tag{3.11}$$

3.6 First-order linear ODEs and PDEs

Since jockeying and abandonment add state-dependent rates to the balance equations, derivative terms appear in the derivations, requiring us to solve first-order linear ODEs and PDEs for the PGF. The three methods below are invoked at specific steps of the analysis.

3.6.1 Integrating factor

This method is standard for solving a *first-order linear ODE in x*, for each fixed y. Such an equation has standard form

$$\frac{dP}{dx} + p(x;\, y)\,P \;=\; q(x;\, y), \tag{3.12}$$

where the semicolon separates the active variable x from the parameter y, a convention used throughout. The *integrating factor* $\mu_I(x; y) = \exp\!\big(\int_0^x p(t; y)\,dt\big)$ converts Equation (3.12) to $\frac{d}{dx}[P \cdot \mu_I] = q \cdot \mu_I$. Integrating from 0 to x gives the general solution

$$P(x, y) \;=\; \frac{1}{\mu_I(x;\, y)}\left[C_0(y) + \int_0^x q(t;\, y)\,\mu_I(t;\, y)\,dt\right], \tag{3.13}$$

where $C_0(y)$ is an arbitrary function of y, determined by a boundary condition.

3.6.2 Method of Characteristics

This method is standard for solving a *first-order linear PDE in both x and y* simultaneously. It reduces the PDE to a family of ODEs along the so-called *characteristic curves* in the (x, y)-plane; along each curve, the PDE collapses to a single-variable ODE.

A PDE of the form

$$a(x,y)\frac{\partial P}{\partial x} + b(x,y)\frac{\partial P}{\partial y} = c(x,y)\,P + d(x,y), \tag{3.14}$$

in which all four coefficients may depend on x and y, is solved by tracing the characteristic curve $(x(t), y(t))$ satisfying

$$\frac{dx}{dt} = a(x,y), \qquad \frac{dy}{dt} = b(x,y). \tag{3.15}$$

By the chain rule, $\frac{d}{dt}P(x(t), y(t)) = a\,\frac{\partial P}{\partial x} + b\,\frac{\partial P}{\partial y}$, so Equation (3.14) reduces to

$$\frac{dP}{dt} = cP + d, \tag{3.16}$$

a first-order linear ODE in t. Its homogeneous part $\frac{dP}{dt} = cP$ is solved at once by separation of variables; the inhomogeneous term d is then absorbed either by the integrating factor (§3.6.1) or, once that homogeneous solution is in hand, by variation of parameters (§3.6.3). When a and b are constants, the characteristics are straight lines and the quantity

$$\xi \;:=\; b\,x - a\,y$$

is preserved along each curve —it is the *first integral* that labels each characteristic. The integration constant of Equation (3.16) therefore varies between characteristics: it is an arbitrary function $F(\xi)$. Re-expressing the solution in the original variables (x, y) via Equation (3.15) gives the general solution of Equation (3.14). In the PGF setting, F is identified with a boundary generating function, and analyticity of $P(x, y)$ for $|x|, |y| \le 1$ uniquely determines it.

3.6.3 Variation of parameters

The integrating factor of §3.6.1 builds a solution from scratch by constructing μ_I. Frequently, however, we *already* hold a nonzero solution of the homogeneous equation —most importantly, the one the Method of Characteristics (§3.6.2) produces along each characteristic curve. *Variation of parameters* is the technique that turns any such known homogeneous solution directly into the general solution of the inhomogeneous equation, with no further integrating factor required. We describe it in the form in which it is used later.

Consider a first-order linear ODE written as

$$\frac{dP}{dx} = f(x)\,P + h(x), \tag{3.17}$$

and suppose a nonzero homogeneous solution P_h is known, that is, $\frac{dP_h}{dx} = f(x)\,P_h$ —the subscript h marks the homogeneous solution, not a derivative. The homogeneous equation then has general solution $C\,P_h(x)$ for an arbitrary constant C. To solve the inhomogeneous equation (3.17), we *vary the constant*: we promote C to a function $C(x)$ and look for a solution of the form

$$P(x) = C(x)\,P_h(x).$$

Differentiating by the product rule and using $P_h' = f\,P_h$,

$$\begin{aligned}\frac{dP}{dx} &= C'(x)\,P_h(x) + C(x)\,P_h'(x)\\ &= C'(x)\,P_h(x) + f(x)\,\big[C(x)\,P_h(x)\big] = C'(x)\,P_h(x) + f(x)\,P(x).\end{aligned}$$

Substituting this back into Equation (3.17), the common term $f(x)\,P(x)$ cancels from both sides and only the derivative of the varied constant survives:

$$C'(x)\,P_h(x) = h(x) \qquad \Longrightarrow \qquad C'(x) = \frac{h(x)}{P_h(x)}.$$

Integrating from a base point x_0 gives $C(x) = F + \int_{x_0}^{x} h(t)/P_h(t)\,dt$ with F an arbitrary constant, and hence the general solution

$$P(x) = P_h(x)\left[F + \int_{x_0}^{x}\frac{h(t)}{P_h(t)}\,dt\right]. \tag{3.18}$$

The known homogeneous solution P_h thus plays two roles: it is the overall prefactor and, as $1/P_h$, the weight inside the integral. The constant F is fixed by a boundary condition. Taking in particular $P_h = 1/\mu_I$ reproduces the integrating-factor formula (3.13) of §3.6.1.

This is exactly the step that completes the Method of Characteristics for an *inhomogeneous* PDE. Once §3.6.2 has reduced the PDE to an ODE of the form (3.17) along a characteristic and supplied its homogeneous solution P_h, formula (3.18) delivers the full solution along that characteristic, the constant F becoming an arbitrary function $F(\xi)$ of the characteristic label. Model-B (§6) is solved by precisely this pairing.

4 The general model and its fundamental equation

This section defines the general two-class system with both jockeying and abandonment, which we call *Model-X*; the simpler variants studied in the later sections are recovered from it as special cases.

The model is an $M/M/1$ queue with two priority classes. Customers are served First-Come, First-Served (FCFS) within their own class, and class-1 (the priority class) has non-preemptive priority over class-2: a service in progress is never interrupted, so a class-2 customer in service is not displaced by an arriving class-1 customer. The server is work-conserving: it is never idle while a customer is present. Beyond these basic dynamics, Model-X adds two queue-leaving mechanisms. First, *jockeying*: a waiting customer leaves its own queue to *join the other queue*, at per-customer rate γ_1 for direction $1 \to 2$ and γ_2 for $2 \to 1$, so the total number of customers is preserved. Second, *abandonment* (reneging): a waiting customer leaves the *system* at per-customer rate θ_i, decreasing the total by one. Concretely, each waiting class-i customer (not the one in service) holds an independent $\text{Exp}(\theta_i)$ patience clock and abandons when it expires. This is precisely the linear-reneging mechanism of the $M/M/1+M$ queue from §3.4.

In Kendall's notation this is an $M/M/1$-type model: Poisson arrivals, exponentially distributed service times, and a single server. Throughout, the parameters are:

- λ_1, λ_2: Poisson arrival rates of classes 1 and 2, respectively; the total arrival rate is $\lambda = \lambda_1 + \lambda_2$;
- μ: exponential service rate of a customer, regardless of class;
- $\rho_i = \frac{\lambda_i}{\mu}$: system load of class $i \in \{1, 2\}$; the total system load is $\rho = \frac{\lambda}{\mu}$;
- γ_1, γ_2: per-customer jockeying rates from Class $1 \to 2$ and $2 \to 1$, respectively; jockeying transfers a customer between queues, preserving the total number of customers;
- θ_1, θ_2: per-customer abandonment (reneging) rates from classes 1 and 2, respectively; abandonment removes a customer from the system, reducing the total by one.

The two class arrival streams are independent Poisson processes, and they are independent of the system's internal exponential service, jockeying and abandonment clocks, so they satisfy the lack-of-anticipation assumption of the PASTA property (§3.2). By Poisson *superposition* the merged arrival stream is itself Poisson at rate λ, and each class stream is an i.i.d. λ_i/λ *thinning* of the merger, so PASTA applies to the aggregate state and to each class separately.

To define the stochastic processes that drive this system, we introduce the following random variables:

- $\mathcal{B}$: binary indicator for the server being idle or busy, $\mathcal{B} \in \{0, 1\}$;
- N_1, N_2 : number of customers of class 1 and 2, respectively, in their respective queues;
- N : total number of customers in the queues.

Note that N_1, N_2 and N count customers in the *queues*, not the *system*. Because the service rate μ is common to both classes, the *memorylessness* of the exponential service time makes the residual service of the customer in progress $\text{Exp}(\mu)$ regardless of its class; the class of the in-service customer therefore does not influence the future evolution of the queue lengths, and the state descriptor need not record it. Accordingly, we write the idle state as (0) and all other states as (n_1, n_2), so that $(0, 0)$ implies that the server is busy while both queues are empty. Formally,

$$S := \{(0)\} \cup \{(n_1, n_2) \mid n_1, n_2 \geq 0\}. \tag{4.1}$$

For the partial-PGF analysis we additionally use an auxiliary state space $\widetilde{S}$, which tracks only n_2 and the total $n = n_1 + n_2$, subject to $n \geq n_2 \geq 0$:

$$\widetilde{S} := \{(0)\} \cup \{(n_2, n) \mid n \geq n_2 \geq 0\}. \tag{4.2}$$

The usefulness of $\widetilde{S}$ will turn out to be confined to Model-A; the rationale for introducing it, and the reason it fails to simplify any model with an active mechanism, are documented in Remark 1 at the end of this section.

Assuming positive recurrence, so that the limiting distribution exists, we define the long-run probabilities of finding the system in each state:

- π_0 for the idle state (0), it belongs to both spaces S and $\widetilde{S}$;
- $\pi(n_1, n_2)$ for states $(n_1, n_2) \in S$, where $n_1, n_2 \geq 0$;
- $\widetilde{\pi}(n_2, n)$ for states $(n_2, n) \in \widetilde{S}$, where $n \geq n_2 \geq 0$.

The simpler variants are recovered from Model-X by setting the appropriate parameters to zero, as summarised in Table 3:

Model	Jockeying	Abandonment	Parameter values
X	both $(1 \leftrightarrow 2)$	both $(1, 2)$	$\gamma_i \geq 0,\ \theta_i \geq 0$ (general)
A	none	none	$\gamma_1 = \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
B	both $(1 \leftrightarrow 2)$	none	$\gamma_i \geq 0,\ \theta_1 = \theta_2 = 0$
B_2	$1 \to 2$ only	none	$\gamma_1 \geq 0,\ \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
C_2	none	class 1 only	$\theta_1 \geq 0,\ \theta_2 = 0,\ \gamma_1 = \gamma_2 = 0$
B_2^{H}	$1 \to 2$, head-of-line	none	$\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}} > 0,\ \gamma_2 = \theta_1 = \theta_2 = 0$
C_2^{H}	none	class 1, head-of-line	$\theta_1 \mathbf{1}_{\{n_1 \geq 1\}} > 0,\ \gamma_1 = \gamma_2 = \theta_2 = 0$

Table 3: Nested family of models. Model-X is the parent; Models A, B, B_2 and C_2 are obtained by restricting the jockeying directions and the abandoning classes. The two head-of-line variants B_2^{H} and C_2^{H} are listed separately, as their mechanism involves the indicator $\mathbf{1}_{\{n_1 \geq 1\}}$ rather than being proportional to the queue length.

The subscript in the variant names records which rate of a mirrored pair is switched off: Model-B_2 sets $\gamma_2 = 0$, retaining only the $1 \to 2$ jockeying direction, and Model-C_2 sets $\theta_2 = 0$, retaining only class-1 abandonment. The mirrored variants Model-B_1 ($\gamma_1 = 0$) and Model-C_1 ($\theta_1 = 0$), not treated in this thesis, are named symmetrically.

Figure 1 gives a visual overview of Model-X and each variant we consider. It omits the two head-of-line variants, which are less conventional and for which such a diagram would add little:

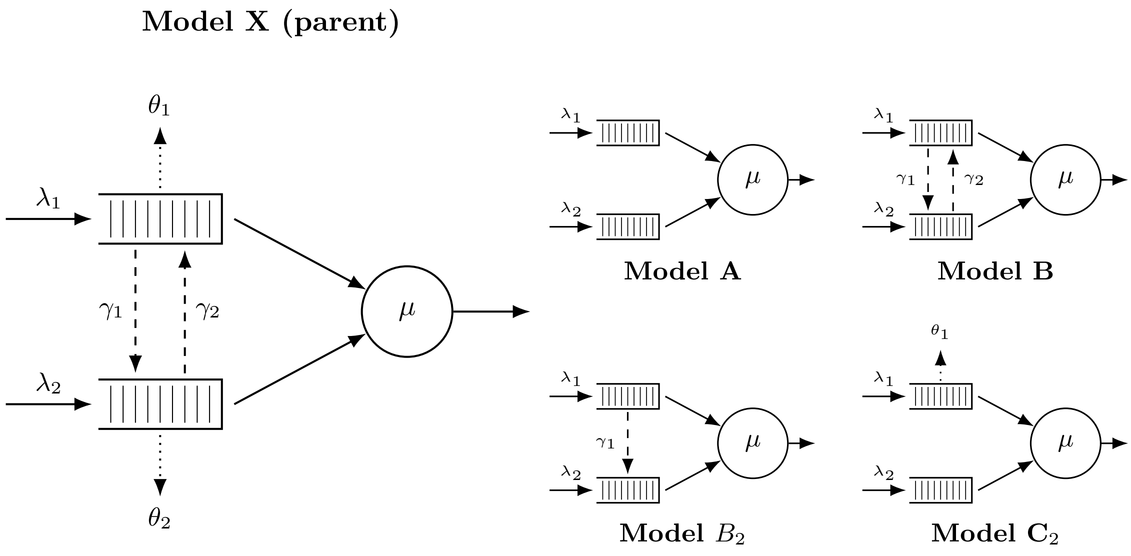

Figure 1: Physical queue layout for Model-X (left) and its four specialisations (right, 2×2 grid). In all models, Queue 1 (top buffer) has non-preemptive priority over Queue 2. Solid arrows are for Poisson arrivals (λ_i) and exponential service (μ); dashed arrows are for jockeying(γ_i); and dotted arrows are for abandonments (θ_i)

Stability. Jockeying does not affect stability (customers are merely relocated). Model-X has abandonments from both classes, so the departure rate $\mu+\theta_1 n_1+\theta_2 n_2$ grows unbounded, ensuring positive recurrence for all loads. In contrast, models without abandonments require $\rho<1$ (they behave like M/M/1). Mixed cases, such as Model-C_2, are discussed separately (§8).

Computation of π_0 and $\pi(0,0)$. The two empty-state probabilities are determined, for the full Model-X, by two exact identities that hold irrespective of the jockeying and abandonment mechanisms: a cut between the idle and busy states, and a renewal–reward identity over busy cycles. Together they reduce the determination of π_0 and $\pi(0,0)$ to a single scalar, the mean busy period $\mathbb{E}[BP]$.

Theorem 1 (Empty-state probabilities of the general model)**.** *Assume the CTMC of Model-X is positive recurrent, so that the stationary distribution exists. Let $\mathbb{E}[BP]$ denote the mean length of a server busy period —the time from an arrival that finds the system empty until the system is next empty. Then*

$$\pi(0,0)\;=\;\rho\,\pi_0, \tag{4.3}$$

and

$$\pi_0\;=\;\frac{1}{1+\lambda\,\mathbb{E}[BP]},\qquad \pi(0,0)\;=\;\frac{\rho}{1+\lambda\,\mathbb{E}[BP]}. \tag{4.4}$$

Proof. Cut identity. Consider the cut separating the idle state (0) from the set of busy states. The only transitions out of (0) are the class-1 and class-2 arrivals, at total rate λ; the only transition into (0) is a service completion from $(0,0)$, at rate μ —from any state with a waiting customer a departure leaves the server busy, and neither jockeying nor abandonment can act when both queues are empty. Equating the probability flux across the cut gives $\lambda\,\pi_0=\mu\,\pi(0,0)$, which is Equation (4.3).

Renewal–reward identity. The idle state is a regeneration point: by the memorylessness of the exponential inter-arrival and service times (the strong Markov property), the evolution after each visit to (0) is an independent probabilistic copy of the process. Successive cycles therefore alternate an idle period of mean $1/\lambda$ (the wait for the next arrival, exponential at rate λ) and a busy period of mean $\mathbb{E}[BP]$. By the renewal–reward theorem, the stationary probability of the idle state equals the long-run fraction of time spent idle,

$$\pi_0\;=\;\frac{1/\lambda}{1/\lambda+\mathbb{E}[BP]}\;=\;\frac{1}{1+\lambda\,\mathbb{E}[BP]},$$

and substituting into Equation (4.3) yields the stated $\pi(0,0)$. □

Both identities are exact for the full model. They reduce the problem to the mean busy period $\mathbb{E}[BP]$, which is elementary when there is no abandonment and is computed model by model once abandonment is present; the resulting π_0 and $\pi(0,0)$ for each tractable model then follow as corollaries (Corollary 1 below for the no-abandonment family; §8 and §10 for the abandonment variants). For the full asymmetric Model-X, $\mathbb{E}[BP]$ itself has no closed form (Remark 2).

The no-abandonment case is concrete enough to state at once; it covers Models A, B, B_2 and B_2^{H} simultaneously.

Corollary 1 (Empty states without abandonment)**.** *Suppose $\theta_1=\theta_2=0$ (Models A, B, B_2 and B_2^{H}) and $\rho<1$. Then the total number in the system is a birth–death process with birth rate λ and death rate μ —an $M/M/1$ queue— for* any *jockeying rates γ_1,γ_2, since jockeying merely relocates a waiting customer and so conserves the total count. Its mean busy period is $\mathbb{E}[BP]=(\mu-\lambda)^{-1}$, and Theorem 1 gives*

$$\pi_0=1-\rho,\qquad \pi(0,0)=\rho(1-\rho). \tag{4.5}$$

4.1 Balance equations

This subsection presents the state-transition diagrams of Model-X on both state spaces S and $\widetilde{S}$, and derives the corresponding sets of balance equations.

4.1.1 Balance equations for state space S

The state-transition diagram of Model-X on S is given in Figure 2.

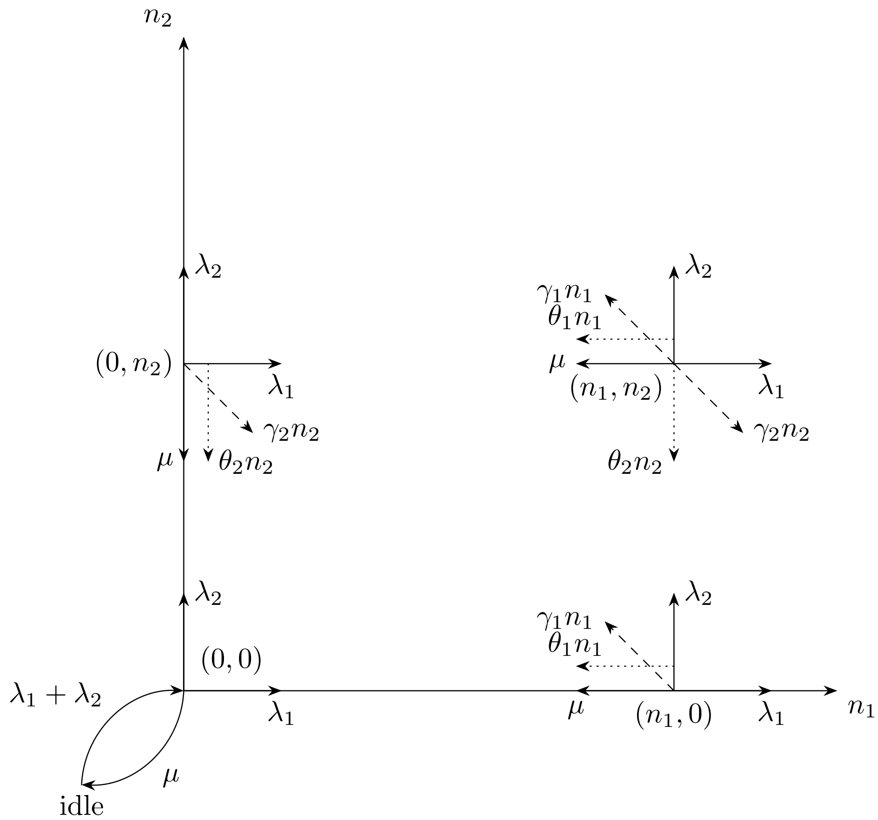

Figure 2: State transition diagram on state space S for Model-X, with jockeying ($\gamma_i n_i$, dashed) and abandonments ($\theta_i n_i$, dotted).

Next, we give the balance equations for the five parts of the state space: the interior ($n_1, n_2 \geq 1$), the two boundaries ($n_1 = 0$ and $n_2 = 0$), the busy state with empty queues $(0, 0)$, and the idle state (0). Recall that each waiting customer carries its own independent jockeying and patience clocks, so in a state with n_i waiting class-i customers the aggregate jockeying and abandonment rates are the linear terms $\gamma_i n_i$ and $\theta_i n_i$.

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1) n_1 + (\gamma_2 + \theta_2) n_2]\, \pi(n_1, n_2) \\
&\quad = \quad \mu \pi(n_1 + 1, n_2) \\
&\qquad + \lambda_1 \pi(n_1 - 1, n_2) \\
&\qquad + \lambda_2 \pi(n_1, n_2 - 1) \\
&\qquad + \gamma_1 (n_1 + 1) \pi(n_1 + 1, n_2 - 1) && \text{for} \quad n_1 \geq 1, n_2 \geq 1 \qquad (4.6) \\
&\qquad + \gamma_2 (n_2 + 1) \pi(n_1 - 1, n_2 + 1) \\
&\qquad + \theta_1 (n_1 + 1) \pi(n_1 + 1, n_2) \\
&\qquad + \theta_2 (n_2 + 1) \pi(n_1, n_2 + 1)
\end{aligned}$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2) n_2]\, \pi(0, n_2) \\
&\quad = \quad \lambda_2 \pi(0, n_2 - 1) \\
&\qquad + \mu \pi(1, n_2) \\
&\qquad + \mu \pi(0, n_2 + 1) && \text{for} \quad n_1 = 0, n_2 \geq 1 \qquad (4.7) \\
&\qquad + \gamma_1 \pi(1, n_2 - 1) \\
&\qquad + \theta_1 \pi(1, n_2) \\
&\qquad + \theta_2 (n_2 + 1) \pi(0, n_2 + 1)
\end{aligned}$$

$$
\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n_1]\,\pi(n_1,0)\\
&\quad=\quad \lambda_1\pi(n_1-1,0)\\
&\qquad+\mu\pi(n_1+1,0)\\
&\qquad+\gamma_2\pi(n_1-1,1)\\
&\qquad+\theta_1(n_1+1)\pi(n_1+1,0)\\
&\qquad+\theta_2\pi(n_1,1)
\end{aligned}
\qquad \text{for} \quad n_1\geq 1, n_2=0 \tag{4.8}
$$

$$
[\lambda_1+\lambda_2]\,\pi(0,0)=(\mu+\theta_1)\pi(1,0)+(\mu+\theta_2)\pi(0,1) \tag{4.9}
$$

$$
(\lambda_1+\lambda_2)\pi_0=\mu\pi(0,0) \tag{4.10}
$$

Note that the idle-state equation (4.10) is precisely the cut identity (4.3) of Theorem 1; it is repeated here so that the balance set is complete.

4.1.2 Balance equations for state space $\widetilde{S}$

Recall the state space $\widetilde{S}$ defined in Equation (4.2), whose states (n_2,n) record the number of waiting class-2 customers n_2 and the total number of waiting customers $n=n_1+n_2$. For readability we denote its limiting probabilities by $\widetilde{\pi}(n_2,n)$, for $(n_2,n)\in\widetilde{S}$. Figure 3 shows the corresponding state transition diagram: jockeying conserves n and moves only n_2 —a horizontal move in the diagram— whereas arrivals increase n by one and services and abandonments decrease it by one.

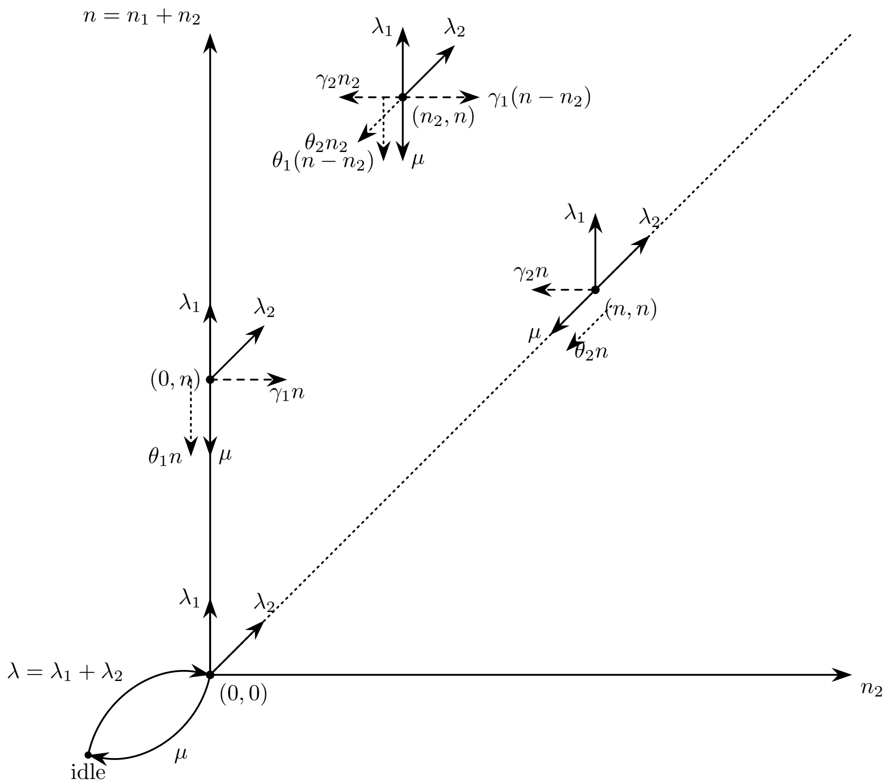

Figure 3: State transition diagram on state space $\widetilde{S}$.

The balance equations, after combining the equations where $\widetilde{\pi}_0$ shows up, are:

$$
\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)(n-n_2)+(\gamma_2+\theta_2)n_2]\,\widetilde{\pi}(n_2,n)\\
&\quad=\quad \lambda_1\widetilde{\pi}(n_2,n-1)\\
&\qquad+\lambda_2\widetilde{\pi}(n_2-1,n-1)\\
&\qquad+\mu\widetilde{\pi}(n_2,n+1)\\
&\qquad+\gamma_1(n-n_2+1)\widetilde{\pi}(n_2-1,n)\\
&\qquad+\gamma_2(n_2+1)\widetilde{\pi}(n_2+1,n)\\
&\qquad+\theta_1(n+1-n_2)\widetilde{\pi}(n_2,n+1)\\
&\qquad+\theta_2(n_2+1)\widetilde{\pi}(n_2+1,n+1)
\end{aligned}
\qquad \text{for} \quad 1\leq n_2\leq n-1,\ n\geq 2 \tag{4.11}
$$

$$
\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_2+\theta_2)n]\,\widetilde{\pi}(n,n)\\
&\quad= \quad \lambda_2\widetilde{\pi}(n-1,n-1)\\
&\qquad+\mu\widetilde{\pi}(n+1,n+1)\\
&\qquad+\mu\widetilde{\pi}(n,n+1) && \text{for} \quad n_2=n,\ n\geq 1 && (4.12)\\
&\qquad+\gamma_1\widetilde{\pi}(n-1,n)\\
&\qquad+\theta_1\widetilde{\pi}(n,n+1)\\
&\qquad+\theta_2(n+1)\widetilde{\pi}(n+1,n+1)\\
&[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n]\,\widetilde{\pi}(0,n)\\
&\quad= \quad \lambda_1\widetilde{\pi}(0,n-1)\\
&\qquad+\mu\widetilde{\pi}(0,n+1)\\
&\qquad+\gamma_2\widetilde{\pi}(1,n) && \text{for} \quad n_2=0,\ n\geq 1 && (4.13)\\
&\qquad+\theta_1(n+1)\widetilde{\pi}(0,n+1)\\
&\qquad+\theta_2\widetilde{\pi}(1,n+1)\\
&[\lambda_1+\lambda_2]\,\widetilde{\pi}(0,0)\\
&\quad= \quad (\mu+\theta_1)\widetilde{\pi}(0,1)+(\mu+\theta_2)\widetilde{\pi}(1,1) && && (4.14)\\
&(\lambda_1+\lambda_2)\widetilde{\pi}_0=\mu\widetilde{\pi}(0,0) && && (4.15)
\end{aligned}
$$

As on S, the idle-state equation (4.15) coincides with the cut identity (4.3) of Theorem 1.

4.2 Equations for the (partial) probability generating functions

To study the limiting behaviour of this two-class system, we define the bivariate Probability Generating Function (PGF) of the joint queue lengths N_1 and N_2:

$$P(x,y)=\mathbb{E}[x^{N_1}y^{N_2}]=\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}.$$

The derivation also uses the boundary terms, where at least one queue is empty:

$$
\begin{aligned}
P_x(x)=P(x,0)&=\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}\\
P_y(y)=P(0,y)&=\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}.
\end{aligned}
$$

Note that both terms include the busy-but-empty state $\pi(0,0)$, whereas the fully idle state is treated separately.

4.2.1 Equation for PGF in state space S

Lemma 1 (Fundamental equation for the PGF on S)**.** *Consider the two-class non-preemptive priority queue on the state space $S=\{(n_1,n_2):n_1,n_2\geq 0\}$, with Poisson arrival rates λ_1,λ_2, exponential service rate μ, jockeying rates γ_1,γ_2 and abandonment rates θ_1,θ_2, where n_1 and n_2 count the class-1 and class-2 customers* waiting *in the queues. Assume the associated CTMC is positive recurrent, so that the stationary distribution $\{\pi(n_1,n_2)\}$ exists and the joint PGF $P(x,y)$ defined above is analytic on the closed unit polydisc $|x|\leq 1,\ |y|\leq 1$. Then $P(x,y)$, together with the boundary function $P_y(y)=P(0,y)$, satisfies the first-order linear partial differential equation*

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0).
\end{aligned}
\tag{4.16}
$$

The structure of Equation (4.16) is worth reading term by term, since it dictates the entire analysis that follows. The bracketed algebraic coefficient multiplying $P(x,y)$ is exactly the kernel of the baseline Model-A: the flux balance between the arrival contributions $\lambda_1 x^2 y$ and $\lambda_2 x y^2$, the service contribution μy, and the total outflow $(\lambda_1 + \lambda_2 + \mu)xy$.

The coefficients of the two partial derivatives encode the added mechanisms. Consider the coefficient $xy\,[\gamma_1(x-y) + \theta_1(x-1)]$ of the derivative in x. Its jockeying factor $(x-y)$ vanishes on the diagonal $x = y$: the analytic signature that jockeying merely relocates a customer and so conserves the total count. Its abandonment factor $(x-1)$ vanishes at $x = 1$: the signature that abandonment respects normalisation ($P(1,1) = 1$) yet removes a customer and thereby destroys conservation. The coefficient $xy\,[\gamma_2(y-x) + \theta_2(y-1)]$ of the derivative in y states the symmetric facts for the class-2 mechanisms, with $(y-x)$ vanishing on the diagonal and $(y-1)$ at $y = 1$.

The right-hand side involves the two boundary unknowns that any solution must determine: the boundary function $P_y(y) = P(0,y)$, the generating function of the states with an empty priority queue, and the scalar $\pi(0,0)$, the probability that the server is busy with both queues empty.

The position of these vanishing factors fixes, depending on the model, where the operative singularity lies and hence which technique applies. With both derivative terms absent (Model-A, §5) the equation is algebraic and $P_y(y)$ is recovered from the kernel root $x^*(y)$ inside the unit disc. With one-way jockeying alone (Model-B_2, §7) the factor $(x-y)$ places the singularity at the interior point $x = y$ with a negative exponent, and $P_y(y)$ is determined by analyticity there. With class-1 abandonment alone (Model-C_2, §8) the factor $(x-1)$ places it on the boundary $x = 1$ with a positive exponent, and $P_y(y)$ is determined by mere boundedness. Restricting a mechanism to the head of the queue removes the derivative term, reducing the equation to the algebraic Model-A form (Models B_2^{H} and C_2^{H}, §9). The mirrored variants Model-B_1 (jockeying $2 \to 1$ only) and Model-C_1 (class-2 abandonment only), not treated in this thesis, place the derivative in y instead; as for Model-B, the equation then couples the unknown boundary function $P_y(y)$ to the PGF along the integration path and reduces to a Volterra-type integral equation (§6). These regimes are gathered side by side in Table 5 of §11.2, and are synthesised, together with the two irreducible obstructions, as the obstruction taxonomy of §12.

Proof of Lemma 1. We multiply each of the balance equations (4.6)–(4.9) by $x^{n_1} y^{n_2}$ and sum over its domain. Two intermediate transforms arise along the way, describing the boundary dynamics when one queue holds exactly one waiting customer:

$$P_y(y \,|\, n_1{=}1) := \sum_{n_2=0}^{\infty} \pi(1,n_2)\, y^{n_2}, \qquad P_x(x \,|\, n_2{=}1) := \sum_{n_1=0}^{\infty} \pi(n_1,1)\, x^{n_1}.$$

Despite the conditional-looking bar, these are joint (unnormalised) transforms of the states with exactly one waiting class-1 (respectively class-2) customer; the derivation eliminates them in favour of the primary unknowns $P_x(x)$, $P_y(y)$ and $\pi(0,0)$.

From the interior equation (4.6):

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&\quad=\quad\mu\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)x^{n_1}y^{n_2}\\
&\qquad+\lambda_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)x^{n_1}y^{n_2}\\
&\qquad+\lambda_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)x^{n_1}y^{n_2}\\
&\qquad+\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2-1)x^{n_1}y^{n_2}\\
&\qquad+\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1-1,n_2+1)x^{n_1}y^{n_2}\\
&\qquad+\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2)x^{n_1}y^{n_2}\\
&\qquad+\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1,n_2+1)x^{n_1}y^{n_2}\\
&LHS_1+LHS_2+LHS_3+LHS_4+LHS_5\\
&\quad=\quad RHS_1+RHS_2+RHS_3+RHS_4+RHS_5+RHS_6+RHS_7
\end{aligned}
$$

where LHS_i and RHS_j denote the i-th and j-th summands on the left- and right-hand sides of the display above, in the order written. Evaluating each summand:

$$
\begin{aligned}
&LHS_1\\
&=\quad(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&=\quad(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\pi(n_1,0)x^{n_1}y^{0}\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}\right]\\
&\quad-\quad(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[P(x,y)-P_y(y)-P_x(x)+\pi(0,0)\right]
\end{aligned}
$$

LHS_2

$$
\begin{aligned}
&= \gamma_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_1 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&= \gamma_1 x \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_1 \pi(n_1, n_2) x^{n_1-1} y^{n_2} \\
&= \gamma_1 x \left[\sum_{n_1=1}^{\infty} \sum_{n_2=0}^{\infty} n_1 \pi(n_1, n_2) x^{n_1-1} y^{n_2} - \sum_{n_1=1}^{\infty} n_1 \pi(n_1, 0) x^{n_1-1} \right] \\
&= \gamma_1 x \left[\sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} n_1 \pi(n_1, n_2) x^{n_1-1} y^{n_2} - \sum_{n_1=0}^{\infty} n_1 \pi(n_1, 0) x^{n_1-1} \right] \\
&= \gamma_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right]
\end{aligned}
$$

LHS_3

$$
\begin{aligned}
&= \gamma_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&= \gamma_2 \sum_{n_1=1}^{\infty} \left[\sum_{n_2=0}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2} - 0 \cdot \pi(n_1, 0) x^{n_1} \right] \\
&= \gamma_2 y \left[\sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2 \pi(0, n_2) y^{n_2-1} \right] \\
&= \gamma_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right]
\end{aligned}
$$

LHS_4

$$
\begin{aligned}
&= \theta_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_1 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&= \theta_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right] \qquad \text{(the computation of } LHS_2 \text{ with } \theta_1 \text{ in place of } \gamma_1\text{)}
\end{aligned}
$$

LHS_5

$$
\begin{aligned}
&= \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&= \theta_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right] \qquad \text{(the computation of } LHS_3 \text{ with } \theta_2 \text{ in place of } \gamma_2\text{)}
\end{aligned}
$$

RHS_1

$$
\begin{aligned}
&= \mu \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1+1,n_2)x^{n_1}y^{n_2} \\
&= \mu x^{-1} \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1+1,n_2)x^{n_1+1}y^{n_2} \\
&= \mu x^{-1} \sum_{m=2}^{\infty}\sum_{n_2=1}^{\infty} \pi(m,n_2)x^{m}y^{n_2} \\
&= \mu x^{-1} \sum_{m=2}^{\infty}\left[\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - \pi(m,0)x^m\right] \\
&= \mu x^{-1} \left[\sum_{m=2}^{\infty}\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - \sum_{m=2}^{\infty}\pi(m,0)x^m\right] \\
&= \mu x^{-1} \left[\sum_{m=0}^{\infty}\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - x\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2} - \sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}\right] \\
&- \mu x^{-1} \left[\sum_{m=0}^{\infty}\pi(m,0)x^m - x\pi(1,0) - \pi(0,0)\right] \\
&= \mu x^{-1}\left[P(x,y) - xP_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0)\right]
\end{aligned}
$$

RHS_2

$$
\begin{aligned}
&= \lambda_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1-1,n_2)x^{n_1}y^{n_2} \\
&= \lambda_1 x \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1-1,n_2)x^{n_1-1}y^{n_2} \\
&= \lambda_1 x \sum_{m=0}^{\infty}\sum_{n_2=1}^{\infty} \pi(m,n_2)x^{m}y^{n_2} \\
&= \lambda_1 x \sum_{m=0}^{\infty}\left[\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - \pi(m,0)x^m\right] \\
&= \lambda_1 x \left[\sum_{m=0}^{\infty}\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - \sum_{m=0}^{\infty}\pi(m,0)x^m\right] \\
&= \lambda_1 x\left[P(x,y) - P_x(x)\right]
\end{aligned}
$$

RHS_3

$$
\begin{aligned}
&= \lambda_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1,n_2-1)x^{n_1}y^{n_2} \\
&= \lambda_2 y \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1,n_2-1)x^{n_1}y^{n_2-1} \\
&= \lambda_2 y \sum_{n_1=1}^{\infty}\sum_{m=0}^{\infty} \pi(n_1,m)x^{n_1}y^{m} \\
&= \lambda_2 y \left[\sum_{n_1=0}^{\infty}\sum_{m=0}^{\infty} \pi(n_1,m)x^{n_1}y^{m} - \sum_{m=0}^{\infty}\pi(0,m)y^m\right] \\
&= \lambda_2 y\left[P(x,y) - P_y(y)\right]
\end{aligned}
$$

RHS_4

$$
\begin{aligned}
&= \gamma_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_1+1)\pi(n_1+1, n_2-1) x^{n_1} y^{n_2} \\
&= \gamma_1 y \sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} (n_1+1)\pi(n_1+1, m_2) x^{n_1} y^{m_2} \\
&= \gamma_1 y \sum_{m_1=2}^{\infty} \sum_{m_2=0}^{\infty} m_1 \pi(m_1, m_2) x^{m_1-1} y^{m_2} \\
&= \gamma_1 y \left[\sum_{m_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_1 \pi(m_1, m_2) x^{m_1-1} y^{m_2} - \sum_{m_2=0}^{\infty} \pi(1, m_2) y^{m_2} \right] \\
&= \gamma_1 y \left[\frac{\partial}{\partial x} P(x,y) - P_y(y|n_1=1) \right]
\end{aligned}
$$

RHS_5

$$
\begin{aligned}
&= \gamma_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2+1)\pi(n_1-1, n_2+1) x^{n_1} y^{n_2} \\
&= \gamma_2 \sum_{n_1=1}^{\infty} \sum_{m_2=2}^{\infty} m_2 \pi(n_1-1, m_2) x^{n_1} y^{m_2-1} \\
&= \gamma_2 x \left[\sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1-1, m_2) x^{n_1-1} y^{m_2-1} - \sum_{n_1=1}^{\infty} \pi(n_1-1, 1) x^{n_1-1} \right] \\
&= \gamma_2 x \left[\sum_{m_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(m_1, m_2) x^{m_1} y^{m_2-1} - \sum_{m_1=0}^{\infty} \pi(m_1, 1) x^{m_1} \right] \\
&= \gamma_2 x \left[\frac{\partial}{\partial y} P(x,y) - P_x(x|n_2=1) \right]
\end{aligned}
$$

RHS_6

$$
\begin{aligned}
&= \theta_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_1+1)\pi(n_1+1, n_2) x^{n_1} y^{n_2} \\
&= \theta_1 \sum_{m_1=2}^{\infty} \sum_{n_2=1}^{\infty} m_1 \pi(m_1, n_2) x^{m_1-1} y^{n_2} \\
&= \theta_1 \left[\sum_{m_1=0}^{\infty} \sum_{n_2=1}^{\infty} m_1 \pi(m_1, n_2) x^{m_1-1} y^{n_2} - \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \right] \\
&= \theta_1 \left[\sum_{m_1=0}^{\infty} \sum_{n_2=0}^{\infty} m_1 \pi(m_1, n_2) x^{m_1-1} y^{n_2} - \sum_{m_1=0}^{\infty} m_1 \pi(m_1, 0) x^{m_1-1} \right] \\
&\quad - \theta_1 \left[\sum_{n_2=0}^{\infty} \pi(1, n_2) y^{n_2} - \pi(1,0) \right] \\
&= \theta_1 \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) - P_y(y|n_1=1) + \pi(1,0) \right]
\end{aligned}
$$

RHS_7

$$\begin{aligned}
&= \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2+1)\pi(n_1, n_2+1) x^{n_1} y^{n_2} \\
&= \theta_2 \sum_{n_1=1}^{\infty} \sum_{m_2=2}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} \\
&= \theta_2 \left[\sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{n_1=1}^{\infty} \pi(n_1, 1) x^{n_1} \right] \\
&= \theta_2 \left[\sum_{n_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{m_2=0}^{\infty} m_2 \pi(0, m_2) y^{m_2-1} \right] \\
&\quad - \theta_2 \left[\sum_{n_1=0}^{\infty} \pi(n_1, 1) x^{n_1} - \pi(0,1) \right] \\
&= \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}$$

Substituting the evaluated summands back into the transformed balance equation:

$$\begin{aligned}
&(\lambda_1 + \lambda_2 + \mu) \left[P(x,y) - P_y(y) - P_x(x) + \pi(0,0) \right] \\
&+ \gamma_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right] \\
&+ \gamma_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right] \\
&+ \theta_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right] \\
&+ \theta_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right] \\
&\quad = \quad \mu x^{-1} \left[P(x,y) - x P_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0) \right] \\
&\quad + \lambda_1 x \left[P(x,y) - P_x(x) \right] \\
&\quad + \lambda_2 y \left[P(x,y) - P_y(y) \right] \\
&\quad + \gamma_1 y \left[\frac{\partial}{\partial x} P(x,y) - P_y(y|n_1=1) \right] \\
&\quad + \gamma_2 x \left[\frac{\partial}{\partial y} P(x,y) - P_x(x|n_2=1) \right] \\
&\quad + \theta_1 \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) - P_y(y|n_1=1) + \pi(1,0) \right] \\
&\quad + \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}$$

Grouping the coefficients of each transform yields

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x-\lambda_2 y\right]P(x,y)\\
&+\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_2 y\right]P_y(y)\\
&\qquad+\left[-(\lambda_1+\lambda_2+\mu)+\mu x^{-1}\right]\pi(0,0)\\
&\qquad+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+\mu\pi(1,0)\\
&\qquad+\theta_1\pi(1,0)\\
&\qquad+\theta_2\pi(0,1)
\end{aligned}
$$

and multiplying through by xy to clear the x^{-1} factors:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad-\left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad+\theta_2 xy\,\pi(0,1)
\end{aligned}
\tag{4.17}
$$

The x-boundary equation (4.8), treated analogously, gives:

$$
\begin{aligned}
&\sum_{n_1=1}^{\infty}\left[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n_1\right]\pi(n_1,0)x^{n_1}\\
&\quad=\quad \lambda_1\sum_{n_1=1}^{\infty}\pi(n_1-1,0)x^{n_1}+\mu\sum_{n_1=1}^{\infty}\pi(n_1+1,0)x^{n_1}\\
&\qquad+\gamma_2\sum_{n_1=1}^{\infty}\pi(n_1-1,1)x^{n_1}+\theta_1\sum_{n_1=1}^{\infty}(n_1+1)\pi(n_1+1,0)x^{n_1}+\theta_2\sum_{n_1=1}^{\infty}\pi(n_1,1)x^{n_1}
\end{aligned}
$$

Extracting the rate prefactors and shifting the summation indices:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}+(\gamma_1+\theta_1)x\sum_{n_1=1}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad=\quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^m+\mu x^{-1}\sum_{m=2}^{\infty}\pi(m,0)x^m\\
&\qquad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^m+\theta_1\sum_{m=2}^{\infty}m\pi(m,0)x^{m-1}+\theta_2\sum_{m=1}^{\infty}\pi(m,1)x^m
\end{aligned}
$$

Completing each sum to start at zero and subtracting the added terms:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right]+(\gamma_1+\theta_1)x\sum_{n_1=0}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad=\quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^m\\
&\qquad+\mu x^{-1}\left[\sum_{m=0}^{\infty}\pi(m,0)x^m-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^m\\
&\qquad+\theta_1\left[\sum_{m=0}^{\infty}m\pi(m,0)x^{m-1}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}\pi(m,1)x^m-\pi(0,1)\right]
\end{aligned}
$$

Recognising the transforms defined in §4.2:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[P_x(x)-\pi(0,0)\right]+(\gamma_1+\theta_1)x\frac{d}{dx}P_x(x)\\
&\quad=\quad \lambda_1 xP_x(x)+\mu x^{-1}\left[P_x(x)-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 xP_x(x|n_2=1)+\theta_1\left[\frac{d}{dx}P_x(x)-\pi(1,0)\right]+\theta_2\left[P_x(x|n_2=1)-\pi(0,1)\right]
\end{aligned}
$$

Grouping the coefficients of $P_x(x)$ and of its derivative:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}\right]\pi(0,0)-(\mu+\theta_1)\pi(1,0)-\theta_2\pi(0,1)\\
&\qquad+\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)
\end{aligned}
$$

and finally multiplying through by xy, ready for substitution into Equation (4.17):

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad-(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad-\theta_2 xy\,\pi(0,1)\\
&\qquad+xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)
\end{aligned}
\tag{4.18}
$$

The y-boundary equation (4.7) gives:

$$\begin{aligned}
&\sum_{n_2=1}^{\infty} [\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2) n_2] \, \pi(0, n_2) y^{n_2} \\
&= \quad \lambda_2 \sum_{n_2=1}^{\infty} \pi(0, n_2 - 1) y^{n_2} \\
&\quad + \mu \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\quad + \mu \sum_{n_2=1}^{\infty} \pi(0, n_2 + 1) y^{n_2} \\
&\quad + \gamma_1 \sum_{n_2=1}^{\infty} \pi(1, n_2 - 1) y^{n_2} \\
&\quad + \theta_1 \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\quad + \theta_2 \sum_{n_2=1}^{\infty} (n_2 + 1) \pi(0, n_2 + 1) y^{n_2}
\end{aligned}$$

Extracting the rate prefactors and shifting the summation indices:

$$\begin{aligned}
&(\lambda_1 + \lambda_2 + \mu) \sum_{n_2=1}^{\infty} \pi(0, n_2) y^{n_2} \\
&+ (\gamma_2 + \theta_2) y \sum_{n_2=1}^{\infty} n_2 \pi(0, n_2) y^{n_2 - 1} \\
&= \quad \lambda_2 y \sum_{m=0}^{\infty} \pi(0, m) y^{m} \\
&\quad + \mu \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\quad + \mu y^{-1} \sum_{m=2}^{\infty} \pi(0, m) y^{m} \\
&\quad + \gamma_1 y \sum_{m=0}^{\infty} \pi(1, m) y^{m} \\
&\quad + \theta_1 \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\quad + \theta_2 \sum_{m=2}^{\infty} m \pi(0, m) y^{m-1}
\end{aligned}$$

Completing each sum and subtracting the added terms:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}-\pi(0,0)\right]\\
&+(\gamma_2+\theta_2)y\sum_{n_2=0}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\\
&\quad=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(0,m)y^m\\
&\qquad+\mu\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[\sum_{m=0}^{\infty}\pi(0,m)y^m-y\pi(0,1)-\pi(0,0)\right]\\
&\qquad+\gamma_1 y\sum_{m=0}^{\infty}\pi(1,m)y^m\\
&\qquad+\theta_1\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}m\pi(0,m)y^{m-1}-\pi(0,1)\right]
\end{aligned}
$$

Recognising the transforms:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[P_y(y)-\pi(0,0)\right]+(\gamma_2+\theta_2)y\frac{d}{dy}P_y(y)\\
&\quad=\quad \lambda_2 yP_y(y)+\mu\left[P_y(y|n_1=1)-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[P_y(y)-y\pi(0,1)-\pi(0,0)\right]+\gamma_1 yP_y(y|n_1=1)\\
&\qquad+\theta_1\left[P_y(y|n_1=1)-\pi(1,0)\right]+\theta_2\left[\frac{d}{dy}P_y(y)-\pi(0,1)\right]
\end{aligned}
$$

Grouping the coefficients of $P_y(y)$ and of its derivative:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-(\mu+\theta_1)\pi(1,0)-(\mu+\theta_2)\pi(0,1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}\right]\pi(0,0)
\end{aligned}
$$

Substituting the empty-queues balance equation (4.9) for $(\mu+\theta_1)\pi(1,0)+(\mu+\theta_2)\pi(0,1)$:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-(\lambda_1+\lambda_2)\right]\pi(0,0)
\end{aligned}
$$

which simplifies to

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)+\left[\mu-\mu y^{-1}\right]\pi(0,0)
\end{aligned}
$$

and, multiplying through by xy:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right]P_y(y)+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]xy\,P_y(y|n_1=1)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\tag{4.19}
$$

Substituting Equation (4.18) into Equation (4.17) eliminates all $P_x(x)$ terms:

$$\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1y+\theta_1\right]P_y(y|n_1=1)
\end{aligned}\tag{4.20}$$

Isolating $[\mu+\gamma_1y+\theta_1]\,xy\,P_y(y|n_1=1)$ from Equation (4.19) and substituting into Equation (4.20):

$$\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2xy^2\right]P_y(y)\\
&\qquad-xy\left[\gamma_2y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\;-\;\mu x(1-y)\,\pi(0,0)
\end{aligned}$$

The two $\frac{d}{dy}P_y(y)$ terms cancel and the two $P_y(y)$ coefficients combine into $\mu(x-y)$, leaving

$$\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0)
\end{aligned}$$

This is Equation (4.16), which completes the proof. □

The fundamental equation also delivers a diagonal identity in the no-abandonment case, complementing Corollary 1.

Corollary 2 (Diagonal identity without abandonment). *Suppose $\theta_1=\theta_2=0$ (Models A, B, B_2 and B_2^{H}) and $\rho<1$. Setting $x=y=z$ in the fundamental equation (4.16) annihilates both derivative terms (each carries the factor $x-y$), leaving the algebraic relation*

$$-(1-z)\big(\mu-\lambda z\big)P(z,z)=-\mu(1-z)\,\pi(0,0),$$

whence the diagonal of the PGF and the normalisation read

$$P(z,z)=\frac{\pi(0,0)}{1-\rho z},\qquad P(1,1)=1-\pi_0=\rho.\tag{4.21}$$

Since $\pi(0,0)=\rho(1-\rho)$ by Corollary 1, the diagonal is, up to the busy probability ρ, the PGF of a geometric distribution: conditionally on the server being busy, the total number of waiting customers is geometric, $\mathbb{P}(N=n\mid\mathcal{B}=1)=(1-\rho)\rho^n$ for $n\geq 0$.

4.2.2 Equation for PPGF in state space $\widetilde{S}$

The equation of Lemma 1 transforms both coordinates at once. An alternative device, notably used by Cohen [2], is the **Partial Probability Generating Function (PPGF)**: transforming only a subset of the random variables while keeping the discrete indices of the rest. For a priority queue this is particularly advantageous: it lets us exploit the well-known properties of the priority class (often a standard $M/M/1$ system) and concentrate the analytical effort on the non-trivial PGF of the non-priority class. Cohen developed this framework for multi-server systems and for the state space S; we apply it here to the single-server case, in a modern formulation consistent with the variables defined above, and —in this subsection— on the alternative state space $\widetilde{S}$.

On S, the PPGF with respect to N_2, jointly with the event that the first queue holds exactly n_1 waiting customers, is

$$\widetilde{P}(n_1, y) = \mathbb{E}\left[y^{N_2}\mathbf{1}_{\{N_1=n_1\}}\right] = \sum_{n_2=0}^{\infty} \pi(n_1, n_2)y^{n_2}, \tag{4.22}$$

and these partial transforms recover the joint PGF through $P(x, y) = \sum_{n_1=0}^{\infty} \widetilde{P}(n_1, y)\, x^{n_1}$. Despite the conditional look of the indicator, Equation (4.22) is a joint (unnormalised) transform; this version is used in the Model-A analysis (§5). On $\widetilde{S}$, the analogous object transforms n_2 and keeps the total count n discrete:

$$\widetilde{P}(y, n) = \sum_{n_2=0}^{n} \widetilde{\pi}(n_2, n)\, y^{n_2}, \tag{4.23}$$

a *polynomial* of degree n in y, since $n_2 \leq n$ on $\widetilde{S}$; this finiteness justifies term-by-term differentiation and boundedness arguments without convergence caveats.

Lemma 2 (Fundamental equation for the PPGF on $\widetilde{S}$)**.** *Consider the same system on the state space $\widetilde{S}$ of Equation (4.2), where n is the total number of customers* waiting *in the queues and n_2 the number of waiting class-*2 *customers. Assume positive recurrence, so that the stationary distribution $\{\widetilde{\pi}(n_2, n)\}$ exists, and let $\widetilde{P}(y, n)$ be the partial PGF of Equation (4.23). Then, for every $n \geq 1$, the family $\{\widetilde{P}(y, n)\}_{n\geq 0}$ satisfies the differential–difference equation*

$$\begin{aligned}
&[-(\gamma_2 + \gamma_1 y)(1 - y) + (\theta_2 - \theta_1)y]\,\frac{d}{dy}\widetilde{P}(y, n) + [\lambda_1 + \lambda_2 + \mu + \gamma_1 n(1 - y) + \theta_1 n]\,\widetilde{P}(y, n)\\
&\quad = \quad (\lambda_1 + \lambda_2 y)\widetilde{P}(y, n - 1) + [\mu + \theta_1(n + 1)]\,\widetilde{P}(y, n + 1)\\
&\qquad + (\theta_2 - \theta_1 y)\frac{d}{dy}\widetilde{P}(y, n + 1) + \mu y^n(1 - y)\,\widetilde{\pi}(n + 1, n + 1).
\end{aligned}$$

Proof. Multiply the interior equation (4.11) by y^{n_2} and sum over $1 \le n_2 \le n-1$:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n)y^{n_2} \\
&+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2} \\
&+\gamma_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2} \\
&+\theta_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2} \\
&+\theta_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2} \\
&\quad= \quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2} \\
&\qquad+\lambda_2\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2-1,n-1)y^{n_2} \\
&\qquad+\mu\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n+1)y^{n_2} \\
&\qquad+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2} \\
&\qquad+\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2} \\
&\qquad+\theta_1\sum_{n_2=1}^{n-1}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2} \\
&\qquad+\theta_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}
\end{aligned}
$$

Adding the diagonal equation (4.12) multiplied by y^n extends most summation limits to n; the λ_1- and γ_2-terms on the RHS remain restricted to $n_2 \le n-1$, and an extra term $\mu\widetilde{\pi}(n+1,n+1)y^n$

remains:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&\quad=\quad\lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\qquad+\lambda_2\sum_{n_2=1}^{n}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n+1)y^{n_2}+\mu\widetilde{\pi}(n+1,n+1)y^{n}\\
&\qquad+\gamma_1\sum_{n_2=1}^{n}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\qquad+\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{n}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{n}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}.
\end{aligned}
$$

As in the proof of Lemma 1, we write LHS_i and RHS_j for the i-th and j-th summands on the left- and right-hand sides of this display, in the order written (the two μ-terms together forming RHS_3). Evaluating each summand:

$$
\begin{aligned}
&LHS_1\\
&=\quad(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\widetilde{\pi}(0,n)y^0\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right]\\
&LHS_2\\
&=\quad\gamma_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&=\quad\gamma_1\left[n\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\right]\\
&=\quad\gamma_1\left[n\left(\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\widetilde{\pi}(0,n)y^0\right)-y\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2-1}\right]\\
&=\quad\gamma_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big)-y\frac{d}{dy}\widetilde{P}(y,n)\right]
\end{aligned}
$$

LHS_3

$$
\begin{aligned}
&= \gamma_2 \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2,n) y^{n_2} \\
&= \gamma_2 \left[y \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2,n) y^{n_2-1} \right] \\
&= \gamma_2 y \frac{d}{dy} \widetilde{P}(y,n)
\end{aligned}
$$

LHS_4

$$
\begin{aligned}
&= \theta_1 \sum_{n_2=1}^{n} (n-n_2) \widetilde{\pi}(n_2,n) y^{n_2} \\
&= \theta_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y \frac{d}{dy} \widetilde{P}(y,n) \right]
\end{aligned}
$$

LHS_5

$$
\begin{aligned}
&= \theta_2 \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2,n) y^{n_2} \\
&= \theta_2 y \frac{d}{dy} \widetilde{P}(y,n)
\end{aligned}
$$

RHS_1

$$
\begin{aligned}
&= \lambda_1 \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2,n-1) y^{n_2} \\
&= \lambda_1 \left[\sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n-1) y^{n_2} - \widetilde{\pi}(n,n-1) y^{n} \right] \\
&= \lambda_1 \left[\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1) y^{n_2} - \widetilde{\pi}(n,n-1) y^{n} - \widetilde{\pi}(0,n-1) y^{0} \right] \\
&= \lambda_1 \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1) y^{n} - \widetilde{\pi}(0,n-1) \right]
\end{aligned}
$$

RHS_2

$$
\begin{aligned}
&= \lambda_2 \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1) y^{n_2} \\
&= \lambda_2 \left[y \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1) y^{n_2-1} \right] \\
&= \lambda_2 \left[y \sum_{n_2=0}^{n-1} \widetilde{\pi}(n_2,n-1) y^{n_2} \right] \\
&= \lambda_2 \left[y \left(\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1) y^{n_2} - \widetilde{\pi}(n,n-1) y^{n} \right) \right] \\
&= \lambda_2 y \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1) y^{n} \right]
\end{aligned}
$$

RHS_3

$$
\begin{aligned}
&= \mu \widetilde{\pi}(n+1,n+1) y^{n} + \mu \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1) y^{n_2} \\
&= \mu \left[\widetilde{\pi}(n+1,n+1) y^{n} + \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1) y^{n_2} \right] \\
&= \mu \left[\widetilde{\pi}(n+1,n+1) y^{n} + \sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1) y^{n_2} - \widetilde{\pi}(0,n+1) y^{0} - \widetilde{\pi}(n+1,n+1) y^{n+1} \right] \\
&= \mu \left[\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1) + \widetilde{\pi}(n+1,n+1) y^{n} (1-y) \right]
\end{aligned}
$$

RHS_4

$$
\begin{aligned}
&= \gamma_1 \sum_{n_2=1}^{n} (n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2} \\
&= \gamma_1 \left[y \sum_{n_2=1}^{n} (n-(n_2-1))\widetilde{\pi}(n_2-1,n)y^{n_2-1} \right] \\
&= \gamma_1 \left[y \sum_{k=0}^{n-1} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \\
&= \gamma_1 \left[y \sum_{k=0}^{n} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \quad \text{(since the } k=n \text{ term is 0)} \\
&= \gamma_1 y \left[n \sum_{k=0}^{n} \widetilde{\pi}(k,n)y^{k} - y \sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} \right] \\
&= \gamma_1 y \left[n\widetilde{P}(y,n) - y\frac{d}{dy}\widetilde{P}(y,n) \right]
\end{aligned}
$$

RHS_5

$$
\begin{aligned}
&= \gamma_2 \sum_{n_2=1}^{n-1} (n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2} \\
&= \gamma_2 \sum_{k=2}^{n} k\widetilde{\pi}(k,n)y^{k-1} \quad \text{(letting } k=n_2+1\text{)} \\
&= \gamma_2 \left[\sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} - 1\cdot\widetilde{\pi}(1,n)y^{0} \right] \\
&= \gamma_2 \left[\frac{d}{dy}\widetilde{P}(y,n) - \widetilde{\pi}(1,n) \right]
\end{aligned}
$$

RHS_6

$$
\begin{aligned}
&= \theta_1 \sum_{n_2=1}^{n} (n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2} \\
&= \theta_1 \left[(n+1) \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1)y^{n_2} - \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n+1)y^{n_2} \right] \\
&= \theta_1 \left[(n+1) \left(\sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1)y^{n_2} - \widetilde{\pi}(0,n+1) - \widetilde{\pi}(n+1,n+1)y^{n+1} \right) \right] \\
&\quad - \theta_1 \left[y \left(\sum_{n_2=0}^{n+1} n_2\widetilde{\pi}(n_2,n+1)y^{n_2-1} - (n+1)\widetilde{\pi}(n+1,n+1)y^{n} \right) \right] \\
&= \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1} \right] \\
&\quad - \theta_1 \left[y\frac{d}{dy}\widetilde{P}(y,n+1) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1} \right] \\
&= \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - y\frac{d}{dy}\widetilde{P}(y,n+1) \right]
\end{aligned}
$$

RHS_7

$$\begin{aligned}
&= \theta_2 \sum_{n_2=1}^{n} (n_2+1)\widetilde{\pi}(n_2+1, n+1) y^{n_2} \\
&= \theta_2 \sum_{k=2}^{n+1} k\widetilde{\pi}(k, n+1) y^{k-1} \quad (\text{letting } k = n_2+1) \\
&= \theta_2 \left[\sum_{k=1}^{n+1} k\widetilde{\pi}(k, n+1) y^{k-1} - 1 \cdot \widetilde{\pi}(1, n+1) y^0\right] \\
&= \theta_2 \left[\frac{d}{dy}\widetilde{P}(y, n+1) - \widetilde{\pi}(1, n+1)\right]
\end{aligned}$$

Combining all the terms above:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\right] \\
&+ \gamma_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+ \gamma_2 y \frac{d}{dy}\widetilde{P}(y,n) \\
&+ \theta_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+ \theta_2 y \frac{d}{dy}\widetilde{P}(y,n) \\
&\quad = \lambda_1 \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^n - \widetilde{\pi}(0,n-1)\right] \\
&\quad + \lambda_2 y \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^n\right] \\
&\quad + \mu \left[\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1) + \widetilde{\pi}(n+1,n+1)y^n(1-y)\right] \\
&\quad + \gamma_1 y \left[n\widetilde{P}(y,n) - y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&\quad + \gamma_2 \left[\frac{d}{dy}\widetilde{P}(y,n) - \widetilde{\pi}(1,n)\right] \\
&\quad + \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - y\frac{d}{dy}\widetilde{P}(y,n+1)\right] \\
&\quad + \theta_2 \left[\frac{d}{dy}\widetilde{P}(y,n+1) - \widetilde{\pi}(1,n+1)\right]
\end{aligned}$$

Grouping the common terms yields:

$$\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y) + (\theta_2-\theta_1)y]\frac{d}{dy}\widetilde{P}(y,n) + [\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&= (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1) + [\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1) + (\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\quad + \{[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n]\,\widetilde{\pi}(0,n) - \lambda_1\widetilde{\pi}(0,n-1) - [\mu+\theta_1(n+1)]\,\widetilde{\pi}(0,n+1) \\
&\quad\quad -\gamma_2\widetilde{\pi}(1,n) - \theta_2\widetilde{\pi}(1,n+1)\} \\
&\quad - y^n(\lambda_1+\lambda_2 y)\widetilde{\pi}(n,n-1) + \mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}$$

The y-boundary equation (4.13) makes the bracketed terms vanish. Moreover, $\widetilde{\pi}(n,n-1)=0$, because the state $(n,n-1)$ lies outside $\widetilde{S}$. Hence

$$\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y) + (\theta_2-\theta_1)y]\frac{d}{dy}\widetilde{P}(y,n) + [\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&= (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1) + [\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1) + (\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\quad + \mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}$$

This is the asserted differential–difference equation, which completes the proof. □

Remark 1 (The $\widetilde{S}$ recursion is a failed simplification). The partial-PGF recurrence of Lemma 2 was introduced with the aim that taking a generating function over the class-2 index only, while carrying the total count n as a discrete parameter, would *simplify* the analysis — collapsing the two-dimensional balance on S to a one-dimensional recurrence in n. The opposite turns out to be true. The inhomogeneous term $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ couples each level n of the recurrence to the diagonal probability $\widetilde{\pi}(n{+}1,n{+}1)=\pi(0,n{+}1)$, and these diagonal probabilities are themselves unknown. The recurrence therefore does not close: solving it would require exactly the boundary probabilities it was intended to produce. Only for the baseline Model-A does this coupling term decay fast enough to be neglected, and even there dropping it yields only an approximate solution, accurate in a restricted parameter region (Approximation 1). For every model with an active mechanism the $\widetilde{S}$ route is strictly harder than the S analysis it was meant to replace. We therefore record it as a deliberate negative result: the $\widetilde{S}$ coordinatisation is retained only for Model-A, and the S formulation is used throughout the remainder of this thesis.

5 Analysis of Model-A

Model-A is the baseline model, obtained by setting all jockeying and abandonment parameters to zero ($\gamma_1=\gamma_2=\theta_1=\theta_2=0$), leaving an ordinary two-class $M/M/1$ queue with non-preemptive priority —the classical, extensively studied system of §2. We analyse it in both state spaces S and $\widetilde{S}$: on the former we give two complete proofs, on the latter a conditionally accurate approximation, evaluated in Approximation 1. The starting points of the analysis are the fundamental equations of Lemma 1 for S and Lemma 2 for $\widetilde{S}$, specialised to these parameter values.

Stability. Since there is no abandonment, the system is positive recurrent if and only if the total load satisfies

$$\rho=\frac{\lambda_1+\lambda_2}{\mu}<1. \tag{5.1}$$

Figure 4 shows the queueing diagram of Model-A.

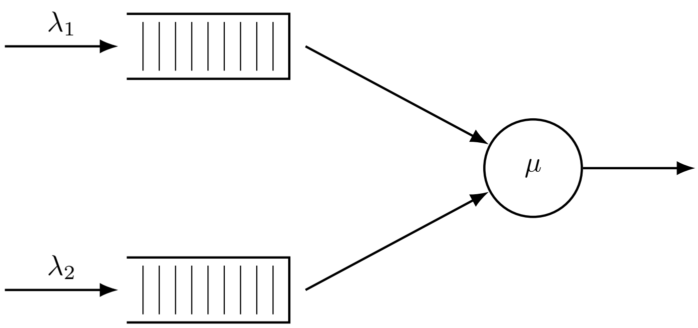

Figure 4: Queue layout for Model-A. Only Poisson arrivals and exponential service are present; the jockeying and abandonment rates are all zero.

5.1 Analysis in state space S

We begin from the fundamental equation of Lemma 1 with $\gamma_1=\gamma_2=\theta_1=\theta_2=0$, in which the boundary function $P_y(y)=P(0,y)$ is still to be determined:

$$\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)=\mu(x-y)P_y(y)-\mu x(1-y)\pi(0,0). \tag{5.2}$$

The following theorem summarises the main result of this subsection; two standalone proofs are given, one analytical and one probabilistic.

Theorem 2. *Consider a queueing system with two priority classes with Poisson arrival rates λ_1 and λ_2, where class-1 customers have non-preemptive priority over those of class 2, and exponentially distributed service times with rate μ independently of the class. Under the stability condition $\rho<1$ (5.1), the joint PGF $P(x,y)=\mathbb{E}[x^{N_1}y^{N_2}]$ is*

$$P(x,y)=\left[(1+\rho_1+\rho_2)xy-y-\rho_1x^2y-\rho_2xy^2\right]^{-1}\cdot y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\cdot\rho(1-\rho), \tag{5.3}$$

where

$$x^*(y)=\frac{[\mu+\lambda_1+\lambda_2(1-y)]-\sqrt{[\mu+\lambda_1+\lambda_2(1-y)]^2-4\lambda_1\mu}}{2\lambda_1}. \tag{5.4}$$

Corollary 3. *Under the stability condition $\rho < 1$ (5.1), the empty-state probabilities for Model-A are*

$$\pi_0 = 1 - \rho, \qquad \pi(0,0) = \rho(1-\rho). \tag{5.5}$$

In particular, $P(1,1) = 1 - \pi_0 = \rho$ and the diagonal generating function satisfies $P(z,z) = \pi(0,0)/(1-\rho z)$.

Proof of Corollary 3. Model-A is the no-mechanism case $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$, so the statement is the specialisation of Corollary 1 —equivalently Theorem 1 with the $M/M/1$ mean busy period $\mathbb{E}[BP] = (\mu - \lambda)^{-1}$, giving $\pi_0 = 1 - \rho$ and $\pi(0,0) = \rho(1-\rho)$. The diagonal value $P(z,z) = \pi(0,0)/(1-\rho z)$ and the normalisation $P(1,1) = 1 - \pi_0 = \rho$ are the reduction (4.21) of the Model-A kernel (5.2) on $x = y = z$ (Corollary 2). □

5.1.1 Analytical approach

Proof of Theorem 2, analytical approach. We begin from the fundamental equation for Model-A (5.2) and apply the *Kernel Method*, used in [2]: for each fixed $y \in (0,1]$, we find $x^*(y)$ such that the coefficient of $P(x,y)$ —the *kernel*— vanishes. Since $P(x,y)$ is a PGF, it is bounded, so at any such root the right-hand side of Equation (5.2) must also vanish; this will determine $P_y(y)$ once the root is identified.

To find the roots of the kernel, divide by $y \neq 0$ and set

$$0 = (\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy.$$

Rearranging into a standard quadratic in x:

$$0 = \lambda_1 x^2 - [\mu + \lambda_1 + \lambda_2(1-y)]\, x + \mu,$$

whose roots are

$$x^{\pm}(y) = \frac{[\mu + \lambda_1 + \lambda_2(1-y)] \pm \sqrt{[\mu + \lambda_1 + \lambda_2(1-y)]^2 - 4\mu\lambda_1}}{2\lambda_1}.$$

We label the root with the negative sign $x^*(y)$ and the root with the positive sign $x^+(y)$. At $y = 1$:

$$x^*(1) = 1 \quad \text{and} \quad x^+(1) = \mu/\lambda_1 > 1.$$

We require our root to lie inside the closed unit disc for $y \in [0,1)$. Computing the derivative of $x^*(y)$:

$$\frac{d}{dy}x^*(y) = \frac{\lambda_2}{2\lambda_1}\left[-1 + \frac{\mu + \lambda_1 + \lambda_2(1-y)}{\sqrt{(\mu + \lambda_1 + \lambda_2(1-y))^2 - 4\lambda_1\mu}}\right] > 0, \tag{5.6}$$

since $(\mu+\lambda_1+\lambda_2(1-y))^2 > (\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu$ whenever $\lambda_1\mu > 0$, by squaring numerator and denominator in the inner fraction. So $x^*(y)$ is strictly increasing in $[0,1]$, and since $x^*(1) = 1$ we have $x^*(y) < 1$ for $y \in [0,1)$. We make use of Vieta's formulas to discard the other root for the kernel polynomial $\lambda_1 x^2 - [\mu + \lambda_1 + \lambda_2(1-y)]x + \mu = 0$: the product of the two roots satisfies $x^*(y)\, x^+(y) = \mu/\lambda_1 > 1$, so $x^+(y) = \mu/(\lambda_1 x^*(y)) > 1$ throughout $[0,1)$. So the only root strictly inside the unit disc for $y \in [0,1)$ is $x^*(y)$:

$$x^*(y) = \frac{(\mu + \lambda_1 + \lambda_2(1-y)) - \sqrt{(\mu + \lambda_1 + \lambda_2(1-y))^2 - 4\lambda_1\mu}}{2\lambda_1}. \tag{5.7}$$

Returning to the fundamental equation (5.2) at $x = x^*(y)$, the RHS also equals zero, giving

$$\mu \cdot [(x^*(y) - y)P_y(y) - x^*(y)(1-y)\pi(0,0)] = 0 \quad \Longrightarrow \quad P_y(y) = \frac{x^*(y)(1-y)}{x^*(y) - y}\pi(0,0). \tag{5.8}$$

By Corollary 3, $\pi(0,0) = \rho(1-\rho)$. Substituting this and Equation (5.8) into the fundamental equation (5.2), whose left-hand side stays unchanged while the right-hand side is simplified in stages:

$$\begin{aligned}
\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(x-y)\frac{x^*(y)(1-y)}{x^*(y) - y}\rho(1-\rho) - \mu x(1-y)\rho(1-\rho) \\
&= \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y) - y} - x\right]\rho(1-\rho)
\end{aligned}$$

$$= \mu(1-y)\left[\frac{y(x-x^*(y))}{x^*(y)-y}\right]\rho(1-\rho).$$

Dividing both sides by μ, so that the kernel is expressed in the loads ρ_i:

$$\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right] P(x,y) = y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).$$

Multiplying both sides by $\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1}$ yields $P(x,y)$ as in Theorem 2 and concludes the proof. □

5.1.2 Probabilistic approach

Proof of Theorem 2, probabilistic approach. This proof mirrors the analytical one, but derives the boundary function $P_y(y)$ by means of probabilistic arguments; the starting point is again the fundamental equation for Model-A (Equation (5.2)).

Class-2 conditional distribution via $M/G/1$. Since class 1 has non-preemptive priority, class-2 customers are not taken into service whenever the priority queue is not empty ($N_1 > 0$); their *effective service time* B is therefore the duration of a complete busy period of the single-class ordinary $M/M/1$ sub-queue with arrival rate λ_1 and service rate μ. From class-2's viewpoint the system is thus an $M/G/1$ queue with arrival rate λ_2 and generally distributed service times B and mean $\mathbb{E}[B]$. The LST and mean of B are presented in §3.1 (Equations (3.1)–(3.2)), now with λ_1 instead of the generic λ:

$$\widetilde{B}(s) = \frac{(\mu+\lambda_1+s) - \sqrt{(\mu+\lambda_1+s)^2 - 4\mu\lambda_1}}{2\lambda_1},$$
$$\mathbb{E}[B] = \frac{1}{\mu(1-\rho_1)}.$$

Applying the Pollaczek–Khinchine (PK) formula (§3.5, Equation (3.11)) to this $M/G/1$ system, with $\widetilde{B}(\lambda_2(1-y)) = x^*(y)$ (cf. (5.7)) and $\lambda_2\mathbb{E}[B] = \rho_2/(1-\rho_1)$:

$$L_2(y) = \frac{(1-\lambda_2\mathbb{E}[B])(1-y)\widetilde{B}(\lambda_2(1-y))}{\widetilde{B}(\lambda_2(1-y)) - y} = \frac{1-\rho}{1-\rho_1}\cdot\frac{x^*(y)(1-y)}{x^*(y)-y}.$$

The identification of L_2 with the conditional law of N_2 given the event $\{\text{busy}, N_1 = 0\}$ deserves a proof, and we give one by *censoring*: watch the chain only while it is in the set $E = \{\text{busy}, N_1 = 0\}$, splicing out the excursions away from E. The watched process is again a Markov chain, and its stationary law is exactly the conditional stationary law given E. While the system is in E, the class-2 queue grows at rate λ_2 and, when $N_2 \geq 1$, shrinks at rate μ: no class-1 customer waits, so each service completion admits the head of Queue 2. The set E is left in two ways, and neither disturbs N_2 on re-entry. From $(0,0)$ a completion empties the system; no customer waits while the server is idle, and the next arrival of either class enters service directly, so the chain re-enters E with N_2 unchanged. From any state of E, a class-1 arrival (rate λ_1) opens a *clearing spell*: by *memorylessness* the residual service is $\text{Exp}(\mu)$, so the waiting count N_1 performs a first passage from 1 to 0 of the birth–death chain with rates (λ_1,μ) —one class-1 busy period B— during which the priority rule admits no class-2 customer into service and the class-2 queue accumulates a batch of $\text{Poisson}(\lambda_2 B)$ arrivals. Successive spells are i.i.d. by the strong Markov property. Watched on E, the class-2 queue is therefore a birth–death chain with an additional batch stream: single arrivals at rate λ_2, single departures at rate μ, and, at rate λ_1, batches of size K with $\mathbb{E}[y^K] = \mathbb{E}[e^{-\lambda_2(1-y)B}] = \widetilde{B}(\lambda_2(1-y)) = x^*(y)$. Writing $G(y)$ for the stationary PGF of the watched chain and $p_0 = G(0)$, balance of probability flow gives

$$\mu\,\frac{1-y}{y}\big(G(y)-p_0\big) \;=\; \big[\lambda_2(1-y) + \lambda_1\big(1-x^*(y)\big)\big]\,G(y),$$

so that

$$G(y) \;=\; \frac{\mu(1-y)\,p_0}{\mu(1-y) - \lambda_2\,y(1-y) - \lambda_1\,y\big(1-x^*(y)\big)}.$$

Since $x^*(y)$ is a root of the kernel quadratic $\lambda_1 x^2 - [\mu+\lambda_1+\lambda_2(1-y)]\,x + \mu = 0$ (§5.1.1), the quadratic rearranged at its root factorises as $\big(1-x^*(y)\big)\big(\mu-\lambda_1 x^*(y)\big) = \lambda_2\,x^*(y)\,(1-y)$. This identity reduces the denominator above, giving $G(y) = p_0\big(\mu - \lambda_1 x^*(y)\big)/\big(\mu - \lambda_1 x^*(y) - \lambda_2 y\big)$; the normalisation $G(1) = 1$ (with $x^*(1) = 1$) fixes $p_0 = (1-\rho)/(1-\rho_1)$, and a second application of the same identity turns the expression into

$$G(y) \;=\; \frac{1-\rho}{1-\rho_1} \cdot \frac{x^*(y)\,(1-y)}{x^*(y)-y} \;=\; L_2(y),$$

exactly the Pollaczek–Khinchine law built above. Hence $\mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] = L_2(y)$: the clearing spells are where the busy period B enters, and they make the $M/G/1$ reading of the construction precise.

By the law of total expectation,

$$P_y(y) = \mathbb{P}(\text{busy}, N_1 = 0) \cdot \mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] = \mathbb{P}(\text{busy}, N_1 = 0) \cdot L_2(y).$$

The factor $\mathbb{P}(\text{busy}, N_1 = 0)$ follows from a cut argument that makes precise the sense in which the priority queue operates autonomously. Only two kinds of transition change N_1. A class-1 arrival while the server is busy raises it by one, at rate λ_1 (an arrival finding the server idle enters service directly and leaves $N_1 = 0$); a service completion while $N_1 \geq 1$ lowers it by one, at rate μ, since by *memorylessness* every service in progress —of either class— completes at rate μ, and the freed server then admits the head of Queue 1. Transitions between idle and busy occur only while $N_1 = 0$ and cross no level. Balance across the cut between $\{N_1 \leq k\}$ and $\{N_1 \geq k+1\}$ therefore reads

$$\lambda_1\,\mathbb{P}(\text{busy},\, N_1 = k) \;=\; \mu\,\mathbb{P}(N_1 = k+1), \qquad k \geq 0,$$

and since $N_1 \geq 1$ implies a busy server, the recursion is geometric,

$$\mathbb{P}(\text{busy},\, N_1 = k) \;=\; \rho_1^k\,\mathbb{P}(\text{busy},\, N_1 = 0).$$

Summing over $k \geq 0$ gives $\mathbb{P}(N_1 = 0 \mid \text{busy}) = 1 - \rho_1$ —during busy time the priority queue exhibits exactly the geometric law of the ordinary $M/M/1$ queue with rates (λ_1, μ). Combined with $\mathbb{P}(\text{busy}) = \rho$, we have

$$\mathbb{P}(\text{busy}, N_1 = 0) = \rho(1-\rho_1);$$

and therefore

$$P_y(y) = \rho(1-\rho_1) \cdot L_2(y) = \rho(1-\rho_1) \cdot \frac{1-\rho}{1-\rho_1} \cdot \frac{x^*(y)(1-y)}{x^*(y)-y} = \frac{x^*(y)(1-y)}{x^*(y)-y} \cdot (1-\rho)\rho \tag{5.9}$$

Substituting Equation (5.9) and $\pi(0,0) = \rho(1-\rho)$ (Corollary 3) into the fundamental equation (5.2), and simplifying the right-hand side in stages as in the analytical proof:

$$\begin{aligned}
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho) - \mu x(1-y)\rho(1-\rho) \\
&= \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y} - x\right]\rho(1-\rho) \\
&= \mu(1-y)\left[\frac{y(x-x^*(y))}{x^*(y)-y}\right]\rho(1-\rho).
\end{aligned}$$

Dividing both sides by μ:

$$\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right] P(x,y) = y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).$$

Multiplying both sides by $\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1}$ to isolate $P(x,y)$ yields Theorem 2 and concludes the proof. □

5.1.3 Partial-PGF approach

This proof follows Cohen [2]'s method. We keep the class-1 index n_1 as a discrete parameter and take transforms only with respect to N_2, via the PPGF $\widetilde{P}(n_1, y) = \sum_{n_2 \geq 0} \pi(n_1, n_2)\, y^{n_2}$ from Equation (4.22). Multiplying the balance equations by y^{n_2} and summing over n_2 turns them into a second-order linear recurrence in n_1, whose characteristic roots fix the geometric structure of the solution in the priority-class index.

Proof. We start from the Model-A balance equations, obtained by setting $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$ in the interior equation (4.6) and the x-boundary equation (4.8):

$$\begin{aligned}(\lambda_1+\lambda_2+\mu)\,\pi(n_1,n_2) &= \mu\,\pi(n_1+1,n_2)+\lambda_1\,\pi(n_1-1,n_2)+\lambda_2\,\pi(n_1,n_2-1), && n_1\geq 1,\ n_2\geq 1,\\ (\lambda_1+\lambda_2+\mu)\,\pi(n_1,0) &= \mu\,\pi(n_1+1,0)+\lambda_1\,\pi(n_1-1,0), && n_1\geq 1,\ n_2=0.\end{aligned} \tag{5.10}$$

Now we take transforms by summing all over their states, but only with respect to n_2, and we substitute our PPGF $\widetilde{P}(n_1,y)=\sum_{n_2=0}^{\infty}\pi(n_1,n_2)y^{n_2}$:

$$\begin{aligned}&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(n_1,n_2)y^{n_2}\\ &=\quad \mu\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)y^{n_2}+\lambda_1\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)y^{n_2}+\lambda_2\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)y^{n_2}\\ &LHS_1 \quad = \quad RHS_1+RHS_2+RHS_3\end{aligned}$$

where LHS_1 and RHS_j denote the terms of the display above, in the order written. Evaluating each summand:

$$\begin{aligned}&LHS_1\\ &=\quad (\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(n_1,n_2)y^{n_2}\\ &=\quad (\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)y^{n_2}-\pi(n_1,0)y^0\right]\\ &=\quad (\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(n_1,y)-\pi(n_1,0)\right]\\ &RHS_1\\ &=\quad \mu\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)y^{n_2}\\ &=\quad \mu\left[\sum_{n_2=0}^{\infty}\pi(n_1+1,n_2)y^{n_2}-\pi(n_1+1,0)y^0\right]\\ &=\quad \mu\left[\widetilde{P}(n_1+1,y)-\pi(n_1+1,0)\right]\\ &RHS_2\\ &=\quad \lambda_1\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)y^{n_2}\\ &=\quad \lambda_1\left[\sum_{n_2=0}^{\infty}\pi(n_1-1,n_2)y^{n_2}-\pi(n_1-1,0)y^0\right]\\ &=\quad \lambda_1\left[\widetilde{P}(n_1-1,y)-\pi(n_1-1,0)\right]\\ &RHS_3\\ &=\quad \lambda_2\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)y^{n_2}\\ &=\quad \lambda_2 y\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)y^{n_2-1}\\ &=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(n_1,m)y^{m}\\ &=\quad \lambda_2 y\,\widetilde{P}(n_1,y)\end{aligned}$$

Combining the terms above:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\big[\widetilde{P}(n_1,y)-\pi(n_1,0)\big]\\
&\quad=\quad \mu\big[\widetilde{P}(n_1+1,y)-\pi(n_1+1,0)\big]\\
&\qquad+\ \lambda_1\big[\widetilde{P}(n_1-1,y)-\pi(n_1-1,0)\big]\\
&\qquad+\ \lambda_2 y\,\widetilde{P}(n_1,y)
\end{aligned}$$

Rearranging, with the transforms on the left and the boundary probabilities on the right:

$$\begin{aligned}
&\big[\mu+\lambda_1+\lambda_2(1-y)\big]\widetilde{P}(n_1,y)-\mu\widetilde{P}(n_1+1,y)-\lambda_1\widetilde{P}(n_1-1,y)\\
&\quad=\quad(\lambda_1+\lambda_2+\mu)\pi(n_1,0)-\mu\pi(n_1+1,0)-\lambda_1\pi(n_1-1,0)
\end{aligned}\tag{5.11}$$

By the x-boundary balance equation (5.10), the right-hand side of Equation (5.11) vanishes completely, giving the homogeneous recurrence

$$[\mu+\lambda_1+\lambda_2(1-y)]\,\widetilde{P}(n_1,y)=\mu\widetilde{P}(n_1+1,y)+\lambda_1\widetilde{P}(n_1-1,y),\qquad n_1\geq 1.\tag{5.12}$$

Characteristic roots and mode selection. Equation (5.12) is a *second-order linear recurrence* in n_1 with constant coefficients, so its general solution is a linear combination of two geometric sequences $[\widetilde{x}^{\pm}(y)]^{n_1}$, where $\widetilde{x}^{\pm}(y)$ solve the characteristic equation

$$\mu\,\widetilde{x}^2-[\mu+\lambda_1+\lambda_2(1-y)]\,\widetilde{x}+\lambda_1=0,\tag{5.13}$$

that is,

$$\widetilde{x}^{\pm}(y)=\frac{\big(\mu+\lambda_1+\lambda_2(1-y)\big)\pm\sqrt{\big(\mu+\lambda_1+\lambda_2(1-y)\big)^2-4\lambda_1\mu}}{2\mu}.$$

The *characteristic polynomial* (Equation (5.13)) is the reciprocal to that of the bivariate PGF approaches (i.e. the reciprocal of $\lambda_1x^2-[\mu+\lambda_1+\lambda_2(1-y)]x+\mu$), thus its roots are the reciprocals of that kernel's roots: the positive root is the reciprocal of the negative one ($\widetilde{x}^{+}(y)=1/x^{*}(y)$), and conversely $\widetilde{x}^{-}(y)=1/x^{+}(y)$. By Vieta's formulas, $\widetilde{x}^{-}(y)\,\widetilde{x}^{+}(y)=\lambda_1/\mu=\rho_1<1$. Recall from §5.1.1 that $x^{*}(y)<1<x^{+}(y)$ for $y\in[0,1)$. We select the negative-sign root because the sequence $\{\widetilde{P}(n_1,y)\}_{n_1\geq 0}$ must be summable:

$$\sum_{n_1=0}^{\infty}\widetilde{P}(n_1,y)=P(1,y)\leq P(1,1)=1-\pi_0=\rho<\infty.$$

At $y=1$, we have

$$\widetilde{x}^{-}(1)=\rho_1\quad\text{and}\quad\widetilde{x}^{+}(1)=1,$$

the latter making the geometric sequence $[\widetilde{x}^{+}(y)]^{n_1}$ non-summable. We therefore define

$$\widetilde{x}^{*}(y):=\widetilde{x}^{-}(y)=\frac{\big(\mu+\lambda_1+\lambda_2(1-y)\big)-\sqrt{\big(\mu+\lambda_1+\lambda_2(1-y)\big)^2-4\lambda_1\mu}}{2\mu}=\rho_1\,x^{*}(y).\tag{5.14}$$

The last identity holds because $\widetilde{x}^{*}$ and x^{*} share the same numerator and differ only in the denominator (2μ versus $2\lambda_1$). Next,

$$\widetilde{P}(n_1,y)=\widetilde{P}(0,y)\,[\widetilde{x}^{*}(y)]^{n_1},\qquad n_1\geq 0,\tag{5.15}$$

where the relation also holds at $n_1=0$ trivially. Recall that at $y=1$, we have $\widetilde{x}^{*}(1)=\lambda_1/\mu=\rho_1$. Thus, $\widetilde{P}(n_1,1)=\widetilde{P}(0,1)\,\rho_1^{n_1}$ and $\widetilde{P}(0,1)=P_y(1)=\rho(1-\rho_1)$. The constant $\widetilde{P}(0,y)$ is resolved by its definition

$$\widetilde{P}(0,y)=\sum_{n_2=0}^{\infty}\pi(0,n_2)\,y^{n_2}=P(0,y)=P_y(y),$$

as determined in the bivariate PGF approaches. Together with $P_y(y)$, the geometric law (5.15) already characterises the full joint distribution; what remains is to verify that it reproduces the closed form of Theorem 2. Summing the geometric law (Equation (5.15)) over $n_1\geq 0$ for $x\in(0,1]$ recovers

$$P(x,y)=\sum_{n_1=0}^{\infty}\widetilde{P}(n_1,y)\,x^{n_1}=P_y(y)\sum_{n_1=0}^{\infty}\big[\widetilde{x}^{*}(y)\,x\big]^{n_1}=\frac{P_y(y)}{1-\widetilde{x}^{*}(y)\,x}.\tag{5.16}$$

To bring this to the form of Theorem 2, we define the kernel of the fundamental equation, divided by μ:

$$K(x,y) := (1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2 = -y\rho_1\big(x - x^*(y)\big)\big(x - x^+(y)\big), \tag{5.17}$$

where $x^*(y)$ and $x^+(y)$ are the roots found in §5.1.1. By Vieta's product formula, $x^*(y)\,x^+(y) = \mu/\lambda_1 = 1/\rho_1$. This, together with $\widetilde{x}^*(y)$ in Equation (5.14), gives $\widetilde{x}^*(y) = \rho_1 x^*(y) = 1/x^+(y)$. Using the factorisation of $K(x,y)$ (5.17), the denominator of the reconstructed PGF (5.16) becomes

$$1 - \widetilde{x}^*(y)\,x = 1 - \frac{x}{x^+(y)} = \frac{x^+(y) - x}{x^+(y)} = \frac{K(x,y)}{y\rho_1\, x^+(y)\big(x - x^*(y)\big)}.$$

Substituting this into our reconstructed $P(x,y)$ (Equation (5.16)), using $P_y(y) = \frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho)$ and also noting that $\rho_1\, x^*(y)\, x^+(y) = 1$, we obtain

$$\begin{aligned}
P(x,y) &= \frac{x^*(y)(1-y)}{x^*(y)-y}\,\rho(1-\rho)\cdot\frac{y\rho_1\, x^+(y)\big(x-x^*(y)\big)}{K(x,y)} \\
&= \underbrace{\rho_1\, x^*(y)\, x^+(y)}_{=1}\;\frac{1}{K(x,y)}\; y(1-y)\,\frac{x - x^*(y)}{x^*(y)-y}\,\rho(1-\rho) \\
&= \big[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\big]^{-1}\cdot y(1-y)\,\frac{x-x^*(y)}{x^*(y)-y}\,\rho(1-\rho),
\end{aligned}$$

which recovers Equation (5.3) from Theorem 2 and concludes the proof. $\square$

5.2 The partial-PGF recursion on $\widetilde{S}$: scope and limitations

Setting $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$ in the fundamental $\widetilde{S}$ equation of Lemma 2 —equivalently, transforming the $\widetilde{S}$ balance equations (4.11)–(4.15) with the PPGF $\widetilde{P}(y,n) = \sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n)\, y^{n_2}$— yields the inhomogeneous recurrence:

$$\begin{aligned}
&[\lambda_1+\lambda_2+\mu]\,\widetilde{P}(y,n) \\
&= \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1) + \mu\widetilde{P}(y,n+1) + \mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned} \tag{5.18}$$

The forcing term in Equation (5.18) is controlled by the diagonal probabilities $\widetilde{\pi}(n+1,n+1) = \pi(0,n+1)$, whose decay we first establish from the exact S-solution.

Lemma 3 (Geometric decay of the boundary probabilities)**.** *For Model-A the boundary probabilities $\pi(0,n)$ satisfy $\pi(0,n) = O(r^{-n})$ for every $r < 1/\rho$; equivalently they decay geometrically with rate ρ.*

Proof. By definition $P_y(y) = P(0,y) = \sum_{n\geq 0}\pi(0,n)\, y^n$ is the generating function of the coefficients $\pi(0,n)$, and from Equation (5.8) with $\pi(0,0) = \rho(1-\rho)$,

$$P_y(y) = \rho(1-\rho)\,\frac{x^*(y)(1-y)}{x^*(y)-y}.$$

Evaluating the kernel quadratic (5.4) at $x = y$ gives $\lambda_1 y^2 - [\mu+\lambda_1+\lambda_2(1-y)]y + \mu = (\lambda y - \mu)(y-1)$ with $\lambda = \lambda_1+\lambda_2$, so $x^*(y) = y$ holds only at $y = 1$ and $y = \mu/\lambda = 1/\rho$. At $y = 1$ the simple zeros of $(1-y)$ and $(x^*(y) - y)$ cancel and P_y is analytic with $P_y(1) = \rho$; the branch point of x^* lies beyond $1/\rho$. Hence the singularity of P_y nearest the origin is the simple pole at $y = 1/\rho > 1$, so P_y is analytic on the disc $|y| < 1/\rho$. By the Cauchy coefficient estimates, for every $r < 1/\rho$ there is $C_r < \infty$ with $\pi(0,n) \leq C_r\, r^{-n}$, i.e. geometric decay at rate ρ. $\square$

Approximation 1 (Heuristic PPGF on $\widetilde{S}$)**.** *Neglecting the boundary-diagonal forcing term $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ in Equation (5.18) —which decays geometrically in n by Lemma 3— the homogeneous recurrence admits the geometric solution*

$$\widetilde{P}(y,n) \;\approx\; \rho(1-\rho)\,[\widetilde{y}^*(y)]^n,$$

where

$$\widetilde{y}^*(y) = \frac{(\lambda+\mu) - \sqrt{(\lambda+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}, \qquad \lambda = \lambda_1+\lambda_2.$$

At $y = 1$ this recovers $\widetilde{P}(1,n) = (1-\rho)\rho^{n+1}$, consistent with the steady-state marginal of the $M/M/1$ queue with rates λ and μ for n customers waiting.

Justification of Approximation 1. The inhomogeneous term $\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)$ prevents the direct application of a geometric-sequence ansatz. The diagonal probabilities $\widetilde{\pi}(n,n) = \pi(0,n)$ correspond to states where all n waiting customers are class 2, and by Lemma 3 they decay geometrically in n at rate ρ; the forcing term is therefore exponentially small *as n grows* —the recurrence holds for every n, and for small n the term need not be negligible— and neglecting it throughout yields an approximation whose accuracy is quantified below. The resulting homogeneous recurrence is

$$(\lambda_1 + \lambda_2 + \mu)\widetilde{P}(y,n) = (\lambda_1 + \lambda_2 y)\widetilde{P}(y,n-1) + \mu\widetilde{P}(y,n+1).$$

The characteristic polynomial of this recurrence is

$$\mu[\widetilde{y}(y)]^2 - (\lambda_1 + \lambda_2 + \mu)\widetilde{y}(y) + (\lambda_1 + \lambda_2 y) = 0,$$

with roots

$$\widetilde{y}^{\pm}(y) = \frac{(\lambda_1 + \lambda_2 + \mu) \pm \sqrt{(\lambda_1 + \lambda_2 + \mu)^2 - 4\mu(\lambda_1 + \lambda_2 y)}}{2\mu}.$$

At $y = 1$ the roots are $\widetilde{y}_+(1) = 1$ and $\widetilde{y}_-(1) = \rho$. The root $\widetilde{y}_+ = 1$ would produce a non-summable geometric series as $n \to \infty$, so only the negative-sign root is admissible:

$$\widetilde{y}^*(y) = \frac{(\lambda_1 + \lambda_2 + \mu) - \sqrt{(\lambda_1 + \lambda_2 + \mu)^2 - 4\mu(\lambda_1 + \lambda_2 y)}}{2\mu}.$$

Note that $\widetilde{y}^*(y)$ differs from the kernel roots $x^*(y)$ and $\widetilde{x}^*(y)$ of the S-space analyses: there the transform variable enters through the linear coefficient $\lambda_2(1-y)$ of the characteristic polynomial, here through its constant term $\lambda_1 + \lambda_2 y$. The geometric ansatz gives $\widetilde{P}(y,n) = \widetilde{P}(y,0) \cdot [\widetilde{y}^*(y)]^n$. Using $\widetilde{P}(y,0) = \widetilde{\pi}(0,0) = \pi(0,0) = \rho(1-\rho)$ (Corollary 3):

$$\widetilde{P}(y,n) = \rho(1-\rho) \cdot [\widetilde{y}^*(y)]^n.$$

Since n counts customers *waiting* (the customer in service is not counted), $n+1$ is the total number in the system, so evaluating at $y = 1$:

$$\widetilde{P}(1,n) = \rho(1-\rho) \cdot \rho^n = (1-\rho)\rho^{n+1},$$

which matches the stationary probability of $n+1$ customers in the $M/M/1$ queue with rates λ and μ [1]. □

Failure mechanism. The neglected forcing $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ feeds in the diagonal states, in which all waiting customers are class 2. Using $\widetilde{\pi}(n,n) = \pi(0,n)$ and the rate proved in Lemma 3, this term decays in n like ρ^n (modulated by y^n), whereas the homogeneous solution decays like $[\widetilde{y}^*(y)]^n$. The approximation is therefore accurate exactly when the forcing decays faster, i.e. $\rho < \widetilde{y}^*(y)$ on the relevant range of y; this holds when the priority class carries most of the load (ρ_1/ρ large, ρ moderate), precisely the empirical validity region of the approximation.

The obstruction is structural and recurs throughout this thesis: an unknown boundary/diagonal trace —here the sequence $\pi(0,n)$— couples into an otherwise solvable recursion, exactly as the unknown diagonal $P(z,z)$ enters the Volterra forcing of Models B and X (§6 and Remark 2). A rigorous closure would retain the forcing and solve Equation (5.18) by variation of constants in n, which reintroduces the very diagonal sequence $\{\pi(0,n)\}$ as an unknown to be determined self-consistently; see §12.3.

6 Analysis of Model-B

In this section, we derive a general solution to the fundamental equation of Model-B, expressed in terms of the unknown boundary function $P_y(y)$. Determining $P_y(y)$ itself reduces to a singular Volterra-type integral equation that we do not solve; whether it admits a closed form, or a tractable solution at all, is left open. The section thus documents the precise obstacle that derivative terms in y pose in our fundamental equation, and thereby also justifies not carrying out an analogous derivation for Model-X. Moreover, since jockeying breaks our Poisson arrivals and renewal structure, the probabilistic argument explained in § 5.1.2 is no longer available.

Model-B introduces bidirectional jockeying ($\gamma_1, \gamma_2 > 0$) and no abandonment ($\theta_1 = \theta_2 = 0$). Figure 5 shows the queue layout.

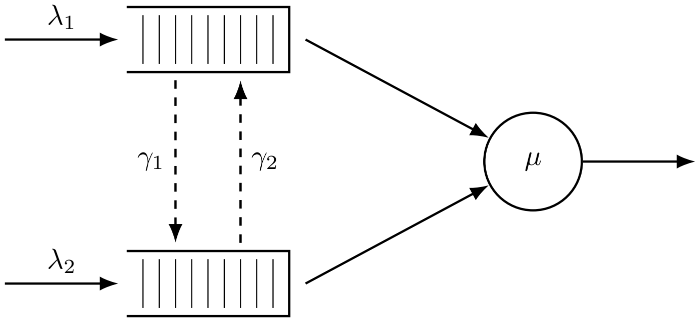

Figure 5: Queue layout for Model-B. Dashed arrows indicate bidirectional jockeying: γ_1 moves a class-1 customer down to Queue 2 and γ_2 moves a class-2 customer up to Queue 1. Both transfers conserve the total n.

Stability. The system is stable if and only if $\rho = (\lambda_1 + \lambda_2)/\mu < 1$: jockeying preserves the total customer count and does not affect this condition, as the arriving customers can also be re-allocated.

Setting $\theta_1 = \theta_2 = 0$ in the general fundamental equation (4.16) (Lemma 1) removes both abandonment factors $(x-1)$ and $(y-1)$ and leaves the *Model-B fundamental equation*

$$\begin{aligned}\big[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big]\, P(x,y)&\\ + \gamma_1\, xy(x-y)\,\frac{\partial P}{\partial x}(x,y) + \gamma_2\, xy(y-x)\,\frac{\partial P}{\partial y}(x,y)&\\ = \mu(x-y)\,P_y(y) - \mu\, x(1-y)\,\pi(0,0).&\end{aligned} \tag{6.1}$$

This equation governs the entire analysis of the section, and we refer to it throughout in place of the general equation (4.16). Its only derivative terms are the two jockeying terms, both carrying the diagonal factor $(x-y)$ that makes the Method of Characteristics applicable.

The result below is stated as a corollary, so that it sits at the same hierarchy level as the corresponding results for the other models. Since Model-B is not solved, however, no theorem is attached to it.

Corollary 4. *The empty-system probability π_0 and the empty-queues probability $\pi(0,0)$ of Model-B coincide with those of the Model-A baseline,*

$$\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho), \tag{6.2}$$

and the diagonal of the PGF is $P(z,z) = \pi(0,0)/(1-\rho z)$.

The proof follows immediately by setting $x = y = z$ in the Model-B fundamental equation (6.1): both jockeying terms, which carry the factor $(x-y)$, vanish, and the term $\mu(x-y)P_y(y)$ vanishes with them. The remaining diagonal equation is therefore *identical* to that of Model-A with $x = y = z$, so Corollary 4 is the no-abandonment instance of Theorem 1: jockeying conserves the total count, leaving an $M/M/1$ process of mean busy period $\mathbb{E}[BP] = (\mu-\lambda)^{-1}$ (Corollary 1).

For Model-B we do not provide a solution; however, working out the problem exposes the precise analytical obstacle, which we record as a claim. For the sake of readability, we will *not* define here the auxiliary functions; they are introduced throughout the proof. The following claim is made:

Claim 1. *Assume stability $\rho < 1$. Along the characteristic lines $\gamma_2 x + \gamma_1 y = C_1$ of the Model-B fundamental equation (6.1), parametrised by $y(x) = (C_1 - \gamma_2 x)/\gamma_1$, the bivariate PGF of Model-B admits the following integral representation:*

$$P\big(x,y(x)\big) \;=\; P_h\big(x,y(x)\big)\int_0^x \frac{h\big(t;C_1\big)}{P_h\big(t,y(t)\big)}\,dt, \tag{6.3}$$

where P_h is the homogeneous solution and the forcing function h depends linearly on the unknown boundary trace $P_y(y) = P(0,y)$. Evaluating Equation (6.3) along the diagonal $x = y = z$, where $P(z,z) = \pi(0,0)/(1-\rho z)$ is known from Corollary 4, reduces the determination of P_y to a Volterra integral equation of the first kind

$$\mathcal{F}(z) \;=\; \int_0^z \mathcal{K}(z,t)\,P_y\big(y(t;z)\big)\,dt. \tag{6.4}$$

Proof of Claim 1. The proof consists of five steps: (i) homogeneous solution via the Method of Characteristics, (ii) sign analysis of the exponent function E, (iii) particular solution via the Method of Variation of Parameters, (iv) determining the arbitrary function from the boundary condition, and (v) evaluation at $x = y$ yielding a Volterra-type equation.

Step (i): Homogeneous solution via the Method of Characteristics.

The Model-B fundamental equation (6.1) is a first-order linear PDE in $P(x, y)$. Setting the right-hand side to zero and dividing by y yields the homogeneous PDE

$$\gamma_1 x(x-y)\,\frac{\partial P}{\partial x} - \gamma_2 x(x-y)\,\frac{\partial P}{\partial y} = -\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P.$$

Following the Method of Characteristics (§3.6.2), we associate to this PDE the characteristic system

$$\frac{dx}{\gamma_1 x(x-y)} = \frac{dy}{-\gamma_2 x(x-y)} = \frac{dP}{-\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P}.$$

Taking the ratio of the first two differentials, the $x(x-y)$ factors cancel, giving $dx/\gamma_1 = dy/(-\gamma_2)$. Integrating yields our characteristic constant:

$$C_1 = \gamma_2 x + \gamma_1 y.$$

The characteristic curves are therefore the straight lines of slope $-\gamma_2/\gamma_1$ in the (x, y)-plane. We parametrise y by setting $y(x) = (C_1 - \gamma_2 x)/\gamma_1$. To extract the equation for P along this curve, we need to define our coefficient $A(x, y)$:

$$A(x,y) := (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy. \tag{6.5}$$

Substituting $y(x) = (C_1 - \gamma_2 x)/\gamma_1$ so that $\lambda_2 x\, y(x) = (\lambda_2 C_1/\gamma_1)x - (\lambda_2\gamma_2/\gamma_1)x^2$, we obtain, collecting by powers of x,

$$\begin{aligned} A\big(x,y(x)\big) &= (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \frac{\lambda_2 C_1}{\gamma_1}\,x + \frac{\lambda_2\gamma_2}{\gamma_1}\,x^2 \\ &= -\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1}\,x^2 + \frac{\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1}{\gamma_1}\,x - \mu. \end{aligned}$$

Similarly, $x - y(x) = [(\gamma_1+\gamma_2)x - C_1]/\gamma_1$, so $\gamma_1 x(x - y(x)) = x[(\gamma_1+\gamma_2)x - C_1]$. The dP/P equation from the characteristic system therefore now reads as follows:

$$\frac{dP}{P} = \frac{-A(x,y(x))}{\gamma_1 x(x-y(x))}\,dx = \frac{(\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1\big]x + \gamma_1\mu}{\gamma_1 x\big[(\gamma_1+\gamma_2)x - C_1\big]}\,dx.$$

To integrate this, we decompose the integrand via partial fractions. Writing the integrand as $\frac{1}{\gamma_1}I(x)\,dx$,

$$I(x) = K + \frac{M}{x} + \frac{N(C_1)}{(\gamma_1+\gamma_2)x - C_1};$$

satisfying

$$\begin{aligned} &K(\gamma_1+\gamma_2)x^2 + \big[M(\gamma_1+\gamma_2) + N - KC_1\big]x - MC_1 \\ =\quad &(\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1\big]x + \gamma_1\mu. \end{aligned}$$

The x^2 and x^0 coefficient matches give K and M immediately; the x^1 match determines N after simplification:

$$K = \frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1+\gamma_2}, \qquad M = -\frac{\gamma_1\mu}{C_1}, \qquad N(C_1) = \gamma_1\left[\frac{\lambda_1+\lambda_2}{\gamma_1+\gamma_2}C_1 - (\lambda_1+\lambda_2+\mu) + \frac{\mu(\gamma_1+\gamma_2)}{C_1}\right]. \tag{6.6}$$

For N, the x^1 coefficient match gives

$$\begin{aligned} N &= -\gamma_1(\lambda_1+\lambda_2+\mu) + \lambda_2 C_1 + KC_1 - M(\gamma_1+\gamma_2) \\ &= -\gamma_1(\lambda_1+\lambda_2+\mu) + \lambda_2 C_1 + \frac{(\gamma_1\lambda_1 - \lambda_2\gamma_2)\,C_1}{\gamma_1+\gamma_2} + \frac{\gamma_1\mu(\gamma_1+\gamma_2)}{C_1} \\ &= -\gamma_1(\lambda+\mu) + C_1\,\frac{\lambda_2(\gamma_1+\gamma_2) + \gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1+\gamma_2} + \frac{\gamma_1\mu(\gamma_1+\gamma_2)}{C_1} \\ &= -\gamma_1(\lambda+\mu) + \frac{\gamma_1\lambda}{\gamma_1+\gamma_2}\,C_1 + \frac{\gamma_1\mu(\gamma_1+\gamma_2)}{C_1}; \end{aligned}$$

factoring γ_1 out recovers $N(C_1)$ from (6.6).

Integrating $dP/P = (1/\gamma_1)\, I(x)\, dx$ term by term gives

$$\ln P = \frac{K}{\gamma_1}x - \frac{\mu}{C_1}\ln x + \frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)}\ln\big[(\gamma_1+\gamma_2)x - C_1\big] + \ln f(C_1), \tag{6.7}$$

where $f(C_1)$ is an arbitrary function of the characteristic constant. Now note that

$$(\gamma_1+\gamma_2)x - C_1 = (\gamma_1+\gamma_2)x - (\gamma_2 x + \gamma_1 y) = \gamma_1(x-y),$$

so $\ln[(\gamma_1+\gamma_2)x - C_1] = \ln\gamma_1 + \ln(x-y)$.

Exponentiating Equation (6.7) term by term,

$$P \;=\; f(C_1)\,\exp\!\Big(\tfrac{K}{\gamma_1}x\Big)\cdot x^{-\mu/C_1}\cdot \gamma_1^{N(C_1)/(\gamma_1(\gamma_1+\gamma_2))}\,(x-y)^{\,N(C_1)/(\gamma_1(\gamma_1+\gamma_2))};$$

the factor $\gamma_1^{N(C_1)/(\gamma_1(\gamma_1+\gamma_2))}$ depends on x and y only through C_1 and can be absorbed, together with f, into a redefined arbitrary function $F(C_1)$:

$$P \;=\; F(C_1)\cdot\exp\!\Big(\tfrac{K}{\gamma_1}x\Big)\cdot x^{-\mu/C_1}\cdot (x-y)^{\,N(C_1)/(\gamma_1(\gamma_1+\gamma_2))}.$$

Reading off the exponent of $(x-y)$, with $C_1 = \gamma_2 x + \gamma_1 y$, defines

$$E(x,y) := \frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)} = \frac{\lambda_1+\lambda_2}{(\gamma_1+\gamma_2)^2}(\gamma_2 x + \gamma_1 y) - \frac{\lambda_1+\lambda_2+\mu}{\gamma_1+\gamma_2} + \frac{\mu}{\gamma_2 x + \gamma_1 y}. \tag{6.8}$$

The general homogeneous solution is therefore $F(C_1)\,P_h(x,y)$, with F an arbitrary, differentiable function of the characteristic constant C_1 and P_h the particular homogeneous solution

$$P_h(x,y) = \exp\!\left(\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1(\gamma_1+\gamma_2)}x\right)\cdot x^{-\mu/(\gamma_2 x+\gamma_1 y)}\cdot (x-y)^{E(x,y)} \tag{6.9}$$

that will serve as the integrating factor for variation of parameters in Step (iii). We will see in Step (iv) that F must vanish.

Step (ii): The exponent $E(x,y)$ is strictly positive in $(0,1)^2$.

Recall $\lambda = \lambda_1 + \lambda_2$. Along any fixed characteristic, $C_1 = \gamma_2 x + \gamma_1 y$ is constant, so the exponent function (6.8) depends only on C_1, which ranges over $(0, \gamma_1+\gamma_2)$ as $(x,y)\in(0,1)^2$:

$$E(C_1) = \frac{\lambda}{(\gamma_1+\gamma_2)^2}\,C_1 - \frac{\lambda+\mu}{\gamma_1+\gamma_2} + \frac{\mu}{C_1}.$$

Showing that $E(C_1) > 0$ on $C_1 \in (0,\gamma_1+\gamma_2)$, follows from two facts. First, E vanishes at the right endpoint,

$$E(\gamma_1+\gamma_2) = \frac{\lambda - (\lambda+\mu) + \mu}{\gamma_1+\gamma_2} = 0.$$

Second, E is strictly decreasing in $(0,\gamma_1+\gamma_2]$, using $u \le \gamma_1+\gamma_2$ and $\mu > \lambda$ (i.e. $\rho < 1$):

$$\begin{aligned} E'(u) &= \frac{\lambda}{(\gamma_1+\gamma_2)^2} - \frac{\mu}{u^2} \\ &\le \frac{\lambda}{(\gamma_1+\gamma_2)^2} - \frac{\mu}{(\gamma_1+\gamma_2)^2} = \frac{\lambda-\mu}{(\gamma_1+\gamma_2)^2} < 0, \end{aligned}$$

Since E decreases strictly to 0 at its right endpoint, it is strictly positive at every point to its left, so:

$$E(C_1) > E(\gamma_1+\gamma_2) = 0 \qquad \text{for all } C_1 \in (0,\gamma_1+\gamma_2),$$

that is, $E(x,y) > 0$ for all $(x,y)\in(0,1)^2$. For this proof, the crucial boundary behaviour is that $E \to 0$ as $(x,y)\to(1,1)$, which drives the singular balance of Step (v). The blow-up caused by μ/C_1 as $C_1 \to 0^+$ is irrelevant here: every characteristic used below has C_1 fixed and strictly positive, and the behaviour near the $x=0$ endpoint enters only through the power $x^{-\mu/C_1}$, which is handled separately in Step (iv).

Step (iii): Particular solution via variation of parameters.

This step consists in upgrading the homogeneous solution P_h (Equation (6.9)) with its inhomogeneous part, along the characteristic. We start by rewriting our fundamental equation for Model-B (Equation (6.1)). The coefficient of P on the LHS can be written as $yA(x,y)$, with $A(x,y)$ as defined in Equation (6.5) of Step (i). We define the RHS of the fundamental equation as follows:

$$G(x,y) := \mu(x-y)\,P_y(y) - \mu\,x(1-y)\,\pi(0,0); \tag{6.10}$$

this forcing involves the unknown boundary trace $P_y(y) = P(0,y)$ together with the known constant $\pi(0,0) = \rho(1-\rho)$ of Corollary 4. Separating the terms that involve derivatives and the ones that do not, we have

$$\gamma_1 xy(x-y)\,\partial_x P + \gamma_2 xy(y-x)\,\partial_y P = -\,y\,A(x,y)\,P + G(x,y),$$

which is the form (3.14) treated by the Method of Characteristics. So, along the characteristic $C_1 = \gamma_2 x + \gamma_1 y$ found in Step (i) (parametrised by $y(x) = (C_1 - \gamma_2 x)/\gamma_1$), §3.6.2 reduces it to a first-order linear ODE of the form (3.17),

$$\frac{dP}{dx} = f(x;C_1)\,P + h(x;C_1). \tag{6.11}$$

The homogeneous part $\frac{dP}{dx} = f(x;C_1)\,P$ is P_h (Equation (6.9)), so there is no need to write f out; and its forcing function is the right-hand side divided by the coefficient of $\partial_x P$,

$$h(x;C_1) = \frac{G\big(x,y(x)\big)}{\gamma_1\,x\,y(x)\,\big(x-y(x)\big)}. \tag{6.12}$$

Equation (6.11) is an instance of Equation (3.17) with the known homogeneous solution P_h. So the variation-of-parameters formula (3.18) of §3.6.3 delivers its general solution, with the arbitrary constant becoming a function $F(C_1)$ of the characteristic label:

$$P(x,y(x)) = P_h(x,y(x))\left[F(C_1) + \int_{x_0}^{x} \frac{h(t;C_1)}{P_h(t,y(t))}\,dt\right]. \tag{6.13}$$

Step (iv): Pinning the free function from the boundary condition at $x=0$.
Along any characteristic with constant $C_1 > 0$, the endpoint at $x=0$ is $y(0) = C_1/\gamma_1$, giving

$$P\left(0,\,\frac{C_1}{\gamma_1}\right) = P_y\left(\frac{C_1}{\gamma_1}\right),$$

which is finite, being a value of the boundary generating function P_y. However, P_h (Equation (6.9)) contains the factor $x^{-\mu/C_1} \to +\infty$ as $x \to 0^+$ (since $\mu/C_1 > 0$ for $C_1 > 0$). Take the base point $x_0 = 0$ in Equation (6.13): the integral is then well defined, since its integrand behaves as $t^{\mu/C_1 - 1}$ near $t = 0$, and the product $P_h \int_0^x h/P_h\,dt$ remains bounded as $x \to 0^+$ —the diverging power $x^{-\mu/C_1}$ of the prefactor cancels against the x^{μ/C_1} decay of the integral. Therefore

$$\lim_{x\to 0^+} P(x,y(x)) = \lim_{x\to 0^+} P_h(x,y(x))\cdot F(C_1) + O(1) = +\infty\cdot F(C_1) + O(1),$$

which can only be finite if $F(C_1) = 0$; indeed, for fixed C_1 the quantity $F(C_1)$ is a constant, not a function of x, so no cancellation can rescue a nonzero value. Since the characteristic constant $C_1 > 0$ was arbitrary, F vanishes identically. Setting $x_0 = 0$ and $F \equiv 0$ in Equation (6.13) yields the integral representation of Claim 1:

$$P(x,y(x)) = P_h(x,y(x)) \int_0^x \frac{h(t;C_1)}{P_h(t,y(t))}\,dt. \tag{6.14}$$

Step (v): Reduction to the Volterra equation.
By Corollary 4 the empty-state probabilities are $\pi_0 = 1-\rho$ and $\pi(0,0) = \rho(1-\rho)$, with diagonal value $P(z,z) = \pi(0,0)/(1-\rho z)$. We plug those, along with our inhomogeneous part G (Equation (6.10)) into our forcing function (Equation (6.12))

$$\begin{aligned} h(t;C_1) &= \frac{\mu(t-y(t))\,P_y(y(t)) - \mu t(1-y(t))\,\pi(0,0)}{\gamma_1\,t\,y(t)\,(t-y(t))} \\ &= \frac{\mu\,P_y(y(t))}{\gamma_1\,t\,y(t)} - \frac{\mu(1-y(t))\,\pi(0,0)}{\gamma_1\,y(t)\,(t-y(t))}. \end{aligned} \tag{6.15}$$

We will evaluate Equation (6.14) along the diagonal —i.e. $x = y = z$, with points of the form (z, z)—, whose characteristic has $C_1 = (\gamma_1+\gamma_2)z$. To make the dependence on the chosen diagonal point explicit, we extend the parametrisation $y(\cdot)$ of Step (i) with the parameter z: for fixed z, the characteristic through (z, z) is parametrised by

$$y(t; z) = \frac{(\gamma_1 + \gamma_2)z - \gamma_2 t}{\gamma_1}, \tag{6.16}$$

that is, $y(t; z)$ is the value of y at abscissa t along that particular characteristic; it satisfies $y(0; z) = (\gamma_1 + \gamma_2)z/\gamma_1 > z$ and $y(z; z) = z$. Thus $y(t; z) > t$ for all $t \in [0, z)$, which implies $t - y(t; z) < 0$ inside the integration interval. Note the change of sign of the second summand when writing $t - y(t; z) = -(y(t; z) - t)$ in the denominator of Equation (6.15):

$$h\big(t;\, (\gamma_1 + \gamma_2)z\big) = \frac{\mu}{\gamma_1\, t\, y(t; z)}\, P_y\big(y(t; z)\big) + \frac{\mu\,(1 - y(t; z))}{\gamma_1\, y(t; z)\, (y(t; z) - t)}\, \pi(0, 0). \tag{6.17}$$

From Step (ii), $E > 0$, so the factor $(x-y(x; z))^E$ in P_h vanishes as $x \to z^-$, giving $P_h\big(x, y(x; z)\big) \to 0$. At the same time, the integrand in Equation (6.14) has a singularity near $t = z$: the forcing function (6.17) contributes with a $(y(t; z) - t)^{-1}$ factor; and the denominator $P_h\big(t, y(t; z)\big)$, with a $(y(t; z) - t)^{-E}$. So together $h/P_h \sim (y(t; z) - t)^{-1-E}$; since $-1 - E < -1$, the integral diverges as $x \to z^-$. Nevertheless, the product of the two factors is finite: by Step (iv), $P_h\big(x, y(x; z)\big) \int_0^x \frac{h}{P_h}\, dt$ equals the PGF value $P\big(x, y(x; z)\big)$, which converges to $P(z, z)$ as $x \to z^-$. This behaviour aligns with the fact that we are working in the context of PGFs, which are always bounded. By Corollary 4 that value is $\pi(0, 0)/(1-\rho z)$, so the limit of the product —a vanishing prefactor against a diverging integral— resolves to

$$P(z, z) = \frac{\pi(0, 0)}{1 - \rho z} = \lim_{x\to z^-} P_h\big(x, y(x; z)\big) \int_0^x \frac{h\big(t; (\gamma_1 + \gamma_2)z\big)}{P_h\big(t, y(t; z)\big)}\, dt.$$

Substituting Equation (6.17) and separating the term containing the unknown boundary function P_y from the known remainder, this limit splits into

$$\begin{aligned} \frac{\pi(0, 0)}{1 - \rho z} &= \lim_{x\to z^-} \frac{\mu\, P_h\big(x, y(x; z)\big)}{\gamma_1} \int_0^x \frac{P_y\big(y(t; z)\big)}{t \cdot y(t; z) \cdot P_h\big(t, y(t; z)\big)}\, dt \\ &\quad + \lim_{x\to z^-} \frac{\mu\, P_h\big(x, y(x; z)\big)\, \pi(0, 0)}{\gamma_1} \int_0^x \frac{1 - y(t; z)}{y(t; z) \cdot (y(t; z) - t) \cdot P_h\big(t, y(t; z)\big)}\, dt. \end{aligned} \tag{6.18}$$

We evaluate the two limits in Equation (6.18) separately. Three facts make the diagonal characteristic exactly solvable.

First, its geometry is linear:

$$y(t; z) - t = \frac{(\gamma_1 + \gamma_2)z - \gamma_2 t}{\gamma_1} - t = \frac{\gamma_1 + \gamma_2}{\gamma_1}\, (z - t). \tag{6.19}$$

Second, the exponent of Step (ii) is constant along the characteristic ($C_1 = (\gamma_1 + \gamma_2)z$) and can be factored as follows:

$$E_z := E\big((\gamma_1 + \gamma_2)z\big) = \frac{\lambda z^2 - (\lambda + \mu)z + \mu}{(\gamma_1 + \gamma_2)z} = \frac{(\mu - \lambda z)(1 - z)}{(\gamma_1 + \gamma_2)z} > 0, \qquad z \in (0, 1).$$

Third, every factor of P_h in Equation (6.9) except the exponential, the power of x, and $(x - y)^E$ depends on (x, y) only through C_1, which is fixed on the characteristic. The ratio therefore splits nicely into a smooth part and a single power:

$$\frac{P_h\big(x, y(x; z)\big)}{P_h\big(t, y(t; z)\big)} = R(x, t)\left(\frac{z - x}{z - t}\right)^{E_z}, \qquad R(x, t) := e^{\frac{K}{\gamma_1}(x-t)}\left(\frac{t}{x}\right)^{\mu/((\gamma_1+\gamma_2)z)}, \tag{6.20}$$

with R smooth and $R(z, z) = 1$.

We apply that to the *second* limit of Equation (6.18). Substituting Equations (6.20) and (6.19) into Equation (6.18), the combined prefactor and integrand of its second limit behave near $t = z$ as $w(z)\,(z - x)^{E_z}(z - t)^{-1-E_z}$, where $w(z) = \mu\pi(0, 0)(1 - z)/((\gamma_1 + \gamma_2)z)$ collects the non-singular factors evaluated at $t = z$ (namely $R \to 1$, $1 - y(t; z) \to 1 - z$, $y(t; z) \to z$, and $y(t; z) - t =$

$\frac{\gamma_1+\gamma_2}{\gamma_1}(z-t))$. The two singular factors then balance exactly: the prefactor vanishes like $(z-x)^{E_z}$ while the truncated integral diverges like $(z-x)^{-E_z}/E_z$. Hence their product converges to $w(z)/E_z$:

$$\begin{aligned}
&\lim_{x\to z^-}(z-x)^{E_z}\int^x w(z)(z-t)^{-1-E_z}dt \\
&= \quad \frac{w(z)}{E_z} = \frac{\mu\pi(0,0)(1-z)}{(\gamma_1+\gamma_2)z}\cdot\frac{(\gamma_1+\gamma_2)z}{(\mu-\lambda z)(1-z)} = \frac{\mu\pi(0,0)}{\mu-\lambda z} = \frac{\pi(0,0)}{1-\rho z}.
\end{aligned}$$

The *first* limit vanishes: its integrand behaves as $(z-t)^{-E_z}$, a singularity one power milder than the $(z-t)^{-1-E_z}$ of the second limit, so it is killed by the vanishing prefactor $(z-x)^{E_z}$. The *second* limit reproduces the full diagonal value from $\pi(0,0)$. Thus the diagonal fixes no information about P_y at leading order; determining P_y requires the next order in $(z-x)$, which is the open problem.

To state this as an equation for P_y, define the kernel and the datum

$$\mathcal{K}(z,t) := \frac{\mu}{\gamma_1}\frac{R(z,t)}{t\,y(t;z)}(z-t)^{-E_z}, \tag{6.21}$$

$$\mathcal{F}(z) := \frac{\pi(0,0)}{1-\rho z} - \frac{\mu\,\pi(0,0)}{\gamma_1}\int_0^z \frac{R(z,t)\,\big(1-y(t;z)\big)}{y(t;z)\,\big(y(t;z)-t\big)}(z-t)^{-E_z}\,dt. \tag{6.22}$$

With these definitions, the diagonal identity (6.18) becomes exactly the first-kind Volterra equation (6.4), $\mathcal{F}(z) = \int_0^z \mathcal{K}(z,t)\,P_y(y(t;z))\,dt$, which proves the claim. Both integrals —the one inside $\mathcal{F}(z)$ in Equation (6.22) and the Volterra integral of Equation (6.4)— are singular at the endpoint $t=z$, with integrands behaving as $(z-t)^{-1-E_z}$ and $(z-t)^{-E_z}$ respectively (the factor $y(t;z)-t$ in the denominator of Equation (6.22) is itself proportional to $z-t$, cf. (6.19)), and must be read in a regularised, finite-part sense. Upgrading the claim to a theorem would require, first, a rigorous definition of both singular integrals under which Equation (6.4) provably holds, and second, a solution theory for the resulting singular first-kind Volterra equation that determines P_y uniquely. Neither is attempted here; this is precisely the analytical obstruction that leaves Model-B open. □

Model-B is beyond the scope of this thesis and, therefore, left open. Its role is to illustrate the mathematical obstacle —the Volterra-type integral equation. Model-B_2 and Model-C_2 avoid such a problem by shutting down all derivative terms in y. This allows us to take our boundary function $P_y(y)$ outside the integral, as we would only integrate with respect to x, as opposed to $y(x)$.

It is worth spelling out why the obstruction takes precisely this form —first kind, endpoint-singular— rather than some more benign one. The γ_2-jockeying term carries the derivative $\partial P/\partial y$, and integrating the PDE means integrating *along the characteristic* $y(t;z)$: the solution at a point is assembled from data collected along the whole characteristic segment, and among that data sits the boundary term $\mu(x-y)P_y(y)$ evaluated at every intermediate argument $y(t;z)$. The integration runs *through* the unknown function's argument, as anticipated in the Introduction, so P_y appears only *inside* the integral, smeared along the path against the kernel $\mathcal{K}(z,t)$ —an equation of the first kind. This matters because first-kind equations lack the free copy of the unknown outside the integral on which the classical Picard–Neumann iteration theory of second-kind Volterra equations rests; recovering P_y amounts to inverting an integral operator, in general an ill-posed problem. The endpoint singularity compounds the difficulty: the kernel blows up like $(z-t)^{-E_z}$ as $t\to z$ —non-integrably wherever $E_z \geq 1$— and the datum $\mathcal{F}$ carries the harsher $(z-t)^{-1-E_z}$, so the equation itself only holds in a regularised sense. Probabilistically, the smearing *is* the mechanism: γ_2-jockeying moves waiting class-2 customers into Queue 1 at the length-proportional rate $\gamma_2 n_2$, so the empty-priority-queue boundary communicates with the interior not through a single flux but through a separate, n_2-weighted flux from every occupied level of Queue 2 —exactly what the integral against $P_y\big(y(t;z)\big)$ expresses.

Remark 2 (The full Model-X is *a fortiori* intractable). Model-B already shows the central problem of taking this route. As one can imagine, reinstating the abandonment mechanism —$\theta_1,\theta_2>0$— does not help simplify the problem; quite the opposite. It deepens the difficulty in two ways: first, the characteristics cease to be straight lines; second, we lose the diagonal anchor —Corollary 4 no longer holds—, so the diagonal $P(z,z)$ is no longer known in closed form.

7 Analysis of Model-B_2

Model-B_2 keeps one-way jockeying from class-1 to class-2 ($\gamma_1 > 0$, $\gamma_2 = 0$) and no abandonment ($\theta_1 = \theta_2 = 0$). This model is more tractable than Model-B, as all the terms with derivatives in y disappear, leaving us a first-order linear ODE in x, for each fixed $y \in (0, 1]$. We will solve this using the integrating factor (§3.6.1).

Stability. As in Models A and B, the system is stable if and only if $\rho = (\lambda_1 + \lambda_2)/\mu < 1$: jockeying conserves the total customer count and does not affect the stability condition. Figure 6 shows the queue layout.

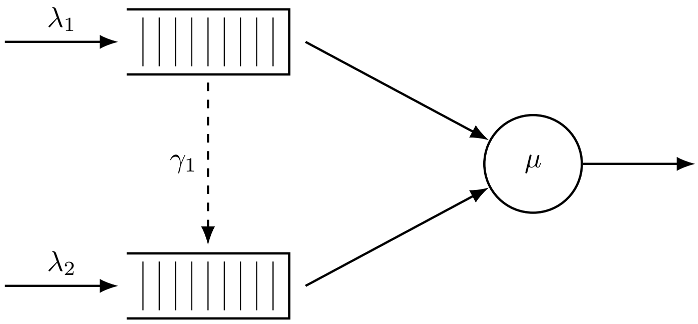

Figure 6: Queue layout for Model-B_2. Only the $1 \to 2$ jockeying direction is active (dashed, downward): class-1 customers may transition to Queue 2, but no customer can move in the reverse direction ($\gamma_2 = 0$).

7.1 Reduced fundamental equation

Setting $\gamma_2 = 0$ and $\theta_1 = \theta_2 = 0$ in the general fundamental equation (4.16) (Lemma 1) annihilates the derivative terms in y, leaving:

$$\begin{aligned}\big[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big] P + \gamma_1 xy(x - y)\,\frac{\partial P}{\partial x} \\ = \mu(x - y)\, P_y(y) - \mu x(1 - y)\, \pi(0, 0). \qquad (7.1)\end{aligned}$$

For each fixed $y \in (0, 1]$ this is a first-order linear ODE in x; the entire analysis of this section is built on it.

Theorem 3. *Consider the two-class non-preemptive priority queueing system of Model-B_2 with per-customer jockeying rate $\gamma_1 > 0$ from Queue 1 to Queue 2 ($\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$). Under the stability condition $\rho = \lambda/\mu < 1$, $\lambda = \lambda_1 + \lambda_2$, the boundary generating function is*

$$P_y(y) = \frac{\mu\, \pi(0, 0)}{\mu - \lambda y} \cdot \frac{{}_1F_1\big(\alpha(y) + 1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}, \qquad (7.2)$$

and the bivariate PGF, for $0 \le x \le y$, is

$$P(x, y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y - x)^{-\beta}}{\gamma_1 y} \Big[P_y(y)\, \mathcal{H}_1(x, y) + (1 - y)\pi(0, 0)\, \mathcal{H}_2(x, y)\Big], \qquad (7.3)$$

where ${}_1F_1$ is Kummer's confluent hypergeometric function, the exponents are

$$\alpha(y) = \frac{\mu}{\gamma_1 y}, \qquad \beta(y) = \frac{(1 - y)(\lambda y - \mu)}{\gamma_1 y}, \qquad b^\star(y) = \alpha + \beta + 1 = \frac{\mu + \lambda(1 - y) + \gamma_1}{\gamma_1}, \qquad (7.4)$$

where the dependence of α, β and $b^\star$ on y is suppressed from the notation where unambiguous, the auxiliary integrals are

$$\mathcal{H}_1(x, y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha - 1}(y - t)^{\beta}\, dt, \qquad (7.5)$$

$$\mathcal{H}_2(x, y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha}(y - t)^{\beta - 1}\, dt, \qquad (7.6)$$

and the empty-state probability $\pi(0, 0) = \rho(1 - \rho)$ is given by Corollary 5 below. On $y < x \le 1$ the same expression (7.3) holds with $\mathcal{H}_1, \mathcal{H}_2$ replaced by their finite-part continuations, and the right-hand side is analytic across the interior point $x = y$; the latter assertion is proved as the smoothness condition in Step (ii) of the proof.

The result below is stated as a corollary, so that it sits at the same hierarchy level as the corresponding results for the other models.

Corollary 5. *The empty-system probability π_0 (server idle, no one waiting) and the both-queues-empty probability $\pi(0,0)$ (server busy, $N_1 = N_2 = 0$) of Model-B_2 are*

$$\pi_0 = 1 - \rho, \qquad \pi(0,0) = \rho(1-\rho),$$

identical to the Model-A baseline.

As in Model-B, the proof of Corollary 5 follows trivially by setting $x = y = z$ in our fundamental equation (Equation (7.1)). The resulting equation is identical to the homologous one in Model-A, so Corollary 5 is the no-abandonment instance of Theorem 1, with the $M/M/1$ mean busy period $\mathbb{E}[BP] = (\mu - \lambda)^{-1}$ (Corollary 1).

The proof of Theorem 3 consists of three steps: (i) obtaining the standard form and the integration factor, (ii) determining $P_y(y)$ by analyticity in $x = y$, and (iii) recovering $P(x, y)$.

Proof of Theorem 3. **Step (i): Standard form and the integrating factor.**

Fix $y \in (0,1]$, cancel a factor y on both sides and divide the reduced fundamental equation (7.1) by the coefficient of the derivative term in x. The equation takes standard form $\frac{dP}{dx} + p(x;y)\,P = q(x;y)$ of §3.6.1 (Equation (3.12)), with

$$p(x;y) = \frac{(\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\gamma_1 x(x-y)}, \tag{7.7}$$

$$q(x;y) = \frac{\mu\, P_y(y)}{\gamma_1 xy} - \frac{\mu(1-y)\,\pi(0,0)}{\gamma_1 y(x-y)}. \tag{7.8}$$

Write $N(x)$ for the numerator of p, a polynomial in x:

$$N(x) = -\lambda_1 x^2 + \big[(\lambda_1 + \lambda_2 + \mu) - \lambda_2 y\big]x - \mu = -\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big),$$

where the roots $x^*(y), x^+(y)$ are those found for the kernel in Model-A (Equation (5.7)), with $\lambda_1 x^* x^+ = \mu$ following from Vieta's formula. We now decompose $p = N(x)/\big[\gamma_1 x(x-y)\big]$ by partial fractions. Since the degree of N equals that of the denominator, the decomposition takes the form

$$p(x;y) = \frac{N(x)}{\gamma_1\, x(x-y)} = \frac{1}{\gamma_1}\left[A + \frac{B}{x} + \frac{C}{x-y}\right],$$

in which A is the ratio of the leading coefficients and B, C are the residues of $N(x)/[x(x-y)]$ at the simple poles $x = 0$ and $x = y$:

$$\begin{aligned}
A &= -\lambda_1,\\
B &= \frac{N(0)}{0-y} = \frac{-\mu}{-y} = \frac{\mu}{y},\\
C &= \frac{N(y)}{y} = \frac{-(\lambda y - \mu)(y-1)}{y} = \frac{(1-y)(\lambda y - \mu)}{y},
\end{aligned}$$

with $N(y) = -\lambda y^2 + (\lambda + \mu)y - \mu = -(\lambda y - \mu)(y-1)$. Thus

$$p(x;y) = \frac{1}{\gamma_1}\left[-\lambda_1 + \frac{\mu/y}{x} + \frac{(1-y)(\lambda y - \mu)/y}{x-y}\right]. \tag{7.9}$$

Since we will have a factor $(x-y)$ due to jockeying, we work on the region $0 \le x \le y \le 1$, on which the difference $x - y$ is non-positive; the complementary region $y < x \le 1$ is recovered later. On this region, a term $\beta \ln|x-y|$ becomes simply $\beta\ln(y-x)$, so integrating the partial fraction (7.9) yields, up to a constant,

$$\begin{aligned}
\int^x p(t;y)\,dt &= \frac{1}{\gamma_1}\Big[-\lambda_1 x + \frac{\mu}{y}\ln x + \frac{(1-y)(\lambda y - \mu)}{y}\,\ln(y-x)\Big]\\
&= -\frac{\lambda_1}{\gamma_1}\,x + \alpha(y)\ln x + \beta(y)\ln(y-x),
\end{aligned}$$

with the exponents $\alpha(y) = \mu/(\gamma_1 y) > 0$ and $\beta(y) = (1-y)(\lambda y - \mu)/(\gamma_1 y)$ of Equation (7.4). Exponentiating, and dropping the multiplicative constant to which an integrating factor is indifferent, gives our integrating factor

$$\mu_I(x;y) = e^{-\lambda_1 x/\gamma_1}\, x^{\alpha(y)}\, (y-x)^{\beta(y)},$$

real-valued in our region because we take the base $(y-x) > 0$ rather than $(x-y)$; its reciprocal $1/\mu_I$ is the homogeneous solution, playing here the role that P_h played for Model-B (§3.6.3). The general solution of §3.6.1 (Equation (3.13)) now applies, and its arbitrary function is fixed immediately: since $\alpha(y) > 0$ we have $\mu_I(x;y) \to 0$ as $x \to 0^+$, so the boundary contribution $P(0,y)\,\mu_I(0;y)$ vanishes and integrating $\frac{d}{dx}[P\,\mu_I] = q\,\mu_I$ over $(0,x]$ leaves

$$P(x,y)\,\mu_I(x;y) = \int_0^x q(t;y)\,\mu_I(t;y)\,dt.$$

Multiplying q from Equation (7.8) by μ_I and applying the identity $(y-t)^\beta/(t-y) = -(y-t)^{\beta-1}$ to the second term, the integrand separates into two pieces,

$$q(t;y)\,\mu_I(t;y) = \frac{\mu}{\gamma_1 y}\Big[P_y(y)\,e^{-\lambda_1 t/\gamma_1}\,t^{\alpha-1}(y-t)^\beta + (1-y)\pi(0,0)\,e^{-\lambda_1 t/\gamma_1}\,t^{\alpha}(y-t)^{\beta-1}\Big],$$

With the auxiliary integrals $\mathcal{H}_1, \mathcal{H}_2$ of Equations (7.5)–(7.6) and dividing through by $\mu_I(x;y) = e^{-\lambda_1 x/\gamma_1}x^\alpha(y-x)^\beta$, the solution on the principal region $0 \le x \le y$ is

$$P(x,y) = \frac{\mu\,e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{H}_2(x,y)\Big], \tag{7.10}$$

which still contains the unknown boundary function $P_y(y)$; Step (ii) determines it.

Step (ii): Determine $P_y(y)$ by analyticity at $x = y$.

Under stability (i.e. $\rho < 1$) $\mu - \lambda y > 0$ for $y \in (0,1)$, so the second exponent is negative:

$$\beta(y) = \frac{(1-y)(\lambda y - \mu)}{\gamma_1 y} < 0, \qquad 0 < y < 1. \tag{7.11}$$

So the prefactor $(y-x)^{-\beta} = (y-x)^{|\beta|}$ in Equation (7.10) *vanishes* as $x \to y^-$, while $\mathcal{H}_1, \mathcal{H}_2$ may *diverge*: the product is an indeterminate $0 \cdot \infty$ form, whose resolution fixes $P_y(y)$.

On the diagonal the value is known in advance. Setting $x = y = z$ in the fundamental equation (7.1) drops the jockeying and boundary terms (each containing a factor $x - y$) and leaves Model A's balance, so

$$P(z,z) = \frac{\rho(1-\rho)}{1-\rho z}, \tag{7.12}$$

the diagonal value of Corollary 3.

Smoothness condition and $P_y(y)$. Pull the analytic, non-vanishing prefactor of Equation (7.10) into $\Psi(x) = \mu\,e^{\lambda_1 x/\gamma_1}x^{-\alpha}/(\gamma_1 y)$, so that

$$P(x,y) = \Psi(x)\,(y-x)^{|\beta|}\,\big[\,P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{H}_2(x,y)\,\big], \qquad |\beta| = -\beta > 0.$$

We need only the leading behaviour of the two integrals as $x \to y^-$, not a full expansion. Splitting each integral at the endpoint, $\mathcal{H}_i(x,y) = \mathcal{H}_i(y,y) - \int_x^y(\cdots)\,dt$, and integrating the behaviour of the integrands near $t = y$ —$\mathcal{H}_1$ carries $(y-t)^\beta$, $\mathcal{H}_2$ carries $(y-t)^{\beta-1}$— gives, with constants c_1, c_2 whose values are immaterial,

$$\mathcal{H}_1(x,y) = \mathcal{H}_1(y,y) + c_1\,(y-x)^{\beta+1} + \cdots, \qquad \mathcal{H}_2(x,y) = \mathcal{H}_2(y,y) + c_2\,(y-x)^{\beta} + \cdots.$$

Upon multiplication by $(y-x)^{-\beta}$, the c_1-remainder becomes a smooth $O(y-x)$ term and the c_2-remainder the constant regular value (7.12); only the endpoint values $\mathcal{H}_i(y,y)$ retain the bare, non-integer power $(y-x)^{|\beta|}$ ($|\beta| \notin \mathbb{Z}$ generically). Since $P(\cdot,y)$ is a power series in x, it is smooth at the interior point $x = y < 1$, whereas that power is not; the coefficient of the power must therefore vanish:

$$P_y(y)\,\mathcal{H}_1(y,y) + (1-y)\,\pi(0,0)\,\mathcal{H}_2(y,y) = 0, \tag{7.13}$$

the endpoint integrals being read, since $\beta < 0$, as the Euler–Beta continuations found next. To evaluate these endpoint integrals, substitute $t = ys$ ($dt = y\,ds$) in Equations (7.5)–(7.6), which rescales the two integrands and pulls out a common $y^{\alpha+\beta}$,

$$t^{\alpha-1}(y-t)^\beta\,dt = y^{\alpha+\beta}\,s^{\alpha-1}(1-s)^\beta\,ds,$$

$$t^{\alpha}(y-t)^{\beta-1}\,dt = y^{\alpha+\beta}\,s^{\alpha}(1-s)^{\beta-1}\,ds,$$

so that, with $c := \lambda_1 y/\gamma_1$,

$$\mathcal{H}_1(y,y) = y^{\alpha+\beta}\int_0^1 s^{\alpha-1}(1-s)^{\beta}e^{-cs}\,ds, \qquad \mathcal{H}_2(y,y) = y^{\alpha+\beta}\int_0^1 s^{\alpha}(1-s)^{\beta-1}e^{-cs}\,ds.$$

These are Euler integrals (3.4), $\int_0^1 t^{a-1}(1-t)^{b-1}e^{zt}\,dt = B(a,b)\,{}_1F_1(a;a+b;z)$, with $(a,b,z) = (\alpha,\beta{+}1,-c)$ for $\mathcal{H}_1$ and $(\alpha{+}1,\beta,-c)$ for $\mathcal{H}_2$, so $a{+}b = b^\star(y)$ in both. As $\beta < 0$ by Equation (7.11), the right-hand side of the Euler identity above is read as its analytic continuation in b beyond the convergence range $b > 0$, giving

$$\mathcal{H}_1(y,y) = B(\alpha,\,\beta+1)\,y^{\alpha+\beta}\,{}_1F_1\big(\alpha;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big), \tag{7.14}$$

$$\mathcal{H}_2(y,y) = B(\alpha+1,\,\beta)\,y^{\alpha+\beta}\,{}_1F_1\big(\alpha+1;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big). \tag{7.15}$$

Solving the smoothness condition (7.13) for $P_y(y)$ and inserting the endpoint values (7.14)–(7.15), the common factor $y^{\alpha+\beta}$ cancels and

$$P_y(y) = -(1-y)\,\pi(0,0)\,\frac{\mathcal{H}_2(y,y)}{\mathcal{H}_1(y,y)} = -(1-y)\,\pi(0,0)\,\frac{B(\alpha+1,\,\beta)}{B(\alpha,\,\beta+1)}\,\frac{{}_1F_1\big(\alpha+1;\,b^\star;\,-\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha;\,b^\star;\,-\lambda_1 y/\gamma_1\big)}. \tag{7.16}$$

The Beta ratio reduces, via $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ and $\Gamma(\beta+1) = \beta\Gamma(\beta)$, to

$$\begin{aligned}\frac{B(\alpha+1,\,\beta)}{B(\alpha,\,\beta+1)} &= \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}\cdot\frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta+1)}\\ &= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)}\cdot\frac{\Gamma(\beta)}{\Gamma(\beta+1)} = \frac{\alpha}{\beta},\end{aligned} \tag{7.17}$$

and the surviving prefactor simplifies, using $\alpha/\beta = \mu/\big[(1-y)(\lambda y-\mu)\big]$, as

$$\begin{aligned}-(1-y)\,\frac{\alpha}{\beta} &= -(1-y)\cdot\frac{\mu}{(1-y)(\lambda y-\mu)}\\ &= -\frac{\mu}{\lambda y-\mu} = \frac{\mu}{\mu-\lambda y}.\end{aligned} \tag{7.18}$$

Substituting Equations (7.17)–(7.18) into Equation (7.16) gives the boundary generating function

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu-\lambda y}\cdot\frac{{}_1F_1\big(\alpha(y)+1;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\,b^\star(y);\,-\lambda_1 y/\gamma_1\big)}, \tag{7.19}$$

confirming Equation (7.2). The prefactor equals the regular value $P(y,y)$ of Equation (7.12), and the Kummer ratio is the boundary correction. As $y \to 0^+$ the argument $-\lambda_1 y/\gamma_1 \to 0$, so the ratio tends to 1 and $P_y(0) = \pi(0,0)$, consistent with $P_y(0) = \sum_{n_2}\pi(0,n_2)\cdot 0^{n_2} = \pi(0,0)$.

Step (iii): Recovery of $P(x,y)$.

Substituting Equation (7.19) into the expression (7.10) gives, for $0 \le x \le y$,

$$\begin{aligned}P(x,y) = \frac{\mu\,e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Bigg[&\frac{\mu\,\pi(0,0)}{\mu-\lambda y}\cdot\frac{{}_1F_1\big(\alpha+1;\,b^\star;\,-\frac{\lambda_1 y}{\gamma_1}\big)}{{}_1F_1\big(\alpha;\,b^\star;\,-\frac{\lambda_1 y}{\gamma_1}\big)}\cdot\mathcal{H}_1(x,y)\\ &+ (1-y)\pi(0,0)\cdot\mathcal{H}_2(x,y)\Bigg],\end{aligned}$$

which is the statement of the theorem (Equation (7.3)). The smoothness condition (7.13) guarantees that this extends analytically across $x = y$, with diagonal value $P(y,y) = \rho(1-\rho)/(1-\rho y)$ as computed in Equation (7.12). On $y < x \le 1$ the same formula holds with $\mathcal{H}_1, \mathcal{H}_2$ read as their finite-part continuations; $\pi(0,0) = \rho(1-\rho)$ from Corollary 5 makes the solution fully explicit. The recovery of Model-A in the limit $\gamma_1 \to 0^+$ is deferred to Remark 3 below, as the limit is singular and merits separate discussion. ☐

Remark 3 (The singular limit $\gamma_1 \to 0^+$). The no-jockeying limit $\gamma_1 \to 0^+$ is a *singular* perturbation: the coefficient $\gamma_1 xy(x-y)$ of the highest derivative $\partial_x P$ vanishes. For the equation to degenerate to the Model-A form it is not enough that this coefficient tends to zero: the derivative $\partial_x P$ could in principle blow up as $\gamma_1 \to 0^+$, keeping the product $\gamma_1 xy(x-y)\,\partial_x P$ alive in the limit. We therefore check that the product itself disappears. Each P is a probability generating function, so $|P| \le 1$ on the closed unit polydisc uniformly in γ_1, and Cauchy's estimate then bounds $\partial_x P$ uniformly on every compact subset of the open polydisc; hence $\gamma_1 xy(x-y)\,\partial_x P \to 0$ there. The fundamental equation (7.1) therefore degenerates to the algebraic Model-A equation (5.2), and the uniformly bounded family $\{P\}_{\gamma_1}$, being normal, converges (in full, by uniqueness of the analytic solution) to the Model-A PGF (5.3). In particular the boundary function reduces to the Model-A value (5.8),

$$P_y(y) \xrightarrow[\gamma_1 \to 0^+]{} \pi(0,0)\,\frac{x^*(y)\,(1-y)}{x^*(y)-y}, \tag{7.20}$$

with $x^*(y)$ the Model-A kernel root (5.7); this needs no special-function asymptotics.

The same limit can be read off the closed form (7.19). Its prefactor $\mu\,\pi(0,0)/(\mu-\lambda y) = \pi(0,0)/(1-\rho y)$ is γ_1-independent, so Equation (7.20) is equivalent to the Kummer ratio tending to

$$\frac{{}_1F_1\big(\alpha+1;\, b^\star;\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha;\, b^\star;\, -\lambda_1 y/\gamma_1\big)} \xrightarrow[\gamma_1 \to 0^+]{} (1-\rho y)\,\frac{x^*(y)\,(1-y)}{x^*(y)-y}. \tag{7.21}$$

The factor $(1-\rho y)$ shows the ratio does *not* tend to the bare $x^*(y)(1-y)/(x^*(y)-y)$; the discrepancy is exactly the prefactor. This matches the first-parameter three-term recurrence for ${}_1F_1$ [5, §13.3]: as $\gamma_1 \to 0^+$ the parameters α, $b^\star$ and $c = \lambda_1 y/\gamma_1$ all grow like $1/\gamma_1$, and the recurrence reduces at leading order to a quadratic whose admissible root is Equation (7.21).

8 Analysis of Model-C_2

In this section we derive $P(x,y)$ for the model in which only class-1 customers abandon, with no class-2 abandonment and no jockeying. Figure 7 shows the queue layout.

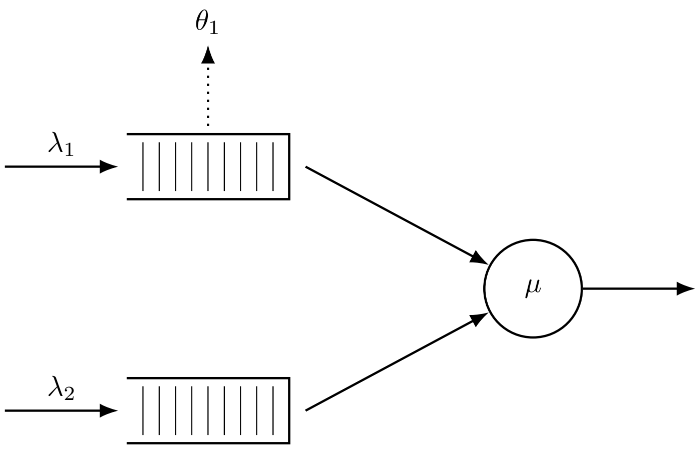

Figure 7: Queue layout for Model-C_2. Class-1 customers abandon the system at rate θ_1 per waiting customer (dotted arrow, upward from Queue 1). Class-2 never abandons ($\theta_2 = 0$). There is no jockeying.

This choice of parameters is motivated by two key aspects of the resulting model: first, class-2 still sees Poisson arrivals, so the probabilistic argument of Model-A carries over; second, substituting these values into the general fundamental equation (4.16) leaves a first-order linear ODE in x with no y-derivatives, so the boundary function $P_y(y)$ comes out of the integral, sparing us the Volterra equation that left Model-B open (§6). The mirrored choice —class-2 abandonment only, $\theta_2 > 0 = \theta_1$, a potential Model-C_1— would instead leave an ODE in y, whose forcing involves the unknown P_y evaluated at the integration variable; it therefore runs into the same Volterra-type obstruction as Model-B and is not pursued.

Stability. Unlike the models without abandonment, stability here is non-trivial and rests on two key observations that also drive the analysis below. First, the class-1 queue alone is an $M/M/1{+}M$ (Erlang-A) subsystem with rates λ_1, μ and θ_1, which is stable for *every* load (§3.4); its busy period B_C is always well defined. Second, class-2 sees an $M/G/1$ queue whose service time is one such class-1 busy period B_C (made precise in §8.3). The only constraint on stability comes from this second queue, which is stable exactly when its utilisation stays below one:

$$\lambda_2\,\mathbb{E}[B_C] \;<\; 1 \qquad\Longleftrightarrow\qquad \rho_2\cdot{}_1F_1\Big(1;\,\tfrac{\mu}{\theta_1}+1;\,\tfrac{\lambda_1}{\theta_1}\Big) \;<\; 1. \tag{8.1}$$

The second form follows from the expression (3.7) for $\mathbb{E}[B_C]$ —here with $(\lambda,\theta)=(\lambda_1,\theta_1)$— via the contiguous identity ${}_1F_1(1;a;c)=1+(c/a)\,{}_1F_1(1;a+1;c)$, read off the series (3.3) with $a=\mu/\theta_1$ and $c=\lambda_1/\theta_1$.

8.1 Reduced fundamental equation

Setting $\theta_1>0$ and $\gamma_1=\gamma_2=\theta_2=0$ in the general fundamental equation (4.16) clears most of the derivative terms, with the class-1 abandonment term as the only survivor. What remains is, for each fixed y, a first-order linear ODE in x:

$$\begin{aligned}&\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\big]\,P(x,y)+\theta_1\,xy(x-1)\,\frac{\partial}{\partial x}P(x,y)\\ &=\quad \mu(x-y)\,P_y(y)-\mu x(1-y)\,\pi(0,0).\end{aligned}\tag{8.2}$$

Equation (8.2) has two unknowns: $P(x,y)$ and its boundary trace $P_y(y)$.

The following theorem states the closed form. Its proof, given immediately afterwards in §8.2, is purely analytical; §8.3 then re-derives the boundary function $P_y(y)$ probabilistically as an independent method to determine the boundary function.

Theorem 4. *Consider the two-class non-preemptive priority system of Model-C_2: Poisson arrivals at rates λ_1,λ_2, common exponential service at rate μ, and per-customer abandonment of waiting class-1 customers at rate θ_1. Under the stability condition (8.1), the PGF is*

$$\begin{aligned}P(x,y) &= \frac{e^{\lambda_1x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\,x^{\mu/\theta_1}(1-x)^{\lambda_2(1-y)/\theta_1}\big(\widetilde{B}_C(\lambda_2(1-y))-y\big)}\\ &\quad\times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{H}_1(x,y)-\big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{H}_2(x,y)\Big],\end{aligned}$$

where

$$\begin{aligned}\mathcal{H}_1(x,y) &= \int_0^x e^{-\lambda_1t/\theta_1}\,t^{\,\mu/\theta_1-1}(1-t)^{\,\lambda_2(1-y)/\theta_1}\,dt,\\ \mathcal{H}_2(x,y) &= \int_0^x e^{-\lambda_1t/\theta_1}\,t^{\,\mu/\theta_1}(1-t)^{\,\lambda_2(1-y)/\theta_1-1}\,dt,\end{aligned}$$

$\widetilde{B}_C(s)$ is the LST of the busy period of the class-1 $M/M/1{+}M$ subsystem, given by Equation (3.6) with $(\lambda,\theta)=(\lambda_1,\theta_1)$, and $\pi(0,0)$ is given by Corollary 6 below.

The following corollary is invoked in the theorem; its proof is deferred to the end of §8.2. At first sight its form differs from the renewal–reward expression of Theorem 1, which may seem remarkable; the proof shows that the two agree, with whole-system mean busy period $\mathbb{E}[BP]=\mathbb{E}[B_C]/(1-\lambda_2\,\mathbb{E}[B_C])$.

Corollary 6. *The limiting probability $\pi(0,0)$ of the busy state with both queues empty, and the empty-system probability π_0, are*

$$\pi(0,0) \;=\; \frac{(\lambda_1+\lambda_2)\big(1-\lambda_2\,\mathbb{E}[B_C]\big)}{\mu\,\big(1+\lambda_1\,\mathbb{E}[B_C]\big)} \;=\; (\rho_1+\rho_2)\,\frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{8.3}$$

$$\pi_0 \;=\; \frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{8.4}$$

where $\mathbb{E}[B_C]$ is the mean class-1 busy period (3.7). As $\theta_1\to0^+$, $\mathbb{E}[B_C]\to(\mu-\lambda_1)^{-1}$ and these recover the Model-A baseline $\pi(0,0)=\rho(1-\rho)$, $\pi_0=1-\rho$.

Note that the stability condition (8.1) is visible in Equation (8.4): $\pi_0>0$ holds precisely when $\lambda_2\,\mathbb{E}[B_C]<1$.

8.2 Analytical derivation of $P(x, y)$

This subsection proves Theorem 4 analytically and closes with the proof of Corollary 6. The proof of the theorem consists of three steps: (i) obtaining the standard form and the integrating factor, (ii) determining $P_y(y)$ by boundedness at $x = 1$, and (iii) recovering $P(x, y)$.

Proof of Theorem 4. **Step (i): standard form and integrating factor.**

The abandonment factor $(x-1)$, the only derivative coefficient remaining, places the operative singularity on the boundary $x = 1$. Dividing Equation (8.2) by the coefficient $\theta_1 xy(x-1)$ of the derivative puts it in the standard form (Equation (3.12)):

$$\frac{\partial P}{\partial x} + \underbrace{\frac{(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\theta_1\, x(x-1)}}_{p(x;\,y)} P = \underbrace{\frac{\mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0)}{\theta_1\, xy(x-1)}}_{q(x;\,y)}.$$

The numerator of $p(x; y)$ is the quadratic $A(x, y) := (\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy$, the kernel polynomial common to all models considered so far. We decompose $A(x; y)/(x(x-1))$ by partial fractions,

$$\frac{A(x;y)}{x(x-1)} = D + \frac{B}{x} + \frac{C}{x-1},$$

in which D is the ratio of the leading coefficients and B, C are the residues at the simple poles $x = 0$ and $x = 1$:

$$\begin{aligned}
&D = -\lambda_1,\\
&B = A(0,y)/(0-1) = (-\mu)/(-1) = \mu,\\
&C = A(1,y) = \lambda_2(1-y);
\end{aligned}$$

so,

$$p(x;y) = \frac{1}{\theta_1}\cdot\frac{A(x;y)}{x(x-1)} = \frac{1}{\theta_1}\left[-\lambda_1 + \frac{\mu}{x} + \frac{\lambda_2(1-y)}{x-1}\right]. \tag{8.5}$$

Integrating term by term, $\int p\,dx = \theta_1^{-1}\big[-\lambda_1 x + \mu\ln x + \lambda_2(1-y)\ln(1-x)\big]$, which exponentiates to the integrating factor

$$\mu_I(x;y) = e^{-\lambda_1 x/\theta_1}\, x^{\alpha}\,(1-x)^{\beta(y)}, \tag{8.6}$$

with

$$\alpha = \frac{\mu}{\theta_1}, \qquad \beta(y) = \frac{\lambda_2(1-y)}{\theta_1}. \tag{8.7}$$

Both exponents are non-negative ($\alpha > 0$ for any parameter values; $\beta(y) \geq 0$ for $y \in [0, 1]$), so μ_I vanishes at $x = 0$ and at $x = 1$. By the integrating-factor solution (3.13) —with the boundary term $P(0, y)\,\mu_I(0; y)$ killed by $\mu_I(0; y) = 0$— we obtain

$$P(x,y) = \frac{1}{\mu_I(x;y)}\int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{8.8}$$

Step (ii): determination of $P_y(y)$ by boundedness at $x = 1$.

We determine $P_y(y)$ by using the boundedness of $P(x, y)$ as $x \to 1$ —automatic for a PGF— for fixed $y \in [0, 1)$: we have $\beta(y) > 0$, so $\mu_I(x; y) \to 0$. Since the left-hand side of Equation (8.8), $P(x, y)$, stays bounded while $1/\mu_I(x; y) \to \infty$, the integral must vanish in the limit:

$$\int_0^1 q(t;y)\,\mu_I(t;y)\,dt = 0. \tag{8.9}$$

Substituting the explicit q and μ_I, the integrand is

$$q(t;y)\,\mu_I(t;y) = \frac{\mu\big[(t-y)\,P_y(y) - t(1-y)\,\pi(0,0)\big]}{\theta_1\, y}\cdot\frac{e^{-\lambda_1 t/\theta_1}\, t^{\alpha}(1-t)^{\beta(y)}}{t\,(t-1)},$$

where $t^\alpha/t = t^{\alpha-1}$ and $(1-t)^{\beta(y)}/(t-1) = -(1-t)^{\beta(y)-1}$ flips the sign and lowers the exponent. The constant prefactor $-\mu/(\theta_1 y)$, which is nonzero for $y \in (0, 1]$, can be divided out of Equation (8.9), leaving

$$\int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\alpha-1}(1-t)^{\beta(y)-1}\big[(t-y)\,P_y(y) - t(1-y)\,\pi(0,0)\big]\,dt = 0.$$

Introduce the auxiliary integrals

$$\begin{aligned}
\mathcal{J}_0(y) &= \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\, dt,\\
\mathcal{J}_1(y) &= \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha}(1-t)^{\,\beta(y)-1}\, dt.
\end{aligned}$$

Distributing the bracket over $t^{\alpha-1}$, we get the term $t\cdot t^{\alpha-1}=t^{\alpha}$ in $\mathcal{J}_1$ and $t^{\alpha-1}$ in $\mathcal{J}_0$, becoming

$$\begin{aligned}
0 &= P_y(y)\int_0^1 e^{-\lambda_1 t/\theta_1}\big(t^{\alpha}-y\,t^{\alpha-1}\big)(1-t)^{\beta(y)-1}\,dt \;-\; (1-y)\,\pi(0,0)\int_0^1 e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)-1}\,dt\\
&= P_y(y)\big[\mathcal{J}_1(y)-y\,\mathcal{J}_0(y)\big] \;-\; (1-y)\,\pi(0,0)\,\mathcal{J}_1(y),
\end{aligned}$$

so

$$P_y(y) \;=\; \pi(0,0)\,(1-y)\,\frac{\mathcal{J}_1(y)}{\mathcal{J}_1(y)-y\,\mathcal{J}_0(y)}. \tag{8.10}$$

It remains to evaluate the ratio $\mathcal{J}_1/\mathcal{J}_0$, which will express Equation (8.10) in terms of the busy-period LST $\widetilde{B}_C$ —the form in which the probabilistic route of §8.3 independently confirms it. Setting $a=\alpha=\mu/\theta_1$, $b=s/\theta_1$, $c=\lambda_1/\theta_1$, the Euler representation (3.4) writes each $\mathcal{J}_i$ as a Beta multiple of a ${}_1F_1$ value, and the closed-form summation (3.10) of §3.4 identifies their ratio with the busy-period LST:

$$\widetilde{B}_C(s) \;=\; \frac{\displaystyle\int_0^1 t^{a}(1-t)^{b-1}e^{-ct}\,dt}{\displaystyle\int_0^1 t^{a-1}(1-t)^{b-1}e^{-ct}\,dt} \;=\; \frac{B(a+1,b)\,{}_1F_1(a+1;\,a+1+b;\,-c)}{B(a,b)\,{}_1F_1(a;\,a+b;\,-c)}. \tag{8.11}$$

Taking $s=\lambda_2(1-y)$ —so that $b=\beta(y)$ matches the exponent in Equation (8.7)— gives $\widetilde{B}_C(\lambda_2(1-y))=\mathcal{J}_1(y)/\mathcal{J}_0(y)$. Dividing the numerator and denominator of Equation (8.10) by $\mathcal{J}_0(y)$ yields the boundary generating function

$$P_y(y) \;=\; \pi(0,0)\,\frac{(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y))-y}. \tag{8.12}$$

This provides $P_y(y)$ analytically through the Beta-integral (Equation (8.11)) alone, without requiring the probabilistic argument in §8.3.

Step (iii): recovery of $P(x,y)$.

We return to the ODE (Equation (8.2)) and substitute $P_y(y)$ (Equation (8.12)) into its rhs. Writing $\zeta(y)\equiv\widetilde{B}_C(\lambda_2(1-y))$ for brevity,

$$\begin{aligned}
\text{RHS} &= \mu(x-y)\,\pi(0,0)\,\frac{(1-y)\,\zeta(y)}{\zeta(y)-y} \;-\; \mu x(1-y)\,\pi(0,0)\\
&= \mu\,\pi(0,0)\,(1-y)\left[\frac{(x-y)\,\zeta(y)}{\zeta(y)-y}-x\right]\\
&= \mu\,\pi(0,0)\,(1-y)\,\frac{(x-y)\,\zeta(y)-x\big[\zeta(y)-y\big]}{\zeta(y)-y}\\
&= \mu\,\pi(0,0)\,y(1-y)\,\frac{x-\zeta(y)}{\zeta(y)-y},
\end{aligned}$$

which has the same form as the RHS of Model-A, now with the more involved service time B_C. With $p(x;y)$ and $\mu_I(x;y)$ as in Equations (8.5) and (8.6), the inhomogeneous term $q(x;y)$ of the standard form becomes

$$q(x;y) \;=\; \frac{\mu\,\pi(0,0)\,(1-y)\,\big[x-\zeta(y)\big]}{\theta_1\,x(x-1)\,\big[\zeta(y)-y\big]},$$

and Equation (8.8) reads $P(x,y)=\mu_I(x;y)^{-1}\int_0^x q(t;y)\,\mu_I(t;y)\,dt$. To carry out the integral, use the partial fraction

$$\frac{t-\zeta(y)}{t(t-1)} \;=\; \frac{\zeta(y)(t-1)+t\big(1-\zeta(y)\big)}{t(t-1)} \;=\; \frac{\zeta(y)}{t}+\frac{1-\zeta(y)}{t-1},$$

and define the auxiliary integrals

$$\mathcal{H}_1(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha-1}(1-t)^{\,\beta(y)}\,dt,$$
$$\mathcal{H}_2(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt,$$

with $\alpha, \beta(y)$ as in Equation (8.7). They are related to the full integrals of Step (ii) as follows: $\mathcal{H}_2$ has the same integrand as $\mathcal{J}_1$, so $\mathcal{H}_2(1,y) = \mathcal{J}_1(y)$ exactly; $\mathcal{H}_1$ differs from $\mathcal{J}_0$ in a single exponent —its $(1-t)$ power is $\beta(y)$ rather than $\beta(y)-1$— so $\mathcal{H}_1(1,y) \neq \mathcal{J}_0(y)$.

Multiplying $q(t;y)$ by $\mu_I(t;y) = e^{-\lambda_1 t/\theta_1} t^{\alpha}(1-t)^{\beta(y)}$ and inserting the partial fraction, the integrand splits into the two pieces defining $\mathcal{H}_1, \mathcal{H}_2$:

$$\begin{aligned} q(t;y)\,\mu_I(t;y) \;&=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\left[\frac{\zeta(y)}{t} + \frac{1-\zeta(y)}{t-1}\right] e^{-\lambda_1 t/\theta_1}\, t^{\alpha}(1-t)^{\beta(y)} \\ &=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\Big[\zeta(y)\,e^{-\lambda_1 t/\theta_1}\, t^{\alpha-1}(1-t)^{\beta(y)} - \big(1-\zeta(y)\big)\,e^{-\lambda_1 t/\theta_1}\, t^{\alpha}(1-t)^{\beta(y)-1}\Big], \end{aligned}$$

the second term again using $(1-t)^{\beta(y)}/(t-1) = -(1-t)^{\beta(y)-1}$, which produces both the minus sign and the exponent shift on $\mathcal{H}_2$. Since $\pi(0,0)$ and $\zeta(y)$ leave the integral,

$$\int_0^x q(t;y)\,\mu_I(t;y)\,dt \;=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\Big[\zeta(y)\,\mathcal{H}_1(x,y) \;-\; \big(1-\zeta(y)\big)\,\mathcal{H}_2(x,y)\Big].$$

Dividing by $\mu_I(x;y)$ and restoring $\zeta(y) = \widetilde{B}_C(\lambda_2(1-y))$, we arrive at the closed form

$$\begin{aligned} P(x,y) \;&=\; \frac{e^{\lambda_1 x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\, x^{\alpha}(1-x)^{\beta(y)}\big(\widetilde{B}_C(\lambda_2(1-y))-y\big)} \\ &\times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{H}_1(x,y) - \big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{H}_2(x,y)\Big], \end{aligned} \tag{8.13}$$

which is the expression in the statement of the theorem. □

The proof of Corollary 6, announced above, comes next.

Proof of Corollary 6. The idle system state, with limiting probability π_0, only communicates to the busy-server-empty-queues state with probability $\pi(0,0)$. No abandonments can take place in either state, as the queues are empty in both cases. By the cut identity of Theorem 1 (Equation (4.10)),

$$(\lambda_1+\lambda_2)\,\pi_0 \;=\; \mu\,\pi(0,0), \tag{8.14}$$

so $\pi(0,0) = (\rho_1+\rho_2)\,\pi_0$. We again use the $M/G/1$ structure already identified in the stability discussion: class-2 sees an $M/G/1$ queue with Poisson input λ_2 whose service time is one class-1 busy period B_C (made precise in §8.3). The long-run fraction of time this $M/G/1$ queue is busy is $\rho_B = \lambda_2\,\mathbb{E}[B_C]$, and "idle" refers exclusively to no class-2 customers being present. So

$$\mathbb{P}(\text{no class-2 present}) \;=\; 1-\lambda_2\,\mathbb{E}[B_C].$$

Conditional on not having class-2 customers, the priority subsystem evolves as the $M/M/1+M$ queue $(\lambda_1, \mu, \theta_1)$ defining B_C, which alternates idle and busy periods in a renewal structure, whose regenerations are the moments it empties. An idle period lasts until the next class-1 arrival takes place, i.e. at an exponential rate λ_1. We have

$$\mathbb{P}(\text{class-1 absent} \mid \text{no class-2 present}) \;=\; \frac{1/\lambda_1}{1/\lambda_1+\mathbb{E}[B_C]} \;=\; \frac{1}{1+\lambda_1\,\mathbb{E}[B_C]}.$$

The fully idle event is the intersection of $A = \{\text{class-1 absent}\}$ and $B = \{\text{no class-2 present}\}$, so $\mathbb{P}(A\cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B)$ (not independence),

$$\pi_0 \;=\; \big(1-\lambda_2\,\mathbb{E}[B_C]\big)\cdot\frac{1}{1+\lambda_1\,\mathbb{E}[B_C]} \;=\; \frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]},$$

which is Equation (8.4); substituting into Equation (8.14) gives Equation (8.3). Equivalently, this is the renewal–reward form $\pi_0 = 1/\big(1+\lambda\,\mathbb{E}[BP]\big)$ of Theorem 1 with whole-system mean busy period $\mathbb{E}[BP] = \mathbb{E}[B_C]/\big(1-\lambda_2\,\mathbb{E}[B_C]\big)$ —the mean busy period of the $M/G/1$ queue in which each class-2 arrival (rate λ_2) carries a service requirement of one class-1 busy period B_C. Theorem 1 thus applies verbatim; the model-specific work lies entirely in computing this mean busy period, which is non-trivial once abandonment is present. □

8.3 Probabilistic derivation of $P_y(y)$

The boundary function $P_y(y)$ admits a second derivation, which mirrors the probabilistic argument of §5.1.2 and makes no use of the boundedness argument of Step (ii). Recall that $P_y(y) = \mathbb{E}[\mathbf{1}\{\text{busy}, N_1 = 0\}\, y^{N_2}]$. By the *law of total expectation*, this factors into the probability of the event $\{\text{busy}, N_1 = 0\}$ and the conditional PGF of N_2 given that event:

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1 = 0)\;\cdot\;\mathbb{E}\big[y^{N_2}\,\big|\,\text{busy},\, N_1 = 0\big]\,. \tag{8.15}$$

We adopt the point of view of a class-2 customer while the server is busy. Unlike jockeying, abandonment removes class-1 customers from the system altogether rather than transferring them into Queue 2, so the class-2 input remains the exogenous Poisson stream of rate λ_2. The class-2 *effective service time* is therefore one full busy period B_C of the class-1 subsystem —an $M/M/1{+}M$ queue with rates λ_1, μ, θ_1, rather than the pure $M/M/1$ of Model-A. So class-2 sees an $M/G/1$ queue with Poisson input λ_2 and service distributed as B_C, whose LST $\widetilde{B}_C(s)$ and mean $\mathbb{E}[B_C]$ come from the first-step analysis of §3.4 (Equations (3.6) and (3.7), with $(\lambda,\theta) = (\lambda_1,\theta_1)$). This $M/G/1$ is stable if and only if its utilisation $\lambda_2\,\mathbb{E}[B_C] < 1$ (Equation (8.1)), a condition strictly weaker than the no-abandonment requirement $\rho = \rho_1 + \rho_2 < 1$.

Applying the Pollaczek–Khinchine formula (3.11) with the identifications $\lambda \mapsto \lambda_2$, $\widetilde{B} \mapsto \widetilde{B}_C$, $\rho_B = \lambda_2\,\mathbb{E}[B_C]$ gives the PGF of the stationary number of class-2 customers in that $M/G/1$:

$$L_2(y) \;=\; \frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}.$$

The identification of this $M/G/1$ law with the conditional law of N_2 given $\{\text{busy}, N_1 = 0\}$ is the exact analogue of the Model-A step, which §5.1.2 proves by watching the chain on the set $\{\text{busy}, N_1 = 0\}$. The same censoring viewpoint applies here, but the clearing spells opened by class-1 arrivals are no longer distributed as B_C —a spell begins with a *waiting*, hence abandonable, class-1 customer, whereas the busy period's initiator is in service and exempt from abandonment— so the one-line collapse of the Model-A computation is not available, and we take the identification over as the structural analogue:

$$\mathbb{E}\big[y^{N_2}\,\big|\,\text{busy},\, N_1 = 0\big] \;=\; L_2(y). \tag{8.16}$$

Substituting Equation (8.16) into the decomposition (8.15) gives

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1 = 0)\,\frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}. \tag{8.17}$$

The unknown $\mathbb{P}(\text{busy}, N_1 = 0)$ is determined by $P_y(0) = \pi(0,0)$: evaluating Equation (8.17) at $y = 0$, where $\widetilde{B}_C(\lambda_2)$ cancels,

$$\begin{aligned}\pi(0,0) = P_y(0) &= \mathbb{P}(\text{busy},\, N_1 = 0)\,\frac{(1-\lambda_2\mathbb{E}[B_C])\cdot 1\cdot \widetilde{B}_C(\lambda_2)}{\widetilde{B}_C(\lambda_2)}\\ &= \mathbb{P}(\text{busy},\, N_1 = 0)\,(1-\lambda_2\mathbb{E}[B_C]),\end{aligned}$$

so $\mathbb{P}(\text{busy}, N_1 = 0) = \pi(0,0)/(1-\lambda_2\mathbb{E}[B_C])$. Substituting back, the factor $(1-\lambda_2\mathbb{E}[B_C])$ cancels and we recover exactly Equation (8.12), in agreement with the analytical route. The result also matches the probabilistic argument for Model-A (Equation (5.8)); the only difference is the effective service time.

Finally, this route shows the Kummer fraction on the probabilistic side. The busy-period LST entering the Pollaczek–Khinchine formula is, from the Erlang-A analysis (§3.4, Equation (3.10)), the ratio of confluent hypergeometric functions

$$\widetilde{B}_C(s) \;=\; \frac{B(a+1,b)\,{}_1F_1(a+1;\, a+1+b;\, -c)}{B(a,b)\,{}_1F_1(a;\, a+b;\, -c)}, \qquad a = \frac{\mu}{\theta_1}, \quad b = \frac{s}{\theta_1}, \quad c = \frac{\lambda_1}{\theta_1}.$$

This is the expression (8.11) that the analytical route produced as the ratio of Beta integrals $\mathcal{J}_1(y)/\mathcal{J}_0(y)$ at $s = \lambda_2(1-y)$. The two derivations therefore meet at this Kummer fraction: what boundedness delivers as a ratio of Euler integrals is precisely the LST of the class-1 busy period.

9 Analysis of Model-B_2^{H}

Model-B_2^{H} restricts the jockeying of Model-B_2 to the head of the queue: the parameter setting is $\gamma_1 > 0$ head-of-line with $\gamma_2 = \theta_1 = \theta_2 = 0$. It is inspired by the work of Devos et al. [4], who analyse a *discrete-time* two-class model in which the server randomly alternates between the two queues. This model —together with Model-C_2^{H}— eliminates the state-dependent jockeying rate for class 1: only the head-of-line customer from class 1 may transition, at a fixed rate γ_1 whenever $n_1 \geq 1$, giving an aggregate jockeying rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$.

Stability. Jockeying conserves the total number of customers; since there is no abandonment, the chain is positive recurrent if and only if the total traffic intensity satisfies

$$\rho = \frac{\lambda_1 + \lambda_2}{\mu} < 1. \tag{9.1}$$

9.1 Balance equations

The jockeying transition $(n_1, n_2) \to (n_1 - 1, n_2 + 1)$ happens at the fixed rate γ_1 for $n_1 \geq 1$. On state space S the global balance equations read

$$\begin{aligned}
[\lambda_1 + \lambda_2 + \mu + \gamma_1]\, \pi(n_1, n_2) &= \mu\, \pi(n_1 + 1, n_2) + \lambda_1 \pi(n_1 - 1, n_2) \\
&\quad + \lambda_2 \pi(n_1, n_2 - 1) + \gamma_1 \pi(n_1 + 1, n_2 - 1), \quad n_1 \geq 1,\ n_2 \geq 1, \quad (9.2)\\
[\lambda_1 + \lambda_2 + \mu + \gamma_1]\, \pi(n_1, 0) &= \mu\, \pi(n_1 + 1, 0) + \lambda_1 \pi(n_1 - 1, 0), \qquad n_1 \geq 1,\ n_2 = 0, \quad (9.3)\\
[\lambda_1 + \lambda_2 + \mu]\, \pi(0, n_2) &= \mu\, \pi(1, n_2) + \gamma_1 \pi(1, n_2 - 1) \\
&\quad + \lambda_2 \pi(0, n_2 - 1) + \mu\, \pi(0, n_2 + 1), \qquad n_1 = 0,\ n_2 \geq 1, \quad (9.4)\\
[\lambda_1 + \lambda_2 + \mu]\, \pi(0, 0) &= (\lambda_1 + \lambda_2)\pi_0 + \mu\, \pi(1, 0) + \mu\, \pi(0, 1), \\
(\lambda_1 + \lambda_2)\pi_0 &= \mu\, \pi(0, 0). \quad (9.5)
\end{aligned}$$

9.2 Fundamental PGF equation

The head-of-line restriction significantly simplifies the model: the constant rate replaces the derivative term in x found in §7 by an algebraic factor $\gamma_1 y(x - y)$, so the fundamental equation remains algebraic and can be solved by the Kernel Method exactly as in Model-A (§5).

Lemma 4 (Fundamental equation for Model-B_2^{H})**.** *Assume positive recurrence, so that the stationary distribution exists and $P(x, y)$ is analytic on the closed unit polydisc. Then $P(x, y)$ and the boundary function $P_y(y) = P(0, y)$ satisfy*

$$\begin{aligned}
&\big[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2 + \gamma_1 y(x - y)\big]\, P(x, y) \\
&= \quad [\mu(x - y) + \gamma_1 y(x - y)]\, P_y(y) - \mu x(1 - y)\, \pi(0, 0).
\end{aligned} \tag{9.6}$$

Proof. Besides the jockeying terms, Equations (9.2)–(9.5) are identical to those of Model-A. Multiplying each balance equation by $x^{n_1} y^{n_2}$, summing over its domain and multiplying through by xy —leaving aside, for the moment, the terms involving γ_1— we obtain the fundamental equation for Model-A (Equation (5.2)):

$$\big[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big]\, P(x, y) = \mu(x - y) P_y(y) - \mu x(1 - y)\, \pi(0, 0),$$

so it remains only to collect the jockeying contribution. The γ_1-term on the left-hand side appears in every state with $n_1 \geq 1$:

$$\gamma_1 \sum_{n_1 \geq 1} \sum_{n_2 \geq 0} \pi(n_1, n_2)\, x^{n_1} y^{n_2} = \gamma_1 \big[P(x, y) - P_y(y)\big],$$

while the γ_1-term on the right-hand side enters every state with $n_2 \geq 1$ from $(n_1 + 1, n_2 - 1)$:

$$\gamma_1 \sum_{n_1 \geq 0} \sum_{n_2 \geq 1} \pi(n_1 + 1, n_2 - 1)\, x^{n_1} y^{n_2} = \gamma_1 \frac{y}{x} \sum_{m_1 \geq 1} \sum_{m_2 \geq 0} \pi(m_1, m_2)\, x^{m_1} y^{m_2} = \gamma_1 \frac{y}{x} \big[P(x, y) - P_y(y)\big],$$

where $m_1 = n_1 + 1$ and $m_2 = n_2 - 1$. The net jockeying contribution is hence $\gamma_1 \big(1 - \frac{y}{x}\big)\big[P(x, y) - P_y(y)\big]$; multiplying by xy gives $\gamma_1 y(x - y)\big[P(x, y) - P_y(y)\big]$. Adding this to the Model-A equation and moving the P_y part to the right-hand side yields Equation (9.6). □

9.3 Closed-form solution for the PGF

The following theorem is the main result of this section; its proof applies the Kernel Method, similarly to the analytical approach for Model-A in §5.1.1.

Theorem 5. *Consider the two-class non-preemptive priority queue with Poisson arrival rates λ_1, λ_2, common exponential service rate μ, and head-of-line class-1 jockeying at rate $\gamma_1 > 0$ (with $\gamma_2 = \theta_1 = \theta_2 = 0$). Under the stability condition (9.1), the joint PGF $P(x, y)$ is the rational function*

$$P(x,y) \;=\; \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*(y)-y\big)\,\big(x^+(y)-x\big)}, \tag{9.7}$$

where $x^*(y)$ and $x^+(y)$ are the two roots of the quadratic equation

$$\lambda_1 x^2 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big]x + (\mu + \gamma_1 y) = 0, \tag{9.8}$$

with $x^*(y)$ the root inside the unit disc.

The result below is stated as a corollary, in parallel with the corresponding results for the other models.

Corollary 7. *Under Equation (9.1), the empty-state probabilities of Model-B_2^{H} coincide with those of Model-A,*

$$\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho).$$

In particular $P(1,1) = 1-\pi_0 = \rho$ and the diagonal generating function satisfies $P(z,z) = \pi(0,0)/(1-\rho z)$.

Corollary 7 is an instance of Corollary 1 —the no-abandonment case of Theorem 1: head-of-line jockeying conserves the total number of customers, so $\mathbb{E}[BP] = (\mu-\lambda)^{-1}$ and the Model-A values hold.

9.3.1 Analytical approach for Model-B_2^{H}

Proof of Theorem 5. We apply the *Kernel Method* to the fundamental equation (9.6). Fix $y \in (0,1]$ and seek the values of x at which the coefficient of $P(x,y)$ vanishes. Since $P(x,y)$ is a PGF it is bounded on the closed unit disc, so at any such root the right-hand side of Equation (9.6) must also vanish. Dividing the kernel by $y \neq 0$ gives

$$0 = (\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy + \gamma_1(x-y),$$

and rearranging into a quadratic equation in x gives exactly Equation (9.8), whose two roots are

$$x^{\pm}(y) = \frac{A(y) \pm \sqrt{A(y)^2 - 4\lambda_1(\mu+\gamma_1 y)}}{2\lambda_1}, \qquad A(y) := \lambda_1 + \lambda_2(1-y) + \mu + \gamma_1.$$

We label the root carrying the negative sign $x^*(y)$ and the other $x^+(y)$. By Vieta's formulas the product of the roots is $x^*(y)\,x^+(y) = (\mu+\gamma_1 y)/\lambda_1$.

To locate the roots relative to the unit disc we evaluate the quadratic polynomial $f(x) := \lambda_1 x^2 - A(y)x + (\mu+\gamma_1 y)$ at $x = 1$:

$$f(1) = \lambda_1 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big] + (\mu + \gamma_1 y) = -(1-y)\,(\lambda_2+\gamma_1).$$

For $y \in [0,1)$ this is strictly negative, so $x = 1$ lies between the two roots and $x^*(y)$ is the unique root inside the unit disc; at $y = 1$ we obtain $f(1) = 0$, with $x^*(1) = 1$ and $x^+(1) = (\mu+\gamma_1)/\lambda_1 > 1$.

For later use, we determine $\frac{d}{dy}x^*(1)$ by implicit differentiation. Regard the kernel equation as $f(x,y) := \lambda_1 x^2 - A(y)x + (\mu+\gamma_1 y) = 0$. Its partial derivatives $f_y = \lambda_2 x + \gamma_1$ and $f_x = 2\lambda_1 x - A(y)$ evaluate at $(x,y) = (1,1)$ to $f_y(1,1) = \lambda_2 + \gamma_1$ and $f_x(1,1) = \lambda_1 - \mu - \gamma_1$, and the latter is nonzero under stability. By the implicit function theorem, $\frac{d}{dy}x^*(1) = -f_y(1,1)/f_x(1,1)$:

$$\frac{d}{dy}x^*(1) = \frac{\lambda_2+\gamma_1}{\mu+\gamma_1-\lambda_1}. \tag{9.9}$$

Compared with Model-A's Equation (5.6) evaluated at $y = 1$, which gives $\lambda_2/(\mu-\lambda_1)$, both numerator and denominator carry an extra γ_1: the constant term $\mu + \gamma_1 y$ of the kernel (9.8) is itself y-dependent here.

Setting $x = x^*(y)$ in Equation (9.6) makes the coefficient of $P(x, y)$ vanish; since $P(x^*(y), y)$ is finite, the right-hand side must vanish as well,

$$\big(x^*(y) - y\big)(\mu + \gamma_1 y)\, P_y(y) = \mu\, x^*(y)(1 - y)\, \pi(0, 0),$$

whence the boundary function is

$$P_y(y) = \frac{\mu\, x^*(y)\,(1 - y)\, \pi(0, 0)}{(\mu + \gamma_1 y)\,\big(x^*(y) - y\big)}. \tag{9.10}$$

Finally, substitute Equation (9.10) and $\pi(0, 0) = \rho(1-\rho)$ (Corollary 7) back into Equation (9.6). Writing the kernel as $-\lambda_1 y\,(x - x^*(y))(x - x^+(y))$ and simplifying the right-hand side,

$$\begin{aligned}\big[\mu(x - y) + \gamma_1 y(x - y)\big] P_y(y) - \mu x(1 - y)\, \pi(0, 0) &= \mu(1 - y)\, \pi(0, 0) \left[\frac{(x - y)\, x^*(y)}{x^*(y) - y} - x\right] \\ &= \mu(1 - y)\, \pi(0, 0)\, \frac{y\,\big(x - x^*(y)\big)}{x^*(y) - y},\end{aligned}$$

where the bracketed expression reduces to $y\big(x - x^*(y)\big)/\big(x^*(y) - y\big)$ by the identity $(x - y)x^*(y) - x\big(x^*(y) - y\big) = y\big(x - x^*(y)\big)$. Dividing by the factored kernel and cancelling the common factor $\big(x - x^*(y)\big)$, we have

$$P(x, y) = \frac{\mu(1 - y)\, \pi(0, 0)\, y\,(x - x^*(y))/(x^*(y) - y)}{-\lambda_1 y\,(x - x^*(y))(x - x^+(y))} = \frac{\mu\, \rho(1 - \rho)\,(1 - y)}{\lambda_1\,\big(x^*(y) - y\big)\,\big(x^+(y) - x\big)},$$

which matches Equation (9.7) and concludes the proof. □

10 Analysis of Model-C_2^{H}

Model-C_2^{H} is the abandonment counterpart of Model-B_2^{H}, and the head-of-line version of the class-1 abandonment model Model-C_2: the parameter setting is $\theta_1 > 0$ head-of-line with $\gamma_1 = \gamma_2 = \theta_2 = 0$. Only the *customer at the head of Queue 1* abandons, at the fixed rate θ_1 whenever $n_1 \geq 1$. As in Model-B_2^{H}, the constant rate $\theta_1\,\mathbf{1}_{\{n_1 \geq 1\}}$ contributes an *algebraic* term to the fundamental equation rather than a derivative term, and the Kernel Method applies directly. Because an abandoning customer leaves the system altogether, the conservation of the total number of customers —which held in every model without abandonment— no longer applies.

Stability. Because only the head-of-line customer from class-1 abandons, the aggregate abandonment rate is bounded by θ_1 regardless of the queue length —unlike in Model-C_2, where it grows proportionally to n_1 and stabilises the priority queue at any load. A non-trivial stability condition is therefore required. As shown in Corollary 8 below, the chain is positive recurrent if and only if $\pi_0 > 0$, that is

$$(\mu + \theta_1)(\mu - \lambda) + \lambda_1\theta_1 \;>\; 0, \qquad \lambda = \lambda_1 + \lambda_2. \tag{10.1}$$

This is *weaker* than $\rho < 1$: if $\rho < 1$ then $\mu - \lambda > 0$ and the condition holds trivially, but it can also hold with $\rho \geq 1$ provided θ_1 is large enough. Moreover, Equation (10.1) implies $\mu + \theta_1 > \lambda_1$, i.e. $\rho_C := \lambda_1/(\mu + \theta_1) < 1$.

10.1 Balance equations

The abandonment transition $(n_1, n_2) \to (n_1 - 1, n_2)$ occurs at the fixed rate θ_1 whenever $n_1 \geq 1$. Since both a service completion and a head-of-line abandonment from $(n_1 + 1, n_2)$ lead to (n_1, n_2), the two mechanisms enter the balance equations through the common in-flow coefficient $\mu + \theta_1$. On S,

$$\begin{aligned}
[\lambda_1 + \lambda_2 + \mu + \theta_1]\, \pi(n_1, n_2) &= (\mu + \theta_1)\pi(n_1 + 1, n_2) + \lambda_1\pi(n_1 - 1, n_2) \\
&\quad + \lambda_2\pi(n_1, n_2 - 1), \qquad\qquad n_1 \geq 1,\; n_2 \geq 1, \qquad (10.2)\\
[\lambda_1 + \lambda_2 + \mu + \theta_1]\, \pi(n_1, 0) &= (\mu + \theta_1)\pi(n_1 + 1, 0) + \lambda_1\pi(n_1 - 1, 0), \qquad n_1 \geq 1,\; n_2 = 0,\\
[\lambda_1 + \lambda_2 + \mu]\, \pi(0, n_2) &= (\mu + \theta_1)\pi(1, n_2) + \lambda_2\pi(0, n_2 - 1) \\
&\quad + \mu\, \pi(0, n_2 + 1), \qquad\qquad n_1 = 0,\; n_2 \geq 1,\\
[\lambda_1 + \lambda_2 + \mu]\, \pi(0, 0) &= (\lambda_1 + \lambda_2)\pi_0 + (\mu + \theta_1)\pi(1, 0) + \mu\, \pi(0, 1),\\
(\lambda_1 + \lambda_2)\pi_0 &= \mu\, \pi(0, 0). \qquad\qquad (10.3)
\end{aligned}$$

Remark 4 (The substitution $\mu \mapsto \mu + \theta_1$). Equations (10.2)–(10.3) are precisely the Model-A balance equations with the service rate μ replaced by $\mu + \theta_1$ in every transition out of a state with $n_1 \geq 1$; transitions out of states with $n_1 = 0$ keep the rate μ, since no waiting class-1 customer is present to abandon. The same substitution acts on the coefficient of $P(x, y)$ in the fundamental equation below, but not on its boundary terms, and it does not carry over to the stability condition (10.1), which instead follows from the busy-period identity of Theorem 1 (see the proof of Corollary 8).

10.2 Fundamental PGF equation

We introduce the following result:

Lemma 5 (Fundamental equation for Model-C_2^{H})**.** *Assume positive recurrence. Then the joint PGF $P(x, y)$ and the boundary function $P_y(y) = P(0, y)$ satisfy*

$$\begin{aligned} &\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 xy^2 + \theta_1 y(x-1)\right] P(x,y) \\ &= \quad [\mu(x-y) + \theta_1 y(x-1)]\, P_y(y) - \mu x(1-y)\, \pi(0,0). \end{aligned} \tag{10.4}$$

Proof. As in Lemma 4, deleting every term that contains θ_1 from Equations (10.2)–(10.3) leaves exactly the Model-A balance equations, which reproduce the Model-A fundamental equation (5.2) after multiplication by $x^{n_1} y^{n_2}$, summation, and multiplication by xy. It remains to collect the terms containing θ_1. The θ_1-term on the left-hand side appears in every state with $n_1 \geq 1$,

$$\theta_1 \sum_{n_1 \geq 1} \sum_{n_2 \geq 0} \pi(n_1, n_2)\, x^{n_1} y^{n_2} = \theta_1 \bigl[P(x,y) - P_y(y)\bigr],$$

while the θ_1-term on the right-hand side enters every state (n_1, n_2) from $(n_1 + 1, n_2)$,

$$\theta_1 \sum_{n_1 \geq 0} \sum_{n_2 \geq 0} \pi(n_1+1, n_2)\, x^{n_1} y^{n_2} = \theta_1\, x^{-1} \sum_{m_1 \geq 1} \sum_{n_2 \geq 0} \pi(m_1, n_2)\, x^{m_1} y^{n_2} = \theta_1\, x^{-1} \bigl[P(x,y) - P_y(y)\bigr],$$

with $m_1 = n_1 + 1$. The net contribution is $\theta_1 \bigl(1 - x^{-1}\bigr)\bigl[P(x,y) - P_y(y)\bigr]$; multiplying by xy gives $\theta_1 y(x-1)\bigl[P(x,y) - P_y(y)\bigr]$. Adding this to the Model-A equation and rearranging yields Equation (10.4). ☐

10.3 Closed-form solution for the PGF

The following theorem is the main result of this section; as for Model-B_2^{H}, its proof applies the Kernel Method of Model-A.

Theorem 6. *Consider the two-class non-preemptive priority queue with Poisson arrival rates λ_1, λ_2, common exponential service rate μ, and head-of-line class-1 abandonment at rate $\theta_1 > 0$ (with $\gamma_1 = \gamma_2 = \theta_2 = 0$). Under the stability condition (10.1), the joint PGF $P(x, y)$ is the rational function*

$$P(x,y) \;=\; \frac{\mu(\mu+\theta_1)\,(1-y)\,\pi(0,0)}{\lambda_1\,\bigl(x^+(y) - x\bigr)\,\bigl[(\mu+\theta_1 y)\,x^*(y) - y(\mu+\theta_1)\bigr]}, \tag{10.5}$$

where $x^*(y)$ and $x^+(y)$ are the two roots of the quadratic equation

$$\lambda_1 x^2 - \bigl[\lambda_1 + \lambda_2(1-y) + \mu + \theta_1\bigr]x + (\mu + \theta_1) = 0, \tag{10.6}$$

with $x^*(y)$ the root inside the unit disc, and $\pi(0,0)$ is given by Corollary 8.

The theorem invokes the following Corollary:

Corollary 8. *Under Equation (10.1), the empty-state probabilities of Model-C_2^{H} are*

$$\pi_0 = \frac{(\mu+\theta_1)(\mu-\lambda) + \lambda_1\theta_1}{\mu(\mu+\theta_1) + \lambda_1\theta_1}, \qquad \pi(0,0) = \frac{\lambda\bigl[(\mu+\theta_1)(\mu-\lambda) + \lambda_1\theta_1\bigr]}{\mu\bigl[\mu(\mu+\theta_1) + \lambda_1\theta_1\bigr]}, \tag{10.7}$$

with $\lambda = \lambda_1 + \lambda_2$. At $\theta_1 \to 0^+$ these reduce to $\pi_0 \to 1 - \rho$ and $\pi(0,0) \to \rho(1-\rho)$, recovering Model-$A$.

As $\theta_1 \to 0^+$, Equation (10.5) likewise recovers the Model-A PGF of Theorem 2 exactly.

Proof of Corollary 8. Two evaluations of the fundamental equation (10.4) suffice. First set $y = 1$: the term $\mu x(1-y)\pi(0,0)$ vanishes, the coefficient of $P(x,1)$ factors as $-(x-1)\big(\lambda_1 x - (\mu+\theta_1)\big)$, and the right-hand side becomes $(x-1)(\mu+\theta_1)P_y(1)$. Cancelling $(x-1)$ we obtain

$$P(x,1) \;=\; \frac{(\mu+\theta_1)\,P_y(1)}{\mu+\theta_1-\lambda_1 x} \;=\; \frac{P_y(1)}{1-\rho_C\,x}. \tag{10.8}$$

This geometric marginal shows that the priority queue clears at the effective rate $\mu+\theta_1$. Evaluating at $x = 1$ and using $P(1,1) = 1-\pi_0$ gives

$$1-\pi_0 \;=\; \frac{P_y(1)}{1-\rho_C} \quad\Longrightarrow\quad P_y(1) \;=\; (1-\pi_0)(1-\rho_C). \tag{10.9}$$

Second, set $x = 1$ in Equation (10.4). The coefficient of $P(1,y)$ collapses to $\lambda_2 y(1-y)$ and the right-hand side to $\mu(1-y)\big[P_y(y)-\pi(0,0)\big]$, yielding the following relation

$$\lambda_2\, y\, P(1,y) \;=\; \mu\big[P_y(y)-\pi(0,0)\big]. \tag{10.10}$$

Letting $y \to 1$, and using Equation (10.9) together with the idle-boundary balance equation (10.3) —$\pi(0,0) = \lambda\pi_0/\mu$, as in Theorem 1— we have

$$\lambda_2(1-\pi_0) \;=\; \mu\big[P_y(1)-\pi(0,0)\big] \;=\; \mu(1-\pi_0)(1-\rho_C)-\lambda\pi_0.$$

Expanding and collecting the π_0 terms,

$$\big[\mu(1-\rho_C)+\lambda-\lambda_2\big]\pi_0 = \mu(1-\rho_C)-\lambda_2, \qquad \text{i.e.} \qquad \pi_0 = \frac{\mu(1-\rho_C)-\lambda_2}{\mu(1-\rho_C)+\lambda_1},$$

where $\lambda-\lambda_2 = \lambda_1$. Substituting $\mu(1-\rho_C) = \mu(\mu+\theta_1-\lambda_1)/(\mu+\theta_1)$ and multiplying numerator and denominator by $\mu+\theta_1$, the numerator becomes $\mu(\mu+\theta_1-\lambda_1)-\lambda_2(\mu+\theta_1) = (\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1$ and the denominator $\mu(\mu+\theta_1-\lambda_1)+\lambda_1(\mu+\theta_1) = \mu(\mu+\theta_1)+\lambda_1\theta_1$, yielding the first identity in Equation (10.7); the second follows directly from $\pi(0,0) = \lambda\pi_0/\mu$. The denominator $\mu(\mu+\theta_1)+\lambda_1\theta_1$ is strictly positive, so $\pi_0 > 0$ is equivalent to the stability condition (10.1). Finally, rewriting the computed π_0 in the renewal–reward form $\pi_0 = 1/\big(1+\lambda\,\mathbb{E}[BP]\big)$ of Theorem 1 identifies the whole-system mean busy period as $\mathbb{E}[BP] = (\mu+\theta_1)/\big[(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1\big]$ —a by-product of the computation rather than an independent argument— which reduces to the $M/M/1$ value $(\mu-\lambda)^{-1}$ as $\theta_1 \to 0^+$. □

Remark 5 (The two evaluations at the singular corner $(1,1)$). At $(x,y) = (1,1)$ every term of Equation (10.4) vanishes —$(1,1)$ is a zero of the kernel and of both right-hand-side coefficients— so evaluating the equation at that point yields only $0 = 0$. Restricting the equation to the line $y = 1$ and to the line $x = 1$, and cancelling the vanishing linear factors $(x-1)$, respectively $(1-y)$, extracts instead two *directional* first-order pieces of information at this singular corner, and they are genuinely different: Equation (10.8) relates the class-1 marginal to $P_y(1)$, while Equation (10.10) relates the class-2 marginal to the whole boundary function. Together with the balance equation (10.3) they pin down the two unknown scalars π_0 and $\pi(0,0)$. The same device applies to the other models: for Model-A, the two evaluations combined with $\pi(0,0) = \lambda\pi_0/\mu$ yield $\pi_0 = 1-\rho$ directly, without invoking Theorem 2.

10.3.1 Analytical approach for Model-C_2^{H}

Proof of Theorem 6. We again apply the Kernel Method to Equation (10.4). Dividing the coefficient of $P(x,y)$ by $y \neq 0$ and rearranging gives the quadratic equation (10.6), whose roots are

$$x^{\pm}(y) = \frac{A(y) \pm \sqrt{A(y)^2 - 4\lambda_1(\mu+\theta_1)}}{2\lambda_1}, \qquad A(y) := \lambda_1 + \lambda_2(1-y) + \mu + \theta_1, \tag{10.11}$$

with product $x^*(y)\,x^+(y) = (\mu+\theta_1)/\lambda_1$ by Vieta's formulas. Evaluating the quadratic polynomial $f(x) := \lambda_1 x^2 - A(y)x + (\mu+\theta_1)$ at $x = 1$,

$$f(1) = \lambda_1 - A(y) + (\mu+\theta_1) = -\lambda_2(1-y).$$

For $y \in [0,1)$ this is strictly negative, so $x = 1$ lies between the two roots and $x^*(y) < 1 < x^+(y)$; at $y = 1$, $x^*(1) = 1$ and $x^+(1) = (\mu+\theta_1)/\lambda_1 > 1$ (the latter by $\rho_C < 1$). Hence, $x^*(y)$ is the unique root inside the unit disc. Differentiating Equation (10.11) at $y = 1$, we obtain

$$\frac{d}{dy}x^*(1) = \frac{\lambda_2}{\mu+\theta_1-\lambda_1}. \tag{10.12}$$

Setting $x = x^*(y)$ in Equation (10.4) makes the coefficient of $P(x, y)$ vanish; since P is bounded on the closed unit polydisc, the right-hand side must vanish as well,

$$\big[\mu(x^*(y) - y) + \theta_1 y(x^*(y) - 1)\big] P_y(y) = \mu x^*(y)(1 - y)\,\pi(0, 0),$$

and, regrouping the bracket as $\mu(x^*(y) - y) + \theta_1 y(x^*(y) - 1) = (\mu + \theta_1 y)\,x^*(y) - y(\mu + \theta_1)$, the boundary function is

$$P_y(y) = \frac{\mu\, x^*(y)\,(1 - y)\,\pi(0, 0)}{(\mu + \theta_1 y)\,x^*(y) - y(\mu + \theta_1)}. \tag{10.13}$$

Write $\Delta(y) := (\mu + \theta_1 y)x^*(y) - y(\mu + \theta_1)$ for the denominator of Equation (10.13); at $y = 0$, $\Delta(0) = \mu\, x^*(0)$ (with $x^*(0) > 0$), so $P_y(0) = \pi(0, 0)$, exactly as for Model-B_2^{H} at Equation (9.10).

Substituting Equation (10.13) into Equation (10.4) and writing the kernel as $-\lambda_1 y\,(x-x^*(y))(x-x^+(y))$, the right-hand side becomes

$$\big[\mu(x-y)+\theta_1 y(x-1)\big] P_y(y)-\mu x(1-y)\,\pi(0,0) = \frac{\mu(1-y)\,\pi(0,0)}{\Delta(y)} \Big\{\big[\mu(x-y)+\theta_1 y(x-1)\big]x^*(y)-x\,\Delta(y)\Big\}.$$

Substituting $\Delta(y)$, the braced expression simplifies to $(\mu + \theta_1)\,y\,(x - x^*(y))$. Dividing by the factored kernel cancels $(x - x^*(y))$, yielding

$$P(x, y) = \frac{\mu(\mu + \theta_1)(1 - y)\,\pi(0, 0)\,y\,(x - x^*(y))/\Delta(y)}{-\lambda_1 y\,(x - x^*(y))(x - x^+(y))} = \frac{\mu(\mu + \theta_1)(1 - y)\,\pi(0, 0)}{\lambda_1\,(x^+(y) - x)\,\Delta(y)},$$

which is Equation (10.5) and concludes the proof. □

11 Numerical verification and comparison

Every closed-form expression derived above is first verified against an exact *continuous-time Markov chain* (CTMC), then used to compare the solved models and quantify the effect of the two waiting-room mechanisms.

11.1 Verification against the CTMC

The method employed is a truncated CTMC: it works by solving the linear system $\pi Q = 0$ directly, with no further approximation than the finite truncation at $N_{\max} = 40$. Each result is therefore verified against the exact limiting distribution at base parameters $\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1.0$, with the active mechanism parameter γ_1 or θ_1 set to 0.5 where applicable.

Table 4 collects the verdict for every result. All results are admitted within a machine tolerance level: lemma residuals and closed-form PGF errors fall between 10^{-7} and 10^{-15}, and the corollary checks —π_0 and $\pi(0, 0)$ against the CTMC, across 60 random parameter configurations each— agree to within 10^{-4}.

Table 4: Verification summary. All mathematical results are accepted. Max errors reflect CTMC truncation, not formula error.

ID	Result	Model(s)	Check	Max error	Verdict
L1	Lemma 1	A, B, B_2, C_2	PGF eq. residual	$< 5 \times 10^{-9}$	Accept
L2	Lemma 2	A, B_2, C_2	PPGF dynamics residual	$< 2 \times 10^{-15}$	Accept
T-A	Theorem 2	A	$P(x, y)$ on 5×5 grid	$< 4 \times 10^{-8}$	Accept
C-A	Corollary 1	A, B, B_2	π_0, $\pi(0, 0)$, 60 configurations	$< 10^{-4}$	Accept
T-B2	Theorem 3	B_2	$P(x, y)$ on 5×5 grid, $x < y$	$< 3 \times 10^{-7}$	Accept
T-C2	Theorem 4	C_2	$P(x, y)$ on 5×5 grid	$< 5 \times 10^{-12}$	Accept
C-C2	Corollary 6	C_2	π_0, $\pi(0, 0)$, 60 configurations	$< 10^{-4}$	Accept
E-B	Theorem 5	B_2^{H}	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-7}$	Accept
E-C	Theorem 6	C_2^{H}	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-11}$	Accept

The PPGF Approximation 1 for Model-A on the state space $\widetilde{S}$ is assessed separately, since its check is of a different kind: it does not verify an exact result but measures the accuracy of an *approximation*. Its sup-norm error ε_∞ over the (ρ_1, ρ_2) plane ranges from 0 to 100%, and the approximation is conditionally accepted, being accurate only in a priority-dominated setup

($\rho_1/\rho \geq 0.6$ and $\rho \leq 0.6$). This was expected, as the approximation only neglected the problematic non-homogeneous term in Lemma 2, coming from the boundary-diagonal term.

Rows C-A and C-C2 combined confirm Theorem 1: the identity $\pi(0,0) = \rho\,\pi_0$ and the busy-period law $\pi_0 = 1/(1 + \lambda\,\mathbb{E}[BP])$ hold to $< 10^{-4}$ across both the no-abandonment regime (Corollary 1, Models A, B, B_2) and the class-1 abandonment regime (Corollary 6, Model-C_2), confirming the single scalar $\mathbb{E}[BP]$ as the only model-dependent input.

11.2 Structural comparison of the solved models

With every closed form verified, we compare the five solved models: Model-A (baseline), Model-B_2 (class-1 jockeying), Model-C_2 (class-1 abandonments) and the two head-of-line variants Model-B_2^{H} and Model-C_2^{H}. The first three are more tractable specialisations of Model-X. The head-of-line models are not special cases of Model-B_2 and Model-C_2 but companion models with a modified —head-of-line— mechanism; it is their governing equations, rather than the models themselves, that are simpler. Each produces a closed-form or integral-form expression for $P(x,y)$. Table 5 collects the key structural facts across all five solved models, alongside the open Model-B: although $P(x,y)$ remains undetermined there, its conservation law, empty-state probabilities and stability condition are known exactly (Corollary 4), so the structural comparison can include it.

Table 5: Structural comparison of the five solved models and the open Model-B. Replacing a length-proportional mechanism rate ($\gamma_1 n_1$ or $\theta_1 n_1$) by a head-of-line rate ($\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$) collapses the governing ODE back to the algebraic Model-A form, so $P(x,y)$ becomes rational again.

	Model-A	**Model-B** (open)	**Model-B_2**	**Model-B_2^{H}**	**Model-C_2**	**Model-C_2^{H}**
Extra mechanism (rate)	none	jockey $1 \leftrightarrow 2$, $\gamma_1 n_1$, $\gamma_2 n_2$	jockey $1 \to 2$, $\gamma_1 n_1$	jockey $1 \to 2$, $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$	abandon, $\theta_1 n_1$	abandon, $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$
Fundamental equation	algebraic	1st-order PDE	1st-order ODE	algebraic	1st-order ODE	algebraic
Pinning of $P_y(y)$	kernel root $x^*(y)$	open — Volterra eq.	analyticity at $x = y$	kernel root $x^*(y)$	boundedness at $x = 1$	kernel root $x^*(y)$
$P(x,y)$ form	rational	integral, unknown P_y	integral $+\,{}_1F_1$	rational	integral $+\,{}_1F_1$	rational
Conserves total N?	yes	yes	yes	yes	no	no
$\pi(0,0)$	$\rho(1-\rho)$	$\rho(1-\rho)$	$\rho(1-\rho)$	$\rho(1-\rho)$	(8.3)	(10.7)
π_0	$1-\rho$	$1-\rho$	$1-\rho$	$1-\rho$	$> 1-\rho$, (8.4)	(10.7)
Stability condition	$\rho < 1$	$\rho < 1$	$\rho < 1$	$\rho < 1$	$\lambda_2 \mathbb{E}[B_C] < 1$	(10.1)

Model-A: the rational baseline

Model-A is classical and well understood, so it acts as the baseline to which every other model should reduce as its active mechanism parameters approach zero. All of the performance measures used in this chapter —the mean queue lengths $\mathbb{E}[N_1]$, $\mathbb{E}[N_2]$, $\mathbb{E}[N]$, the waiting times $\mathbb{E}[W_1]$, $\mathbb{E}[W_2]$, and the empty-state probabilities π_0, $\pi(0,0)$— admit closed forms for Model-A. It also exhibits the structural identities that persist across the models without abandonment —notably conservation of the total number of customers— and against which the deviations of the abandonment models are measured.

Effect of jockeying: Model-A $\to$ Model-B_2

Adding one-way jockeying ($\gamma_1 > 0$, $\gamma_2 = 0$) couples the two queues: a class-1 customer waiting in Queue 1 may transition to the queue for class-2 at rate γ_1. The effect on the two classes is asymmetric.

What is preserved? Jockeying merely reallocates customers, so the total count of customers in the queues is preserved and remains to be that of the standard $M/M/1$ queueing model. Consequently $P(z,z)$, π_0, $\pi(0,0)$, and $\mathbb{E}[N]$ are identical to Model-A for all $\gamma_1 > 0$.

What no longer holds? Jockeying breaks the Poisson structure of the class-2 input: customers join Queue 2 either by an exogenous Poisson arrival or by jockeying from Queue 1, and the latter occurs at a state-dependent rate. The Pollaczek–Khinchine argument used for Model-A —which treats Queue 2 as an $M/G/1$ queue whose service times are class-1 busy periods (§5.1.2)— is therefore not available.

What is new? The PGF satisfies a first-order linear ODE in x, for each fixed y. The solution is found by means of an integrating factor with a singularity at the point $x = y$; therefore the singularity must vanish since $P(x, y)$ is a PGF and, hence, analytic. This narrows it down to a ratio of Kummer functions.

Effect of abandonment: Model-A → Model-C_2

Adding class-1 abandonment ($\theta_1 > 0$, $\theta_2 = 0$, $\gamma_i = 0$) allows each waiting class-1 customer to leave the system at a per-customer rate θ_1.

What is preserved? Class-2 customers still see Poisson arrivals, as abandonment in class-1 does not alter the stream of arrivals to the queue for class-2. Therefore, the PASTA property still applies, and so does the Pollaczek–Khinchine probabilistic approach to derive $P_y(y)$.

What no longer holds? Since abandonment removes customers from the system altogether, the total number of customers is no longer preserved, requiring a more mathematically involved analysis compared to Corollary 3 to determine the stability conditions, the idle system probability π_0 and the busy-server-empty-queues probability $\pi(0, 0)$.

What is new? The PGF satisfies a first-order linear ODE in x, for fixed y, as in Model-B_2. Here the singularity is found at the boundary $x = 1$, with a positive exponent so that the integrating factor μ_I vanishes there and $\int_0^1 q\,\mu_I\,dt = 0$ is imposed by boundedness of the PGF.

Mechanism contrast: Models B_2 and C_2

Both models reduce the fundamental equation to a first-order linear ODE in x and both solutions are expressed in terms of confluent hypergeometric functions; but the mechanism that determines the boundary function $P_y(y)$ is structurally different. In the table below, μ_I denotes the integrating factor of the respective ODE and $\beta(y)$ the exponent with which μ_I vanishes or blows up at the operative singular point, as defined in Equations (7.4) and (8.7).

	Model-C_2 ($\theta_1 > 0$)	**Model-B_2** ($\gamma_1 > 0$)
Derivative-term coefficient	$\theta_1\, x(x-1)$	$\gamma_1\, x(x-y)$
Operative singular point	boundary $x = 1$	interior $x = y$
Exponent at singularity	$\beta(y) = \lambda_2(1-y)/\theta_1 > 0$ $(\mu_I \to 0)$	$\beta(y) = (1-y)(\lambda y-\mu)/(\gamma_1 y) < 0$ $(\mu_I \to \infty)$
Pinning principle	*boundedness*: $\int_0^1 q\,\mu_I\,dt = 0$	*analyticity*: branch coefficient $= 0$
$\pi(0,0)$	$M/G/1$+renewal (conservation broken)	$\rho(1-\rho)$ (conservation holds)
Probabilistic backbone	class-2 is $M/G/1$; P-K with B_C	class-1 is $M/M/1{+}M$ (reneging $= \gamma_1$); no P-K

For Model-C_2 the singularity lies on the boundary $x = 1$ with a positive exponent, so μ_I vanishes and $P_y(y)$ is determined by the PGF being bounded. The arrival stream for class-2 remains Poisson, so the Pollaczek–Khinchine and $M/G/1$ renewal argument yield $P_y(y)$ and $\pi(0, 0)$. In contrast, for Model-B_2 the singularity is found at the interior point $x = y$ with a negative exponent, so μ_I blows up and only analyticity determines $P_y(y)$. Jockeying leaves no room for the Pollaczek–Khinchine identity; the $M/M/1{+}M$ reading of the class-1 marginal is the origin of the ${}_1F_1$ factors.

Common Erlang-A structure

A common object appears in the PGF of both Model-B_2 and Model-C_2:

$$ {}_1F_1\left(1;\, \frac{\mu}{r} + 1;\, \frac{\lambda_1}{r}\right), \tag{11.1} $$

with $r = \gamma_1$ (Model-B_2) or $r = \theta_1$ (Model-C_2). This comes from the Erlang-A substructure: in Model-C_2 this Kummer value equals $\mu\,\mathbb{E}[B_C]$, while in Model-B_2 the same function —with exponent $\alpha(y) = \mu/(\gamma_1 y)$— governs the Kummer ratio in $P_y(y)$. In both cases, the function evaluates a negative argument ($-\lambda_1/r$ in Model-C_2 and $-\lambda_1 y/\gamma_1$ in Model-B_2), and the ratio ${}_1F_1(\alpha+1;\cdot;\cdot)/{}_1F_1(\alpha;\cdot;\cdot)$ acts as a boundary correction to the algebraic formula found in Model-A.

Model-B_2 reduces to Model-A only in the limit $\gamma_1 \to 0^+$, in which the exponents $\alpha(y), \beta(y) \to \infty$ and the Kummer ratio collapses to $x^*(y)(1-y)/(x^*(y)-y)$, the Model-A boundary function (5.8) up to the factor $\pi(0,0)$. Similarly, Model-C_2 degenerates to Model-A uniformly as $\theta_1 \to 0^+$, and all formulas recover their Model-A counterparts.

Head-of-line simplifications: Models B_2^{H} and C_2^{H}

The head-of-line variants of §§9–10 restrict each mechanism to the single customer at the head of the queue for class 1, replacing the length-proportional rate ($\gamma_1 n_1$ or $\theta_1 n_1$) by a constant one ($\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$). As a consequence, the derivative terms in our fundamental equations disappear, leaving behind an algebraic expression. For Model-B_2^{H} the result is structurally identical to Model-A, the only change being $\mu \to \mu + \gamma_1 y$ in the kernel, and conservation is retained ($\pi_0 = 1 - \rho$, $\pi(0,0) = \rho(1-\rho)$). For Model-C_2^{H} the broken conservation of the number of total customers introduces the factor $(\mu+\theta_1)$ and the stability boundary shifts to Equation (10.1). Each head-of-line model may thus be viewed as the $k = 1$ member of a family of models indexed by k, the number of waiting class-1 customers subject to the mechanism (rate $\gamma_1 \min(n_1, k)$ or $\theta_1 \min(n_1, k)$): the family starts at Model-A ($k = 0$), passes through the head-of-line models ($k = 1$), and reaches the full-rate parents Model-B_2 and Model-C_2 as $k \to \infty$. This family is taken up again in the future work of §12.3.

11.3 Quantitative comparison across load

Keeping $\mu = 1$ and the arrival split fixed at $\lambda_1{:}\lambda_2 = 3{:}4$, we vary the offered load ρ over $[0.10, 0.95]$ and compute the exact stationary performance measures of the five solved models with the truncated CTMC. The open Model-B ($\gamma_1 = 0.5$, $\gamma_2 = 0.3$) joins this comparison on the same footing: every value here is computed from the truncated CTMC, so the missing closed form for its PGF is no obstacle to comparing it quantitatively. Figure 8 plots $\mathbb{E}[N]$, $\mathbb{E}[N_1]$, $\mathbb{E}[W_1] = \mathbb{E}[N_1]/\lambda_1$ (Little's Law), and π_0; Table 6 reads the entire set of descriptors at three reference loads.

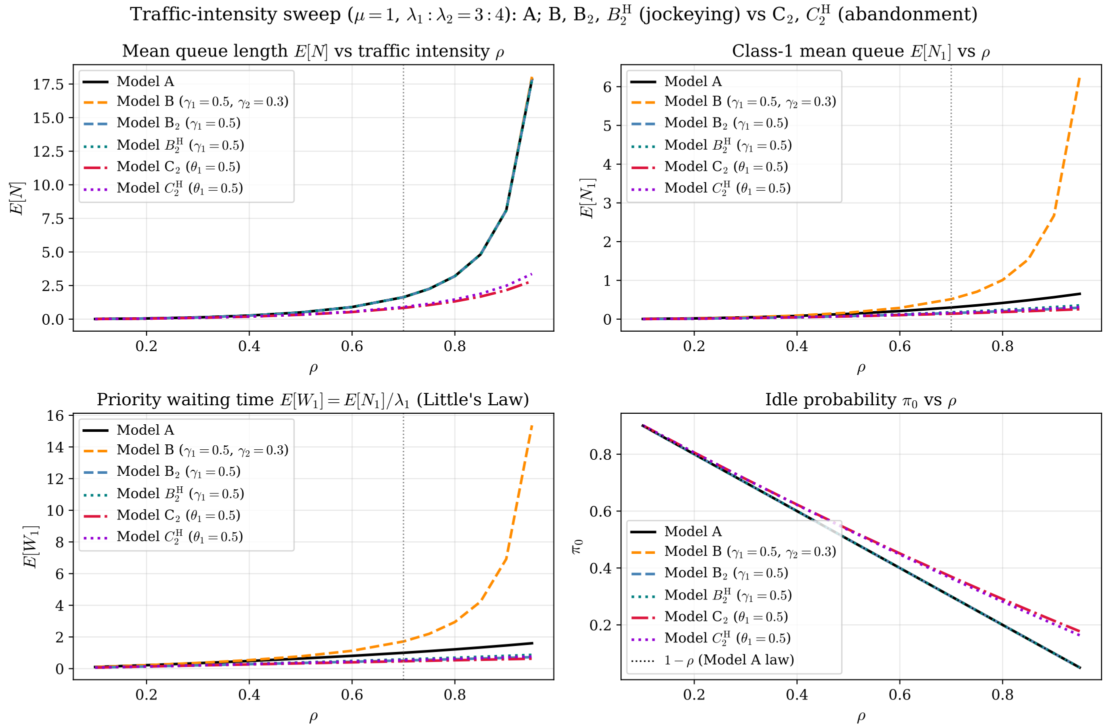

Figure 8: Performance versus traffic intensity ρ for the five solved models and the open Model-B —the baseline A, the jockeying models B ($\gamma_1 = 0.5$, $\gamma_2 = 0.3$), B_2 and B_2^{H} ($\gamma_1 = 0.5$), and the abandonment pair C_2, C_2^{H} ($\theta_1 = 0.5$)— at $\mu = 1$ and $\lambda_1{:}\lambda_2 = 3{:}4$; the vertical line marks the canonical $\rho = 0.70$. *Top-left:* total queue $\mathbb{E}[N]$ —A, B, B_2 and B_2^{H} coincide because jockeying conserves N (Corollaries 3, 4, 7), while C_2 and C_2^{H} lie strictly below because abandonment removes customers (C_2^{H} by less). *Top-right:* priority queue $\mathbb{E}[N_1]$ —the head-of-line variants drain it less than their full-rate counterparts, and the two-way Model-B *lengthens* it. *Bottom-left:* priority waiting time $\mathbb{E}[W_1]$. *Bottom-right:* idle probability π_0, with the $1 - \rho$ law (Models A, B, B_2, B_2^{H}) shown dotted; C_2 and C_2^{H} lie above it for every ρ, as predicted by Corollaries 6, 8.

The two mechanisms differ fundamentally in the way they operate. Jockeying is an internal rearrangement of customers, whereas abandonment removes customers from the system. This distinction has direct consequences. Abandonment reduces the overall queue length and raises the idle probability, whereas jockeying merely redistributes the load.

This is why B_2 and A are virtually indistinguishable: jockeying alone cannot reduce the total number of customers in the system. Their curves for $\mathbb{E}[N]$ and π_0 overlap across all traffic intensities. The same holds for the two-way Model-B: its $\mathbb{E}[N]$, π_0 and $\pi(0,0)$ values coincide with Model-A's at every load, which is a direct numerical confirmation of Corollary 4 for the one model whose PGF remains open. Its per-class split, however, moves in the *opposite* direction to B_2's: because the priority discipline keeps Queue 2 long, the return flow $\gamma_2 n_2$ outweighs the drain $\gamma_1 n_1$ even though $\gamma_2 < \gamma_1$, and the two-way exchange *lengthens* the priority queue —at $\rho = 0.70$, Table 6 shows $\mathbb{E}[N_1]$ rising to 0.51, against 0.3000 for A and 0.1545 for B_2. Model-C_2, by contrast, stands apart. The priority class sheds load before it reaches the server, keeping both the queue shorter and idle time higher.

The numbers back the claim. At $\rho = 0.70$ (Table 6), Model-C_2 achieves $\pi_0 = 0.3696$ compared to 0.3000 for Model-A and Model-B_2, with $\mathbb{E}[N] = 0.8358$ against 1.6333. At higher loads, the gap widens: $\rho = 0.90$ sees the baseline class-2 queue increase to $\mathbb{E}[N_2] = 7.5349$, while Model-C_2 keeps it at 1.9245. The two head-of-line models occupy a middle ground, interpolating between A and their full-rate parents across every metric: Model-B_2^{H} does not alter the queue-conservation property of Model-B_2 but shifts the class-1 share, whereas Model-C_2^{H} sheds less load than Model-C_2 ($\pi_0 = 0.3636$, $\mathbb{E}[N] = 0.9015$ at $\rho = 0.70$).

Two remarks are in order. First, $\mathbb{E}[W_1]$ is smallest for Model-C_2 at every load. Here $\mathbb{E}[W_i] = \mathbb{E}[N_i]/\lambda_i$ is obtained from Little's Law applied to queue i with the full arrival rate λ_i, so it measures the mean time spent waiting in queue i per *arriving* class-i customer, where a customer's wait ends when it enters service *or abandons*. Abandoning customers are thus counted with their time until abandonment, and a smaller $\mathbb{E}[W_1]$ reflects every departure mode rather than faster access to

Table 6: Comparison of Models A, B ($\gamma_1 = 0.5$, $\gamma_2 = 0.3$), B_2 and B_2^{H} ($\gamma_1 = 0.5$), and C_2 and C_2^{H} ($\theta_1 = 0.5$) at three traffic intensities with $\lambda_1{:}\lambda_2 = 3{:}4$, $\mu = 1$. The head-of-line models B_2^{H} and C_2^{H} use the rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$, respectively $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$; Model-$B_2$ and Model-C_2 use the full rates $\gamma_1 n_1$ and $\theta_1 n_1$. Model-B, whose PGF remains open, joins through its truncated-CTMC solution —like every entry of this table— since no closed form is required for the comparison.

ρ	Model	$E[N_1]$	$E[N_2]$	$E[N]$	$E[W_1]$	$E[W_2]$	π_0	$\pi(0,0)$
0.50	A	0.1364	0.3636	0.5000	0.6364	1.2727	0.5000	0.2500
	B	0.1665	0.3335	0.5000	0.7772	1.1671	0.5000	0.2500
	B_2	0.0767	0.4233	0.5000	0.3578	1.4817	0.5000	0.2500
	B_2^{H}	0.0833	0.4167	0.5000	0.3889	1.4583	0.5000	0.2500
	C_2	0.0712	0.2521	0.3233	0.3323	0.8823	0.5356	0.2678
	C_2^{H}	0.0778	0.2593	0.3370	0.3630	0.9074	0.5333	0.2667
0.70	A	0.3000	1.3333	1.6333	1.0000	3.3333	0.3000	0.2100
	B	0.5149	1.1184	1.6333	1.7164	2.7961	0.3000	0.2100
	B_2	0.1545	1.4788	1.6333	0.5151	3.6970	0.3000	0.2100
	B_2^{H}	0.1750	1.4583	1.6333	0.5833	3.6458	0.3000	0.2100
	C_2	0.1392	0.6967	0.8358	0.4639	1.7417	0.3696	0.2587
	C_2^{H}	0.1591	0.7424	0.9015	0.5303	1.8561	0.3636	0.2545
0.90	A	0.5651	7.5346	8.0997	1.4651	14.6506	0.1000	0.0900
	B	2.6711	5.4289	8.1000	6.9250	10.5563	0.1000	0.0900
	B_2	0.2626	7.8371	8.0997	0.6808	15.2388	0.1000	0.0900
	B_2^{H}	0.3115	7.7882	8.0997	0.8077	15.1437	0.1000	0.0900
	C_2	0.2292	1.9245	2.1537	0.5941	3.7421	0.2146	0.1931
	C_2^{H}	0.2760	2.2084	2.4844	0.7157	4.2941	0.2025	0.1823

service. Second, jockeying actively *penalises* class 2: at $\rho = 0.70$, Model-B_2 raises $\mathbb{E}[W_2]$ to 3.6970 from the baseline 3.3333 (+10.9%).

11.4 Convergence to the baseline

Each of the four simplified models must recover the baseline Model-A as their parameters for the active mechanisms approach zero: Model-B_2 and Model-B_2^{H} as $\gamma_1 \to 0^+$, Model-C_2 and Model-C_2^{H} as $\theta_1 \to 0^+$. Some of these limits were already established analytically in the model chapters —e.g. $\pi_0 \to 1 - \rho$ and $\pi(0,0) \to \rho(1-\rho)$ (Corollary 6). Here we confirm both limits numerically, on a single synoptic axis, and quantify the rate of approach.

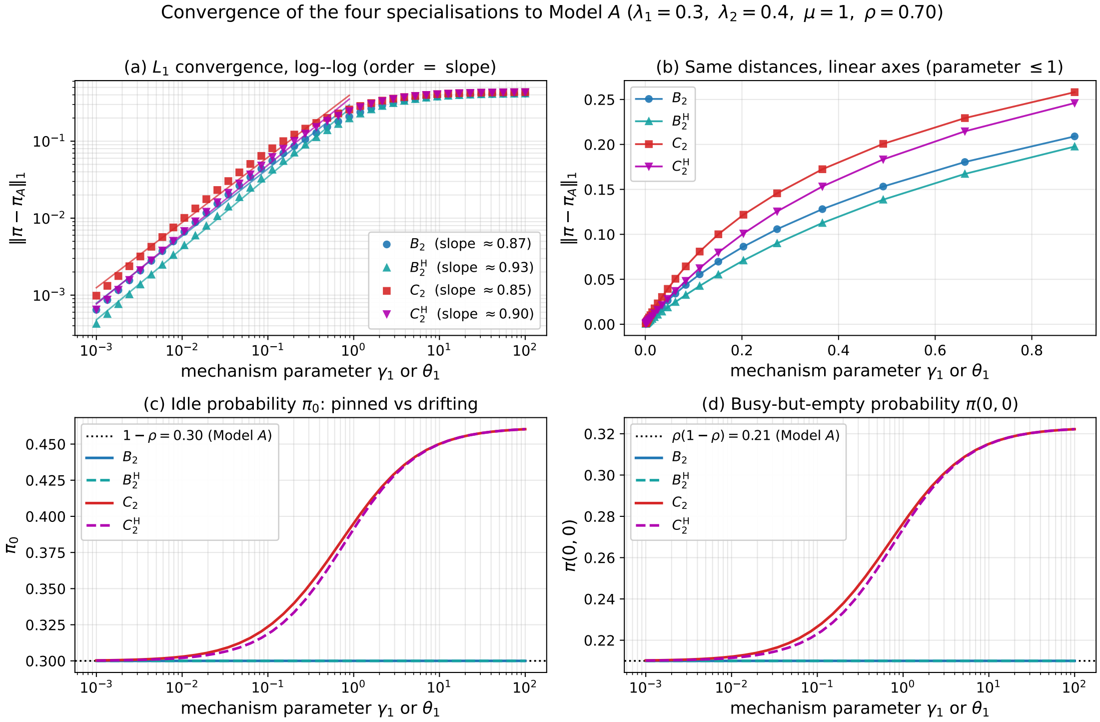

Figure 9: Convergence of the four specialisations to Model-A on a single axis ($\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1$, $\rho = 0.70$). *(a)* Log–log plot of the L_1 distance $\|\pi - \pi_A\|_1$ between the full joint distributions against the distinguishing mechanism parameter (γ_1 for the jockeying pair B_2, B_2^{H}; θ_1 for the abandonment pair C_2, C_2^{H}); all four collapse onto the baseline at first order, with the fitted log–log slopes shown in the legend. *(b)* The same distances on linear axes over the fitted range: the response grows linearly through the origin —the first-order behaviour read off directly, without logarithmic compression— and bends only as the mechanism saturates. *(c)* The idle probability π_0: the conserving models B_2 and B_2^{H} remain pinned to the Model-A value $1-\rho$ (dotted) for every γ_1, so they converge by interior redistribution alone, whereas C_2 and C_2^{H} drift above and relax back only as $\theta_1 \to 0^+$. *(d)* The same contrast for the busy-but-empty probability $\pi(0,0)$ against $\rho(1-\rho)$.

To measure how closely each simplified model approaches Model-A, we use the L_1 distance $\|\pi - \pi_A\|_1$ between their full joint distributions. As each model's distinguishing parameter decreases to zero (jockeying rate γ_1 for Model-B_2, abandonment rate θ_1 for Model-C_2, and so on), all four variants converge back to the baseline. We capture this convergence by plotting the L_1-distance against the parameter on a log–log scale: on these axes the order of convergence appears directly as the slope of the plotted line, which is what we measure. Panel (b) of Figure 9 shows the same distances on linear axes as a complementary view: the linear growth through the origin exhibits the first-order response directly.

Theory expects a slope of exactly 1, i.e. first-order convergence. Indeed, the generator of the truncated chain depends affinely on the mechanism parameter $\varepsilon \in \{\gamma_1, \theta_1\}$, and the stationary distribution of a finite irreducible chain is, by Cramer's rule, a rational —hence differentiable— function of its transition rates; a first-order expansion $\pi(\varepsilon) = \pi_A + \varepsilon\,\pi' + O(\varepsilon^2)$ with $\pi' \neq 0$ then gives $\|\pi(\varepsilon) - \pi_A\|_1 = \Theta(\varepsilon)$. Our measured slopes, shown in Figure 9(a), fall between 0.85 and 0.93, which is acceptable to confirm the theoretical prediction. The small gap below unity is attributable to the truncation of the CTMC: fitting a straight line to finitely many, slightly noisy points on a log–log plot pulls the measured slope down. The underlying convergence rate is indeed linear. Equivalently, the figure can be read as a sensitivity analysis: the slopes quantify the first-order impact of the mechanism parameters γ_1 and θ_1 on the stationary distribution.

The two mechanisms are shown to converge to the baseline along fundamentally different paths, visible in Figure 9(c) and (d) by tracking the idle-state probabilities π_0 and $\pi(0,0)$.

Under jockeying: these probabilities remain fixed at their Model-A values. By Corollaries 5 and 7, we have $\pi_0 = 1-\rho$ and $\pi(0,0) = \rho(1-\rho)$ identically for every $\gamma_1 > 0$, due to conservation of the total number of customers. Therefore, both Model-B_2 and Model-B_2^{H} lie exactly on the Model-A baseline throughout the entire parameter sweep. Their convergence to Model-A is invisible at the level of idle probabilities; instead, it happens entirely in the interior, where mass gradually redistributes between the class-1 and class-2 coordinates.

Under abandonment: the idle-state probabilities do change, because abandonment removes customers from the system. They only recover Model-A as $\theta_1 \to 0^+$. For Model-C_2 and Model-C_2^{H}, convergence is thus driven by both mechanisms: the idle state drifts, and the interior mass redistributes.

This conservation under jockeying versus drift under abandonment is the same conservation dichotomy that distinguishes the two mechanisms throughout the entire comparison (§11.2). In the convergence experiment it is visible along the entire parameter range, not merely in the limit: the conserving models remain pinned to the Model-A idle values for every γ_1, whereas the abandonment models deviate for every $\theta_1 > 0$ and reach those values only as $\theta_1 \to 0^+$.

12 Conclusion and Future Work

This thesis analysed a two-class non-preemptive priority $M/M/1$ queue under two waiting-room mechanisms —jockeying and abandonment. It obtained the exact stationary bivariate PGF $P(x, y)$ in closed or integral form for five models: the baseline Model-A, one-way $(1 \to 2)$ jockeying Model-B_2, class-1 abandonment Model-C_2, and the head-of-line variants Model-B_2^{H} and Model-C_2^{H}.

One common property is shared by all solved models: the form of the fundamental equation that the PGF satisfies —algebraic, or carrying derivative terms— determines the tractability of the model with the methods employed in this work. As a result, the collection of closed-form solutions is smaller than the model hierarchy might suggest: derivative terms in y complicate the problem, as one would need more advanced mathematical tools to determine the boundary function $P_y(y)$.

12.1 The obstruction and the mechanism dichotomy

Read as a competition between an algebraic coefficient and two derivative terms, the fundamental equation exposes a clean dichotomy. Where the equation is solvable, the unknown boundary function $P_y(y)$ is determined by exactly one of four mechanisms. The first is evaluation at a kernel root lying strictly inside the unit disc —the purely algebraic route of the baseline Model-A. The second is interior analyticity: when one-way jockeying survives, the factor $(x - y)$ places the operative singularity at the diagonal point $x = y$ inside the disc, and the boundary function is fixed by requiring analyticity there (Model-B_2). The third is boundary boundedness: when class-1 abandonment survives, the factor $(x - 1)$ moves the singularity to the unit-circle boundary $x = 1$, where mere boundedness of the PGF suffices to determine the unknown (Model-C_2). The fourth is head-of-line algebraic collapse: restricting a mechanism to the head-of-line customer replaces the length-proportional rate with a constant one, eliminates the derivative term in the equation, and recovers the purely algebraic form of Model-A, solvable by the same kernel argument (Models B_2^{H} and C_2^{H}).

The unsolvable cases —Model-B and Model-X— expose structural limits, as the derivative terms in y make the task of determining $P_y(y)$ significantly more challenging than it already was. Model-B requires solving a Volterra-type integral equation whose forcing term depends on the boundary function itself; Model-X further compounds this difficulty with non-linear characteristic curves, although we have not determined the exact difficulty or feasibility of the problem. The hierarchy is therefore not a loose collection of cases, but a complete enumeration: four mechanisms that enable exact solutions and two obstructions that prevent them.

The two mechanisms themselves act in opposite directions. Jockeying *redistributes* delay at fixed total work: relocating a waiting customer conserves the total count, so $\mathbb{E}[N]$, π_0 and $\pi(0, 0)$ are identical to the baseline for every γ_1. The mechanism's entire effect is to move delay between classes —the priority premium $\mathbb{E}[W_2]/\mathbb{E}[W_1]$ inflating to 22.38 at $\rho = 0.90$ against the Model-A value 10.00.

Abandonment instead *reduces* the load: the idle probability rises, every queue shortens, and total waiting times fall.

12.2 Limitations

Several assumptions are intrinsic to the $M/M/1$ framework itself. All results rest on exponential interarrival, service, jockeying, and abandonment times —an idealisation that is most restrictive for the service times. Allowing class-dependent rates $\mu_1 \neq \mu_2$ would invalidate the analysis at its foundation, since the state-space construction and the definition of the limiting probabilities both rely on the server being indifferent to the class in service. Relaxing exponentiality entirely to general service times further compounds the difficulty; the intractability that already forces de Waal's approximate $M/G/1$ treatment illustrates how quickly the problem ceases to admit closed-form solutions.

The partial-PGF result on the alternative state space $\widetilde{S}$ is an approximation, not a theorem, and should be read as a documented negative result. The rationale was as follows. Cohen [2] applied a partial generating function on the standard state space (n_1, n_2) with respect to the class-2 variable, leaving the tractable class-1 subsystem untouched and transforming only in the harder direction. In the presence of jockeying, neither class has simple dynamics individually, but the total queue length $N = N_1 + N_2$ does. The alternative state space (n_2, n) was therefore introduced to exploit this total-count structure. The attempt does not succeed: rather than simplifying the problem, the change of coordinates makes the balance equations harder even for Model-A.

12.3 Future work

Several extensions follow directly from the machinery already built:

1. *Head-of-line versions of the remaining models.* The head-of-line models could be extended with bidirectional active mechanisms —namely Model-B^{H} for jockeying and Model-C^{H} for abandonment— rather than limiting them to strictly one direction, as the derivative terms in the fundamental equation would also disappear. Even a head-of-line model carrying both active mechanisms at the same time, Model-X^{H}, would remain algebraic. These extensions appear to be within reach of the tools already developed here.

2. *The k-customer head-of-line ladder.* The head-of-line variants let only the single customer at the head of the queue for class 1 jockey or abandon; the natural extension lets the first k waiting class-1 customers do so, jockeying at rate $\gamma_1 \min(n_1, k)$ (and likewise abandoning at rate $\theta_1 \min(n_1, k)$). Writing $\min(n_1, k) = \sum_{j=1}^{k} \mathbf{1}_{\{n_1 \geq j\}}$, the bounded rate keeps the fundamental equation *algebraic* at the potential price of k unknown boundary functions. This route has not been explored in this work, so it is unclear what it could yield. One would expect that such models would converge to Model-B_2 and Model-C_2 as $k \to \infty$.

3. *Numerical solution of the Model-B Volterra equation.* Unlike Model-X, Model-B yields a first-kind Volterra equation with an *explicit* kernel and a *known* forcing term. Whether its solution can be characterised further —in closed form or through numerical approximation— is left open; either route lies beyond the scope of this thesis.

4. *Multi-server extension.* Carrying the mechanisms to $M/M/c$ would connect this single-server line to the multi-server setting. With a common service rate for both classes, the non-preemptive priority $M/M/c$ queue admits an exact waiting-time analysis [12], and exact methodology for two priority classes exists even with class-dependent rates under a preemptive discipline [14], so exactness need not be surrendered at the outset; the open question is whether jockeying and abandonment can be added while preserving it. This would also connect to the (approximate) proactive-scheduling results of [9].

5. *Derive mean performance metrics.* The closed-form PGFs obtained for Model-B_2, Model-C_2, Model-B_2^{H}, and Model-C_2^{H} contain the full distributional information, but extracting mean queue lengths and waiting times requires differentiating at $(1, 1)$, which in Model-B_2 and Model-C_2 involves boundary functions given as integrals. A systematic differentiation —applying L'Hôpital where necessary at the singularity $x = 1$— would yield closed-form expressions for $\mathbb{E}[N_1]$, $\mathbb{E}[N_2]$, and, via Little's law, $\mathbb{E}[W_1]$ and $\mathbb{E}[W_2]$, replacing the numerical estimates reported in the results chapter with exact ones.

6. *Calibration.* Fitting γ_1, θ_1 and the loads to real elderly-care waiting-list data [16] would test whether the regimes identified here accurately describe operational reality. Since the transplant waiting-list model of [16] differs from ours in several respects, such a calibration would likely require further model extensions rather than a direct fit of γ_1 and θ_1.

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